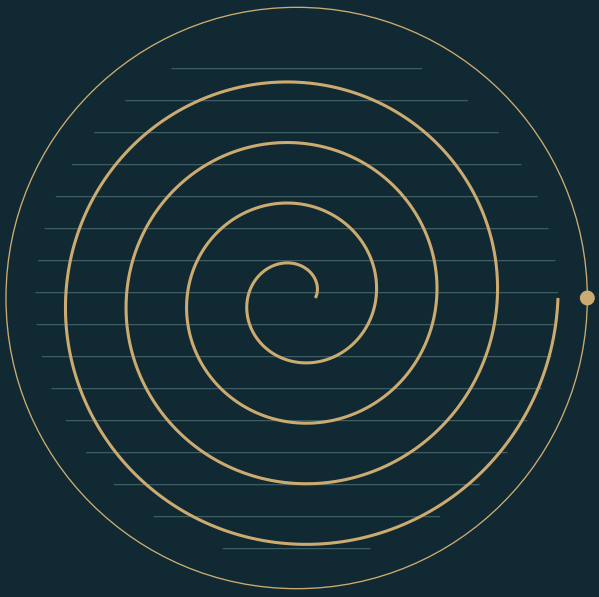
REVISED READING EDITION

The Ledger Principle

Memory, power, and who pays



SHEHZAD AHMED

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Shehzad Ahmed

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Copyright and edition

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The source notes distinguish documented cases, experiments, models, constructed illustrations and forecasts. The seven-day practice is a proposed method; the complete package has not been evaluated as an intervention.

Earlier editions are preserved in the Zenodo record family. The accompanying edition change log identifies the revised chapters.

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Prologue: The Pattern That Borrows Stuff

A leaf enters a whirlpool at the bend of a river. It travels around twice, catches the outward current, and is gone. Another leaf takes its place. You could stand there all afternoon and watch the same shape hold while the water that first made it passed downstream.

A rock sits a few feet below it. The rock keeps most of its material and slowly loses its edge. The whirlpool borrows new material to keep its edge. Take away the current and there is no damaged whirlpool resting on the riverbed. There is no whirlpool.

Your body and mind persist more like the whirlpool.

In 1953, the physicist Paul Aebersold estimated that roughly **98 percent of the atoms in a person are replaced within a year**. It was a whole-body estimate; tissues and elements turn over at different rates. You can feel the movement without a laboratory. You drink, breathe, eat, shed, exhale. Material leaves. More arrives. Yet a street learned in childhood can still guide your feet. A sentence spoken thirty years ago can change your pulse today.

Where did that sentence wait?

Calling it a recording invites us to imagine an original tape in a protected cabinet. But memory faces a more specific problem than whole-body turnover. Much of the molecular machinery at the junctions between neurons is renewed on timescales far shorter than the memories it helps carry. The arrangement has to remain usable as its components change.

Connections strengthen or weaken. Cells build and replace proteins. Activity returns to some paths and lets others fade. Neuroscientists investigate how these processes sustain the effects of learning over

time. A past encounter survives by changing the way a later encounter unfolds.

Think of a wrong turn on the way home. The error does not have to remain as an untouched picture of a corner. It has to make you hesitate there next time. What persists is a difference in what you can do.

Here the river image reaches its limit. The whirlpool keeps its form without remembering the leaf. You maintain a body that carries a history into new situations. You can compare the old warning with the road in front of you. You can also keep obeying it after the danger has passed. Maintenance preserves useful routes and practiced mistakes alike.

Then the pattern makes its great crossing. It leaves the body.

Imagine a house beside a river. After a flood, someone draws a short line on a wooden post to show how high the water reached. Years later, another person is deciding where to set the floor of a new house. The flood is gone. The person who drew the line is gone. The mark remains. The builder sets the floor above it.

One stranger paid for a discovery. Another inherits it without standing in the water.

The scratch can wait through silence. It can be copied, carried elsewhere, compared with other marks, and found by someone its maker never imagined. Experience has acquired a life longer than the life that acquired it.

Now read the mark again. It records the height of **one flood**. The builder takes it for the highest flood the river could ever bring. Nothing beside the line says “one year only.” The trace is faithful; the conclusion is wrong. When a higher flood comes, the family living in the house pays for a condition the record left out.

Everyone used the same mark. They did not use the same meaning.

A date beside the line would help. So would older marks, another village's account, or a person who remembers water above that height. But the correction has to reach the builder before the floor is set. A warning added to a file nobody opens can preserve a perfect account of a preventable loss.

A maintained record becomes a **ledger** when someone consults it to decide what happens next. It can be a line on a post, a recipe, a map, an account, a court docket, a story, or a machine's saved state. Their powers differ. A recipe advises; a court order can compel. Their shared capacity is to make yesterday available to act on today.

Did the line merely report the flood? It selected a place, a height, and an event. It left out a time span. Its reader supplied an interpretation, chose a floor height, and placed a family above the water she expected. The record helped determine who would stand in the next flood. Yesterday's observation had become one of tomorrow's causes.

The builder still made a choice. She could have measured more or asked around. Yet the mark put an observation in front of someone who had never seen the flood. Alter its label and you can alter her choice. A ledger needs no mind or motive of its own to acquire power. It needs keepers, readers, and decisions that continue to consult it.

More information can improve the arrangement. Fill the post with dates. Add photographs. Build a database of every river rise. Each step also places more authority somewhere: with the person who measures, the office that keeps the series, the institution that chooses which height an insurer or lender will use. A complete-looking record may be harder to challenge than a scratch on wood. Someone still has to get a warranted correction into the decision.

The Ledger Principle is this: **Systems that persist do it by maintaining a checkable record, and whoever writes the record decides who pays.**

Here, persistence means carrying learned history into a later action. Checkability means a route by which an entry can meet evidence, be

corrected, and change that action. The crucial political question is who gets to use the route. A keeper may inspect a record closely while the people governed by it struggle to see the page. Control of a record includes control of whose challenge is allowed to count.

The gift is immense. A child inherits tools, numbers, and stories that thousands of dead strangers spent their lives improving. She can begin beyond the limits they paid to reach. She also inherits their omissions. A faulty instruction travels as readily as a useful one. An old judgment can arrive in a new life with all its authority intact.

Civilization makes those judgments enforceable. An entry can say who owns a field, which company owes a debt, or whose injury qualifies for payment. Its placement can let one party collect today while another waits years to discover what remains. The record distributes the bill before anyone opens an argument about fairness.

Culture brings the same power closer. A fable can give a child a model of betrayal before she has met it. The words are portable; her memories and imagination complete them. Later, the remembered story can help her notice a danger - or make an innocent person resemble its villain. The record changes its reader, and the changed reader chooses what to record next.

Now machines write and consult records for other machines. They can pass an efficient code to successors, generate the material from which successors learn, and inspect entries faster than a human clerk. A green light can glow even when the checker read the wrong file. More speed gives the next decision less time to encounter a human objection.

The scale changes. Four questions keep their force: **Who keeps the record? Who can check and amend it? What decision does it guide? Who pays when it is wrong?**

You can follow them through a kitchen, a court, a bank filing, or a software log. They let us test the Ledger's power where it becomes a consequence someone has to live with.

Return to the river. The whirlpool cannot care whether the leaf survives. The flood mark cannot care whether the builder understands it. People can care. We made records so that what one person learned could reach another person in time. Their greatest promise and deepest danger begin at that crossing.

How does a discovery survive its discoverer - and what must the next person learn before the inheritance becomes useful?

Chapter 1: The Mark That Outlived Its Maker

The first move earns the bee nothing.

Inside a clear puzzle box is sugar. A red tab opens the way to it, but the red tab will not move until a blue tab has been pushed aside. Moving blue pays nothing. A bee working alone has every reason to give up at that point. The first action brings no reward, and the sweet solution stays hidden behind the second.

In a 2024 experiment, researchers trained bumblebees to solve this sequence. Even the demonstrators needed help at first: the researchers temporarily rewarded the blue step, then removed that reward. After training, an inexperienced bee could watch a demonstrator. Five of fifteen observers went on to open the box without ever receiving sugar for moving blue.

What did the successful learners get to see?

Some trained bees opened the box by squeezing past both tabs in one continuous turn. Five of ten observers paired with those demonstrators learned. Other demonstrators pushed the blue tab, moved around, then pushed red; none of the five observers paired with them learned. The researchers could not reduce this difference to the number of demonstrations.

The same box, the same reward, two ways of reaching it. In this experiment, the form of the example helped decide whether the solution traveled. A living teacher has to do more than succeed. Its action has to become legible to a learner.

Set this beside the simplest story people tell about human civilization. In that story, an animal discovers a trick, dies, and takes the trick with it. A human discovers a trick, passes it on, and history begins. The bee refuses its assigned part. A behavior that the observers did not work

out alone passed between animals. The laboratory did not give the bees a library. It did show that human beings do not own the basic escape route from the death of a knowledgeable individual.

Pigeons press the point further. In another experiment, a pigeon flew home from the same release point twelve times, learning a route. It was then paired with a bird that did not know the route. After twelve flights together, the more experienced bird was replaced by a new one. The replacements continued through five generations of pairs. The later pairs flew more efficient routes than birds training alone or pairs whose membership never changed. Consecutive generations in a chain took similar paths, evidence that something about the route had traveled from bird to bird.

No pigeon wrote down a turn. The record was in the pair's behavior, in the corrections each flight left available to the next. The route improved as its carriers changed. If a ratchet is a tool that lets a wheel advance one tooth without slipping all the way back, this was a small cultural ratchet made of wings.

So the first easy explanation is gone. Culture, and even some cumulative improvement, cannot be defined simply as whatever humans do. The question worth asking is sharper: what can a pattern do when it no longer needs to be shown by the living body that knows it?

The distance a lesson can travel

Stand beside someone who knows how to make bread. She presses the dough; you press yours. She adds a little water and tells you to stop when the mass yields in a certain way. She can see that your flour is drier than hers. She can interrupt before you make a mistake. If words fail, she can put your hand where the dough should be.

This is a powerful form of memory transfer. It has something a page will never have: a teacher who can respond to the learner in front of her. The bee's demonstrator and the pigeon's experienced partner

offered narrower versions of the same advantage. Their actions were present, not merely reported.

The dependence is its weakness. The right teacher must meet the learner at the right time. If the teacher moves, dies, or has nobody interested in the lesson, a useful distinction can vanish. A later learner may receive most of a practice and miss the one small movement that makes it work. Oral traditions can preserve intricate skills over long spans; they are not failed books waiting to be invented. They do, however, require a chain of people and occasions in which the practice can be performed, noticed, and repaired.

Now put a recipe on a page. The page cannot touch the dough or notice that the oven runs hot. But it can sit unopened for years. It can cross a sea. Someone who never met the writer can place it beside another recipe, spot a difference, and try both. A person yet unborn can argue with its author in the margin.

That change separates the life of an instruction from the life of its maker. It also lets a learner compare absent teachers. In the pigeon experiment, the earlier route reached the newcomer through a living partner. A written route can be opened beside an older version. The disputed turn becomes an object that strangers can inspect together.

The earliest external records were often humbler than a recipe. Before writing as we know it, people in the ancient Near East used small clay counters of different shapes to track goods. Denise Schmandt-Besserat, the archaeologist who studied these objects, describes a progression from loose tokens to tokens held in sealed clay envelopes. Some envelopes carried impressions on the outside that corresponded to the counters inside. A surviving envelope from Susa, in present-day Iran, dates to about 3300 BCE.

Hold the object in your mind. The contents are inside a clay ball. The outside carries marks. To know what is being claimed about the contents, a reader need not break the envelope. A count has acquired

an exterior. It can be checked against what was sealed away. A person can point to the same mark another person sees.

A clay tablet from Godin Tepe, dated to about 3100 BCE, makes the transition more striking. On Schmandt-Besserat's reading, its incised sign names oil and its circular and wedge impressions count thirty-three jars. The tablet does not preserve the jars. It preserves a claim about them. A holder of the tablet and a holder of the oil need not be standing together for the claim to remain available.

Spoken stories, tools, paths, and practiced skills had carried human histories long before those tablets. The clay added an inventory that could survive its counter. A later official could consult it. Two people could disagree over what arrived and point to a count made by someone who was no longer there.

To an untrained reader, thirty-three jars can look like a handful of dents. You can copy a Chinese character exactly without knowing what it says. A reader who has learned the language encounters the same strokes and finds meaning. Training supplies the difference. The tablet's reader needs its signs, the transaction they name, and the conditions under which the count was made. Meaning is not buried in the clay like oil in a jar. The durable surface gives later minds something from which to rebuild it.

An Egyptian stone lets us watch the rebuilding. In 196 BCE, a council of priests issued a decree concerning Ptolemy V. It appeared on stone in three scripts: hieroglyphs, demotic, and Greek. The two Egyptian scripts reached different readers; Greek served the administration. The inscription was made to be understood in its own world. Then the world of its readers changed. Hieroglyphic writing fell out of use, and the ability to read it disappeared. The marks survived their makers *and* the people trained to read them.

In 1799, soldiers at Rashid found a broken copy of the decree. Scholars could still read the Greek. That gave them a familiar account to place beside unfamiliar signs, much as a recipe in a language you know can

help you approach the same recipe in one you do not. But the stone was no automatic dictionary. Thomas Young identified signs spelling the royal name Ptolemy. Jean-François Champollion tested the sound principle more widely, drawing also on Coptic, a later form of Egyptian. His 1822 announcement depended on earlier comparisons and surviving languages as well as on the stone.

The decree had not changed while it lay unread. What changed was the route between a mark and a trained mind. Even this famous parallel record needed living readers who could compare, make a wrong guess, and correct it. Its three scripts were not three identical rows of a modern translation table. A surviving surface made recovery possible; it did not perform the recovery.

What the saved page leaves out

Return to the recipe. Imagine that its first line says to add a fixed amount of water. A later baker follows it and gets a sticky dough that collapses in the oven. The original cook is gone. The printed instruction remains perfectly legible. The later baker pays for the omitted condition in flour, fuel, and an evening's work.

What failed? It might have been the weather. It might have been a different flour. The cook might have relied on a gesture she never thought to write down. The page preserved the instruction faithfully, which is exactly why the failure can repeat. More accurate copying of an incomplete recipe produces more accurately repeated disappointment.

The baker writes a note in the margin: “Begin with less water when the flour already feels damp.” This is a small act, but it changes the inheritance. The next baker can use the original quantity, the amendment, and a fresh batch of dough to test both. If the new note is wrong, the next failure has a place to attach. The book now holds a dispute between attempts rather than a command from the absent cook.

The ink can survive unchanged while the practice fails. A recipe needs a reader with ingredients, skill, and judgment about the dough in front of her. The margin gives that judgment a way to reach the next reader.

Human beings are unusually good at making such surfaces matter. Children sometimes copy an adult's apparently unnecessary move even when a more direct route to the reward is visible. In a well-known puzzle-box experiment, chimpanzees were more likely to drop an irrelevant action when the box made its irrelevance clear; young children often continued to copy it. That can look like foolish obedience. It can also be a sensible wager for a novice: the adult may know a hidden condition, or the move may be part of a rule of the group. Copying the whole sequence keeps an unknown dependency alive until the learner can test it.

The wager has a price. If the adult's extra move is truly useless, faithful copying wastes effort. If the adult's explanation is wrong, it can narrow what the learner tries next. A culture needs enough fidelity to keep a rare working technique and enough freedom to notice when a transmitted step has ceased to earn its place. The recipe's margin is where those two needs meet.

A steel ratchet locks each tooth mechanically. Cultural inheritance can slip, split, or lock the wrong thing in place. A durable record lengthens the interval in which a loss can be noticed and repaired. Somebody still has to notice.

Must we understand what we inherit?

Perhaps the page matters less than the mind that reads it. Humans build exceptional tools because they reason better about causes, this argument says. The recipe is just a convenient way for the clever baker to communicate what she already knows. If every improvement depends on a person who understands the whole mechanism, the true protagonist is still the individual brain.

Give people a wheel design and ask them to make it faster. Let the next learner inherit what the previous one tried. In Maxime Derex and his colleagues' experiment, the wheels improved across successive rounds, while participants' explicit explanations of how they worked did not improve along with them. Passing those explanations on did not noticeably accelerate progress. Inaccurate theories could even constrain later exploration. A working configuration traveled farther than its maker's account of why it worked.

You can ride a bicycle you could not manufacture, obey a medicine schedule without knowing its biochemistry, or use a mathematical method you could not have invented. What you inherit is real competence, assembled across many lives. The fact that you cannot reconstruct its whole discovery path does not make your action unreal.

François Osiurak and colleagues later ran a partial replication and found improvement accompanied by better technical understanding. Their result preserves an important role for reasoning. When the inherited design breaks, understanding can help a learner diagnose, adapt, or escape a bad theory. The question is whether every improvement requires that understanding first.

A 2026 study by Ali Seyhun Saral and colleagues separated those contributions more directly. Across two experiments with **900 participants**, learners inherited and refined solutions in tasks with a physical context, an abstract causal structure, or no useful causal structure. Improvement accumulated in all three conditions. Causal reasoning helped early progress; the physical context added no advantage over the abstract causal task. A successful choice could be preserved before its owner had an explanation worth passing on. Eureka need not precede improvement.

The same is true of the baker. She can use the written quantities before she understands dough. When the loaf fails, her ability to reason about water, flour, and heat becomes precious. The written page lets her begin ahead of the original cook's starting point; it does not spare her

from thought. The copied artifact and the thinking reader make one working system.

A group of skilled teachers can keep a practice alive through demonstration. The page gives that group members who are dead, absent, or unknown. Communities can compare versions without bringing their oldest cooks together. A correction can wait beside the disputed step until a stranger finds it. The inheritance now carries responsibility with knowledge: an omitted condition can charge its cost to someone the writer will never meet.

The baker can write in a margin because she owns the book and knows how to read it. The next person may not. Clay marks need a code, a keeper, and a reader who knows where to find them. The strongest instruction can vanish from use if its language, tools, or teachers disappear. A tablet may last five thousand years while its working meaning disappears for most of that time.

The page has survived. Its readers may not. How many people must still know the signs? How many makers must be able to test the instruction? A record can outlive its last useful reading. To understand that loss, we have to enter the room of maps built around the single mark.

Chapter 2: The Map Room

The point of the bone tool was still imaginable long after the last one vanished.

At sites in Tasmania, archaeologists found worked bone in older layers. In the later layers, the tools became less common, less varied, and then absent. At Rocky Cape, Joseph Henrich reported a ratio of stone to bone tools of three to one about seven thousand years ago. Several millennia later it was fifteen to one. By roughly three and a half thousand years ago, bone tools no longer appear in that record.

Picture the last lesson: a bone split, a point ground, a hand testing the edge. An apprentice sees a sequence she may one day need to show someone else. The archaeological layers preserve the objects. They leave us to reconstruct what happened to the lesson.

One account of the disappearance begins with a rising sea. Tasmania had been connected to mainland Australia. As the sea filled the Bass Strait, the island's people became more isolated from other groups. Henrich proposed that the resulting reduction in the *effective* pool of people from whom one could learn made some techniques harder to maintain. If a difficult procedure is copied imperfectly, the best model available to the next learner matters. With enough models and enough correction, an occasional mistake need not become the next generation's starting point. With fewer, a useful distinction can gradually disappear.

This is more unsettling than a sudden disaster. A tool can be lost while every maker remains a capable human being. No one has to decide to destroy the knowledge. The next apprentice merely inherits a slightly poorer demonstration, then gives a slightly poorer demonstration in turn.

Perhaps the lesson ended because the tool was no longer needed. Food, climate, materials, and other inventions change what deserves

to be made. Archaeologist Krist Vaesen and his colleagues pressed this objection against the demographic account. Tasmanians continued to make canoes, baskets, shelters, and stone tools, including artifacts more demanding than the missing bone points. A record that counts the vanished point and omits the surviving work can manufacture a decline of its own. Henrich's model supplies a mechanism by which reduced contact could erode a technique. The objects alone do not establish that mechanism as the cause of this disappearance.

The objection changes what we must count. A census tells us how many people exist. A transmission map tells us whom an apprentice can watch, which example she selects, what materials she can use, and where she can take a failed attempt. The maintained route is the thing to explain.

The paper plane that would not improve

Take a sheet of paper and try to make it fly as far as possible. At the next table, place the planes made by an earlier group. A new participant can study one, two, or four examples before making a plane for the next group. Which arrangement should produce the longest flight after several rounds?

The easy answer is four. More examples mean more designs and a better chance of encountering an excellent one. In a study of 543 participants, that first part happened: larger sets offered more variation and better available planes. But the expected inheritance did not follow. Plane distance improved across experimental generations in the condition where each participant saw one predecessor's plane. It did not improve across generations in the conditions with two or four models.

The experimenters examined the copies. Participants favored better planes, yet resemblance between one generation's designs and the next weakened as the number of models grew. The investigators proposed limited working memory and the burden of comparison as part of the

explanation. More designs were available. The copies were less faithful. The missing step lay between access and inheritance.

Now imagine a crowded workshop where every maker shows a tool and announces how well it worked. Before your turn, the examples are taken away. The room holds good information; your hands must rebuild it from memory. You may borrow incompatible features from several designs or forget the one fold that mattered.

Bradley Walker and Nicolas Fay put the missing page inside a later experiment. Participants drew virtual arrowheads through ten trials. Each design received a score, but the rule that produced the score was hidden. In a group of six, a player watched the other five designs and scores one by one before making the next arrowhead. Groups of three saw fewer. If every example vanished before the player drew again, the larger group offered more ideas but no measured advantage over the smaller one.

In another condition, the same kinds of designs and scores stayed in a panel while the player drew the next arrowhead. Now the six-person groups outperformed the three-person groups. In those larger groups, performance improved across trials with the panel; it did not improve when everyone had to view all the examples without one. The researchers also found stronger copying of the high-scoring designs when the record was available. An additional maker helped once the next maker could return to an example at the moment of choice.

There is a rival explanation worth keeping: perhaps the panel mainly helped people filter attention, rather than combine several designs. In the researchers' separate first experiment, larger groups also benefited when a player could choose to inspect just one high-scoring model. The team did not directly measure recombination in the record condition. The panel was cleared after each trial, too; this was no proof of an archive surviving centuries. The experiment isolates a smaller, decisive step. More possible teachers become an advantage when a learner can select or re-examine the example that matters.

Keep a second route open

Another experiment tested how learners were connected. Groups of six people tried to discover combinations of virtual ingredients. Some groups could see everybody else's attempts after every round. Others worked mostly in pairs, with occasional movement between pairs. The fully connected groups quickly learned from visible success, but tended to converge on one line of search. The partly connected groups kept different lines alive long enough to combine them. More than half of those groups discovered both prerequisite paths; none of the fully connected groups did.

The productive arrangement combined room to explore with later contact across groups. Too little contact can strand knowledge. Too much imitation of the first visible winner can erase alternatives before anyone knows which will be needed. The cultural record needs both faithful inheritance and a route by which another group can bring it something worth changing.

The bee's demonstration had to be legible. The plane's fold had to survive comparison. The arrowhead had to remain available when its next maker chose what to copy. The ingredient groups had to keep a second possibility alive. Counting minds leaves all four jobs unfinished.

Two maps of one city

A crow feeding beside a bin has chosen a place to eat. It has not necessarily chosen a place to raise chicks.

In Debrecen, Hungary, researchers completed **241 surveys of sixteen city sections over three years**, recording **7,144 crow observations**. During the breeding season, more bins in the wider surrounding area went with more observed birds. Yet the probability that a section would become newly occupied for breeding was lower where bins and restaurants were more numerous. The authors

proposed human disturbance and the presence of nonbreeding birds among the possible explanations.

One city held a map of visits and a map of settlement. They answered different questions. The study measured habitat use; its lesson for us is a measurement analogy. The crowded workshop and the vanished panel now give that distinction a human counterpart. Many encounters with skill do not tell us which motion reached the next apprentice's hands. We need a map of the transfer.

The room around the mind

Take the map room away altogether. Imagine a human newborn alone in a forest, with no caregiver, language, tools, or demonstration. The child would almost certainly die. Even keeping her alive would not supply the inherited practices she could learn in a human community. An ape infant also depends on its mother and group. A developing brain needs a world in which its capacities can become working abilities.

Researchers gave **two-and-a-half-year-old children**, chimpanzees, and orangutans tasks involving the physical and social world. The children and chimpanzees performed similarly on the physical tasks; the children did much better on the social ones. Part of the human biological gift is a remarkable capacity to learn *with* other people. What the forest image takes away is the community through which that gift acquires its contents.

An experienced driver does something strange without noticing it. The car is wider than her body, yet she approaches a narrow gap and says, "I cannot fit." She is judging the gap through a vehicle she has learned to handle. For the duration of the drive, her practical range of movement includes the steering wheel, the car's edges, and the road she can see. Her skin has not expanded to the doors. Her working ability has expanded into a system beyond her skin.

The same thing happens when a student learns to work with a map, a diagram, or mathematical notation. Give two children the same problem. Give one a familiar diagram, a worked example, and a teacher who can show where a calculation went wrong. Give the other the question alone. If the first child succeeds, the child has done the thinking. The marks around her have changed which thoughts she can carry to the end.

The driving image has a limit. A car usually extends one driver's action in the present. A shared notation can extend the action of strangers who lived centuries apart. It can also be taught selectively, guarded, revised, or imposed. The student does not simply *have* the map; she enters a room where other people have decided which maps remain on the shelves and how to read them.

This is what an external memory ledger adds to a capable brain. It gives a person a place to start that previous people paid to reach. The marks do not contain meaning by themselves. As with the clay tablet in the previous chapter, the reader needs the learned code and the circumstances of use. But a readable record can preserve a distinction after its maker dies. An error can be marked in the margin. Two inherited solutions can be set side by side and tested.

A genius can make a move few others could make. But the move begins on terrain no individual built. The algebraic sign, the measuring instrument, the record of a failed experiment, and the teacher who explained an error all enter the next mind as possible moves. Before crediting the navigator, ask which routes she could see. Originality can be a new passage through a map room assembled by the dead.

Michael Muthukrishna, Joseph Henrich, and their colleagues carried the feedback further in the Cultural Brain Hypothesis. They modeled how access to socially learned information, the cost of maintaining a large brain, opportunities to learn from others, group structure, and the long period of human development can evolve together. A brain can make cultural learning possible; a growing store of useful culture

can, in turn, change the value of a brain that can learn from it. The record shapes the organ that reads the record.

The model makes dependence run in both directions. Accumulated knowledge changes what a learner can do and which capacities are valuable to develop. The child inherits a brain and a world arranged for that brain to learn in. Each can alter the future of the other.

Who inherits the map room?

When Jagadish Chandra Bose dedicated his institute in Calcutta on 30 November 1917, he described instruments in cases in the entrance hall. Behind the instruments stood a less visible inheritance: disciples he had been training for ten years. A lecture room would make new discoveries publicly demonstrable. Publications would carry the results to the world. Personal training, he acknowledged, could reach comparatively few.

The different routes mattered. A published result could travel far beyond its maker. Preparing another person to investigate required time, experimental skill, and work with the apparatus. Bose's founding address joined the circulation of discoveries to the cultivation of people who could make the next ones.

More than a century later, the institute's 2026 summer-internship notice made part of that route visible. It advertised six to twelve weeks of faculty mentorship for approximately fifty students. Applicants needed specified academic preparation and an institutional recommendation. The notice also stated that the institute would provide neither a fellowship nor accommodation.

Here is an actual transmission mechanism, with teachers, time, entry conditions, and limited places. It also has a second set of requirements. A selected student must be able to spend those weeks learning; someone must cover living costs and, where necessary, a place to stay. Academic admission and the practical ability to take up a place are separate achievements. The notice cannot tell us how many

students lacked those resources. It tells us where to ask the next question.

The same distinction can be seen at a smaller scale. Imagine a workshop owner offering to teach an apprentice who already has the necessary preparation. The drawings are available; the machines work. The apprentice accepts a copy of the drawings. She cannot accept the supervised morning shifts: they overlap with the paid work that covers her rent. Going alone after hours would mean using unfamiliar machinery without supervision.

Now give the apprenticeship a budget that pays for some of her training time. She can be at the bench when the owner demonstrates a measurement. She makes a part, has it checked, adjusts, and makes another. The allowance has changed which knowledge she can acquire. It pays for the opportunity to practise while keeping her household going.

Remove that support, and the drawings can still reach her. The same opportunity to become a practitioner may not. The workshop's continued existence leaves this difference unresolved.

This is why a surviving university, laboratory, or library cannot by itself settle how well a society is renewing its capabilities. We have to follow a newcomer. Can she learn the code, reach the tools, afford the time, obtain useful criticism, and try again? Preparation and supervised access can protect demanding work. Their value also makes it important to examine who has a route to acquire that preparation. A selection process begins after earlier opportunities have already shaped its applicants.

Nor does successful learning finish the transfer. The trained person still needs somewhere to apply the method. A workshop needs materials and orders. A laboratory needs working instruments and support for another experiment. A city can preserve the evidence of discovery while making the next attempt difficult to fund. It can also sustain excellent training whose graduates find their usable

opportunities elsewhere. These are different breaks in the chain, and they require different repairs.

For each practice, follow the path from an earlier success to a later attempt. What survived? Who can use it? Where can a failed attempt obtain a correction? What enables the corrected attempt to become useful work? When that route holds, a later person can spend her effort beyond the old limit. When it breaks, part of the discovery must be paid for again.

But a map room is never a room of all possible maps. Someone chooses which book to copy, which language to teach, which maker's design to display, and which failure to leave out. A ledger can preserve a useful route and preserve a distortion with the same mechanical fidelity. Access determines who gets to correct the distortion before it becomes an inherited rule.

There is one more inheritance to follow. The person who learns to make the part does not thereby own the workshop, the design, or the proceeds of its sale. Competence makes production possible. Agreements and institutions determine which claims accompany it. A society can transmit the ability to produce while reserving durable ownership for a smaller circle. The next question therefore concerns what the contributor is permitted to retain.

Suppose the map now tells a court who owns the tool, a lender who owes the money, and a worker which company must answer for the loss. The people who keep that record can still decide which distinctions survive and which corrections arrive in time. Now the next reader's mistake is measured in cash, property, and claims against a company that may have been emptied before the claim is heard.

The map room made us powerful. The next question is who gets to own the map - and who receives the bill when its lines are drawn to someone else's advantage.

Chapter 3: Cash Before Cost

On April 23, 2002, the toys did not leave the shelves. The cash did.

KB Toys' operating companies borrowed. Some of the proceeds paid bonuses to more than fifty managers and executives. The rest traveled upward through a chain of holding companies. At the top, KB Holdings used the money to buy back 65 percent of its shares from Bain Capital affiliates and other insiders. A Delaware court later described a liquidity event that let the investors withdraw money from the toy business.

The court put the distribution to the sponsor, affiliates, managers, and others at about $120 million. A later study, drawing on published reporting, attributed an $85 million payout to Bain against an original equity investment of $18 million - about a 370 percent return by the study's calculation. The figures describe different recipients. The common direction is unmistakable: lenders advanced money to the business, money left for owners, and the obligation to repay stayed with the business.

The same day produced a second kind of paper. Robert Feldman, a KB Toys executive, wrote to the general counsel of Big Lots, which had sold the chain and still held a note payable by one of its holding companies. After the share repurchase, Feldman certified, the top company would have a consolidated net worth of at least $20 million. No contract required him to send that assurance. Big Lots later alleged that the letter was meant to keep it from objecting to the transaction.

Put yourself at the other end of the letter. You can read a number, but what can you do with it before the money leaves? The court later found that Big Lots had no contractual right to block the payout. Its complaint did not show that it had surrendered a viable right to seek an injunction because it relied on Feldman's assurance. The written

figure was available on the day the cash moved. It was not a brake on that movement.

Twenty-two months later, on January 14, 2004, KB Toys filed for Chapter 11. Big Lots alleged that the payout had left the toy companies insolvent. Its $45 million note had grown to about $57 million with accrued interest. The court dismissed Big Lots' suit against the payout's participants, largely because company-wide injury belonged to the bankruptcy estate rather than to one creditor suing for itself. The retailer also faced brutal competition and expensive mall leases. The ruling did not establish that the recap caused bankruptcy, or that Feldman's $20 million figure was false. The sequence alone cannot decide which pressure caused the failure.

It does establish something that a later argument cannot undo. The 2002 payout had already happened. The business's ability to earn enough to service the borrowing still lay ahead. If it prospered, creditors could be paid and the early distribution could be a successful return. If sales fell, the distribution did not automatically rewind. Workers, lenders, and the company would meet the shock with a larger debt schedule.

That is the Ledger's first financial move: **turn tomorrow's expected receipts into today's payment, then let the uncertainty resolve on another balance sheet.**

Two clocks on one desk

Imagine a bakery with a reliable stream of customers. Its owner borrows against the income expected over the next several years and pays the borrowed money to herself. The ovens have not improved. No new shop has opened. The owner has cash today; the bakery has a larger loan schedule.

A dividend recapitalization, usually shortened to *dividend recap*, does this without selling the firm. The company borrows. Its owners receive

a distribution. The operating cash that could have paid wages, rent, equipment, or older creditors must now service the new debt as well.

One clock records the closing. Cash arrives from the lender, a distribution reaches the owner, and the fund can report money returned. The second clock records interest dates, maturities, refinancing decisions, and the sales needed to meet them. The transaction changes who holds the cash while the second clock runs.

The bakery image has a boundary. An owner who keeps the business may bear much of the later damage herself. A fund may distribute a company's payout onward to its investors. A sponsor, a fund investor, an employee, and an older lender occupy different positions. We have to follow the actual recipients and obligations. What looks like a withdrawal from one purse is a transfer among several.

The calendar has a price. Receiving a return after one year and receiving it after ten years are different investment results, even when the total dollars are the same. Investors call one measure of this difference the *internal rate of return*, or IRR. It gives weight to when the cash arrives. Pay some of it earlier and the annualized result can improve before the business produces any additional value.

This creates an incentive to move the first clock. The second need not improve with it.

The strongest defense is straightforward: a lender has agreed to the loan, knows where the proceeds will go, and prices the risk. The debt and dividend are visible on the accounts. Why treat an agreed transaction as a problem?

Because that agreement has parties outside it. The new lender can negotiate a price for its own exposure. An employee awaiting next year's wage decision and a creditor holding an older loan may acquire the consequences without negotiating the new payout. Recording the debt correctly does not prevent it from changing which future demands the company can meet.

The defense would be stronger if the borrowing merely identified companies already headed for trouble. We need a comparison that separates the choice of company from the effect of adding debt.

The firms chosen for the payout

Abhishek Bhardwaj, Abhinav Gupta, and Sabrina Howell assembled 61,628 U.S. buyout deals and identified subsequent dividend recaps for 1,572. Recap deals made up 2.6 percent of the sample but 9.3 percent of its buyout volume. The companies chosen were generally larger and healthier than the others.

That finding gives the defense its best starting point. Lenders prefer firms with room to borrow. Owners prefer to take a payout when a company's recent performance makes credit cheap. Over the following ten years, 9.2 percent of recap targets in the study entered bankruptcy or restructuring, against 3.4 percent of the control firms. The gap looks like a verdict until you ask which companies could obtain a recap in the first place. Compare the later outcomes without accounting for that selection and you can confuse the firm's initial strength with the effect of the new debt.

The researchers found a way into the problem through the banks. Some sponsors had relationships with banks whose activity in collateralized loan obligations increased. These pools buy leveraged loans; their expansion could open a cheaper credit channel. Sponsors exposed to that opening were more likely to recap a portfolio company. The authors used this credit-supply variation to estimate what the added borrowing did.

The rival explanation changes shape here. Perhaps those banks were simply finding better companies, and the later results reflect the companies rather than the loan.

The researchers compared firms in similar industries, buyout years, and sponsor-size groups; they also checked whether recap decisions started moving *before* the bank channel opened. Their design asks

what changes when a sponsor gains easier access to debt that can fund a payout.

The extra credit is our way into the comparison. Its decisive assumption is that this particular credit opening affects the company's later distress through the recap, rather than through an unrelated improvement or shock. The paper tests that assumption in several ways. It does not need every recap in the economy to look like the firms moved by this credit channel.

A typical recap increased total company debt by 84 percent. In the paper's April 2025 revision, the preferred estimate was a **22-percentage-point increase in financial distress over the following ten years**, with distress defined as bankruptcy or restructuring. The estimate concerns firms tipped into a recap by that credit opening: roughly twenty-two additional distressed firms per hundred of those deals. It answers the question the raw comparison could not: a firm can be healthy enough to obtain the loan and become substantially more fragile because it borrowed to pay owners.

The result reached beyond the owners. The study found lower wage growth among employees in firms that survived long enough to be observed. Loans held by pre-existing creditors fell in price around recaps. That price association is a separate observation from the main credit-supply estimate, but its location matters. A transaction that delivered cash to owners made an older creditor's claim less attractive to the market.

The return clock behaved as expected. The researchers found higher deal IRR without a statistically clear increase in the average total cash multiple. The timing of money could improve the headline return while the evidence for more money remained weaker. At the fund level, their causal estimates found lower cash multiples and lower returns relative to a public-market comparison. A flattering result for one deal did not make every investor in the wider fund better off.

Recaps also increased the chance of exceptional outcomes, including public offerings and very strong revenue growth among surviving firms. The added leverage widened the range of endings. This makes the argument more consequential: the architecture can reward an early withdrawal across a range of later outcomes. It need not guarantee failure to shift who bears the danger of failure.

KB Toys supplies a documented clock: payout, then bankruptcy twenty-two months later. The larger study supplies a test of additional borrowing after allowing for the stronger firms selected to carry it. One gives the mechanism a face; the other tests whether the face was chosen merely because its ending made a good story.

Selling the future

A dividend recap is one way to move future receipts into the present. Asset-backed securities are another. A lender can assemble credit-card, auto-loan, or other receivables and sell securities whose payments depend on collections from that pool. The originator receives funding now. Investors acquire claims on payments still to come.

The scale is large. SIFMA's 2025 Capital Markets Fact Book records $388.1 billion of U.S. asset-backed issuance in 2024, up 43.3 percent from the previous year. A separate credit-market report, using PitchBook data, counts about $81 billion of institutional leveraged loans used for dividend recaps in 2024. These totals describe two financing channels. They measure present funding exchanged for future obligations, not losses or net sponsor receipts.

Think of selling next year's harvest before it grows. The sale gives the farmer cash now. The terms determine who pays if the harvest disappoints. Finance extends that arrangement through pools, guarantees, retained interests, and many different claimholders. A securitization can fund productive lending and distribute risk. A recap can return capital from a business strong enough to service the loan.

The moral verdict depends on the transaction; the time shift occurs in either case.

To locate the risk, open the record. Who received the advance? Who owes the later payment? Which promise or guarantee remains when collections fall short? An issuance total tells us how much traveled forward. It cannot name the person left carrying the shortfall.

One bank report lets us watch the remaining obligations enter a different field of view.

The morning the boundary moved

On January 1, 2010, Bank of America adopted new U.S. accounting guidance for consolidating certain special-purpose entities. Its annual report says the adoption added a net $100.4 billion in assets and $106.7 billion in liabilities to the consolidated balance sheet. The bank had not gained a hundred billion dollars of fresh deposits overnight. Existing arrangements had crossed a reporting boundary.

Of the added assets, $69.7 billion came from credit-card trusts and $30.7 billion from other special-purpose entities, including multi-seller conduits. Loans that had been accounted for through another arrangement now entered the main consolidated picture.

Other entries carried the credit consequences. The bank increased its allowance for loan and lease losses by $10.8 billion, mostly for credit-card receivables, and recorded a cumulative $6.2 billion after-tax charge to retained earnings. The new guidance and related regulatory rule increased risk-weighted assets and reduced Tier 1 capital ratios.

The asset total measures what was brought inside. The allowance measures expected credit losses; the retained-earnings charge records the cumulative adoption effect. Keeping those entries separate lets us see the event's real power. A changed reporting rule altered the main account from which capital strength was calculated.

Picture a theater with a side room visible through a doorway. An attentive visitor can count its chairs. Remove the wall and every count of the main floor includes them. The building has gained no chairs; the standard count has gained a room. Accounting follows control, risks, and contractual interests rather than physical walls. But the change in what the ordinary count must include is the same.

Before consolidation, information about these arrangements could be found in notes and other disclosures. After it, they changed balance-sheet measures and capital calculations. Disclosure had made the entries available. The rule change made them unavoidable in a different set of decisions.

The bank's economics did not begin on January 1. The mandatory consolidated picture did. That distinction is why a footnote and a balance-sheet line can tell the truth and still exert different powers. Placement decides which reader must confront an entry, and timing decides whether that confrontation comes before the next commitment.

The Ledger can move the receipt of cash, the recognition of exposure, and the date on which that exposure constrains action. These movements have different instruments and different bearers. Following the dates reveals where a completed gain stands ahead of an unresolved obligation.

KB Toys leaves a final clue. Big Lots' note was owed by a holding company whose only asset was stock in another holding company. Creditors of the operating businesses stood nearer the stores and their assets. Cash could move upward through the chain. A creditor's claim did not automatically follow it.

The calendar tells us when the money left. The name on the promise tells us where someone has to go to ask for it back.

Chapter 4: The Company That Could Disappear

The note had a name on it. That was the problem.

When Big Lots sold KB Toys in 2000, it received cash and a $45 million promise to be paid later. The promise was a *pay-in-kind* note: interest could accumulate rather than arrive as cash each year. The company that issued it was Havens Corners Corporation, or HCC. On a chart of the toy business, HCC looked comfortably connected. Above it sat KB Holdings. Below it sat another holding company, then the operating subsidiaries with the stores, inventory, and customers.

The chart made all of these boxes look like one enterprise. The note named one box. HCC's only asset, the Delaware court later explained, was stock in the holding company below it. If the operating subsidiaries could not pay their own creditors, value would not climb the chain to HCC merely because all the boxes carried the same toy-store identity. Big Lots had accepted a claim that was subordinate to creditors nearer the stores.

The first question was *which company owed the money?*

In 2002, the operating subsidiaries borrowed and sent part of the proceeds upward. About $120 million was distributed to the buyout sponsor, affiliates, managers, and others, according to the court's account. The calendar told us when the money left. Now follow its address. The operating businesses took on the borrowing. The cash passed through HCC and up to KB Holdings, where shares were bought from owners. Big Lots' note stayed an HCC obligation.

When KB Toys entered Chapter 11 in January 2004, the note had grown to about $57 million with accrued interest. Big Lots sued the Bain affiliates and individuals, alleging that the refinancing had drained value from the business and left its note unpaid. It asked a

Delaware court to treat the injury as one for which Big Lots could pursue those defendants directly.

The court dismissed the suit. Several of the asserted wrongs, it held, were injuries to the company as a whole. Claims for that injury belonged to the bankruptcy estate, which acts for creditors collectively, rather than to one creditor trying to recover first. Other pleaded claims failed on their own terms. The dismissal closed this attempted route to the recipients of the cash; it left the estate's collective claims to the bankruptcy process.

That ruling served a real purpose. If the largest, best-lawyered unsecured creditor could take a company-wide claim out of bankruptcy and collect for itself, every smaller unsecured creditor would be left behind. The rule protected the collective process. It also closed Big Lots' attempted route to the people who had received the earlier cash.

The name on the key

Imagine a corridor of locked rooms in one building. The sign outside says **KB Toys**. Your key says **HCC**. It opens the HCC room; it does not open every door bearing the same sign. In the HCC room you find shares in a room farther down the corridor. Before you can turn those shares into money, people with claims directly against that lower room must be paid.

The physical picture helps only up to a point. Corporate law is not a set of unbreakable doors. A guarantee can give a creditor a key to a parent. A lien can attach to a named asset. Insurance can fund a claim. A court can set aside a fraudulent transfer, or in some circumstances disregard a company's separate personality. Bankruptcy law may gather claims and property that individual lawsuits cannot. The contract, statute, and judgment decide which passages are open. A logo does not.

This is why the familiar statement “the company has plenty of money” may tell a claimant almost nothing. *Which company?* Does that

company owe this debt? Did it guarantee another's debt? What asset backs the promise? What happens to the promise if the operating company fails? A consolidated balance sheet can add the rooms together for investors. A court still needs the legal route from claimant to asset.

Big Lots was no powerless bystander to its original bargain. The court emphasized that it was a sophisticated seller and had no contractual right to block the 2002 refinancing. It had taken a deeply subordinated note. Perhaps this was a risk Big Lots priced and accepted. That defense explains why its note stood behind other claims. It also makes the architecture plain: a negotiated promise can stay at one address while cash moves among related addresses. The contract gives the creditor a position in the chain, not ownership of the whole chain.

Now reverse the operation. Instead of leaving a creditor at an existing address, create an address for claims already brought against the business. The next question is what travels into that room with the claims.

A company built to hold the claims

Johnson & Johnson's consumer-products business had faced talc lawsuits for years and disputed that its products caused the alleged injuries. For claimants, a second question stood before any award: which entity would answer, and through which forum?

In October 2021, J&J reorganized its consumer business through a divisional merger under Texas law. The old company ended. One successor, New Consumer, received the operating business and most of its assets. Another, LTL Management, received essentially all of the old company's talc-related liabilities. Two days later, on October 14, LTL sought Chapter 11 protection. People who had brought talc claims would now face a company created to carry those claims into bankruptcy.

The brands had one address; the claims had another. But the plan also built a passage between them. LTL received cash, an interest in a royalty business, and, most importantly, a funding agreement. Under that agreement, J&J and New Consumer were jointly responsible for paying LTL's qualifying talc costs up to the value of New Consumer. The court described the right as worth roughly $61.5 billion at the time of filing, with a floor tied to New Consumer's value at the split. That payment right changed what the new address meant.

Follow the promise through three questions.

Who could demand payment? LTL could.

Who had to supply it? Both J&J and New Consumer were responsible.

What happened after the money arrived? LTL did not have to repay it. Outside bankruptcy, the permitted uses included talc costs and ordinary expenses; inside bankruptcy, they included administration and a claimant trust. The conditions were limited. This was a contractual route to cash, rather than a hope that a generous parent might send help.

The distinction matters because a corporate chart can conceal resources in either direction. A liability box can look empty while holding a valuable claim on another box. Count only the cash inside it and you miss the promise. Count the whole group's assets and you miss the conditions under which the box can demand them. The Third Circuit read the payment right and the strength of the parties behind it. A company seeking shelter had to account for the bridge it had built to reassure everyone that its obligations would be funded.

The Third Circuit used that right to turn the company's argument back on itself. LTL had asked bankruptcy to protect it from the cost and disorder of mass litigation. The court asked whether the debtor actually faced the kind of financial distress that justified Chapter 11. A valuable, reliable payment right against J&J and New Consumer made

the answer no on the record before it. In 2023 the court ordered the first LTL bankruptcy dismissed.

The company holding liabilities was separate. The funding promise made the separation permeable. The bridge intended to support a bankruptcy resolution became the reason the court closed its entrance: LTL had resources sufficient to defeat its claim to the required financial distress. A drawn boundary had met an enforceable passage and a court willing to read both.

There was still a valuable prize in the reorganization: control of the route. A mass-tort bankruptcy can gather current and future claims, pause individual suits, value disputed claims, and, if a lawful plan is confirmed, channel payment through a trust. Those powers can help pay many people fairly when separate lawsuits threaten to consume the common pool. They can also replace a claimant's path to an operating company or jury with a slower, collective procedure. Moving the forum is a distributional act even when a funding agreement leaves substantial money on the table.

The next case shows that a similar corridor can remain closed to individual suits while that collective procedure continues.

The door that stayed shut

Georgia-Pacific had faced asbestos claims arising from products sold decades earlier. In 2017 it used a Texas divisional merger to create New GP and Bestwall. New GP received most of the operating company's assets. Bestwall received the asbestos liabilities, about $32 million in cash, litigation-related contracts and insurance, a plaster business valued at roughly $145 million at filing, and a funding agreement under which New GP would cover specified expenses and, if necessary, a bankruptcy trust. Bestwall filed Chapter 11 that November.

Some claimants continued to name New GP in asbestos suits. Bestwall asked the bankruptcy court to halt those suits while it handled claims

in one proceeding. The court granted a preliminary injunction. In 2023, the Fourth Circuit affirmed the injunction against suits involving the nondebtor affiliate. Its majority said the parallel claims against New GP could affect Bestwall's bankruptcy and that the funding agreement could make New GP's assets available through the estate for approved claims.

To understand the disagreement, follow a possible payment around the loop. Bestwall had agreed to indemnify New GP for asbestos liabilities. If a claimant won against New GP, New GP could seek reimbursement from Bestwall. That demand could affect the bankruptcy estate. But New GP had also agreed to fund Bestwall when the relevant resources were insufficient. In the dissent's account, New GP could supply Bestwall with the money Bestwall would use to reimburse New GP. The claimant's lawsuit was being connected to a bankruptcy through agreements drawn by the companies themselves.

The majority saw an estate that could still be affected. Bestwall's own resources came first; New GP's funding served as a backstop. Even an exchange in which New GP supplied the money could change Bestwall's liabilities and the administration of the case. The dissent challenged both the claimed economic effect and the use of constructed relationships to establish the court's reach. These were rival readings of an actual contract, with a different next step for claimants depending on which reading prevailed.

The majority's reading prevailed. The path of suing New GP separately remained blocked by the injunction while the bankruptcy process went on. The promise helped make resources available through the estate, and the order made that estate the route claimants had to use for the enjoined suits.

In August 2025 the Fourth Circuit also affirmed the denial of a separate jurisdictional dismissal challenge. The arrangement had survived two challenges: individual suits against New GP remained redirected, and the bankruptcy continued on the questions presented. The funding promise supplied a passage to resources; the procedure

selected the passage through which claimants could seek them. No final trust or individual recovery was established by those appellate rulings.

LTL and Bestwall met different legal tests in different courts. One funding bridge helped defeat entry into bankruptcy. The other supported a route that kept individual affiliate suits outside it. The decisive question is what a court allows the written boundary and its bridge to do.

The vote that did not become a voice

J&J tried again. Its 2024 Red River Talc plan proposed roughly $9 billion to resolve ovarian and other gynecological cancer claims. It sought votes before the bankruptcy filing, aiming to arrive with the support needed for a prepackaged settlement. The new liability company and its parent presented a plan that, if confirmed, would route claims through a trust and release some claims against others.

One settlement offered finality across tens of thousands of disputed claims and a share for future claimants. Many claimants and firms supported it. The court had to decide whether the recorded support could authorize the plan to bind the people whose rights it changed.

The bankruptcy judge examined the record of the votes. He found serious solicitation and tabulation problems, including too little time for thousands of claimants to make a choice and difficulties establishing who had authority to vote for whom. The court could not certify that the required class had accepted the plan. It also found other barriers to confirmation, including improper nonconsensual releases of claims against parties outside bankruptcy and an overbroad attempt to channel claims against retailers. On March 31, 2025, it denied confirmation and dismissed Red River's case for cause.

The judge's short sentence belongs at the center of this book: “Votes are voices.” A mark in a voting ledger can authorize a settlement that binds people. If a lawyer's mark does not reliably represent a client's

decision, the page may be full while the court has not heard the people whose claims would be changed. The vote count had crossed into authority. A tally could not supply the voices it failed to carry.

J&J said it would return to the tort system rather than appeal. It also announced that it would reverse approximately $7 billion it had reserved for the proposed bankruptcy resolution. That accounting change was neither cash paid to claimants nor a judgment that the injuries had vanished. It was the company's next entry after the chosen route failed. The pending claims still needed a forum and a result.

Here the Ledger's power becomes visible at human scale. A divisional merger chooses the name on the defendant's door. A funding agreement chooses what can pass between rooms. An injunction chooses where a claimant may speak. A ballot chooses whose consent can make a new plan binding. Each written choice changes the next move available to someone else.

Follow the passage, not the logo

The financial report draws a different map of the rooms. In Chapter 3, an accounting rule brought Bank of America's existing securitization vehicles into its main balance sheet. Other bank filings disclose a *maximum exposure to loss* from vehicles they do not consolidate. That number estimates what the **bank** could lose from its interests and commitments to a vehicle. It is not, by itself, the amount a creditor of that vehicle can legally collect from the bank. Accounting consolidation changes reported assets, liabilities, and ratios. A guarantee, lien, contract, statute, or judgment determines legal recourse. A bank's risk of losing money and a claimant's right to reach that bank are different entries.

Three reassuring shortcuts have now failed. “There is money in the group” could not turn Big Lots' HCC note into a claim against every company in the toy-store chain. “The liability company is separate, so there is no money” missed LTL's enforceable funding right. “There is a funding promise, so the claimant's position is secure” missed

Bestwall's change of forum and Red River's attempt to bind claimants through a disputed vote.

What survives those tests is a question we can actually investigate: who can make which payer supply money, on what conditions, through which procedure? An empty company does not settle recovery: follow any enforceable route to assets, insurance, or another payer, then examine the resources and priorities at the end of it. A funded company can still leave its claimants fighting over the route. The entry and the power to use it must be read together.

Corporate boundaries let people invest, lend, insure, and reorganize without every failure consuming every related enterprise. A collective process can protect later claimants from the first plaintiffs taking everything. These are powerful reasons to draw a boundary - and powerful reasons to ask who draws it. The rule that prevents one creditor from jumping the queue can decide which assets enter the queue and which route every creditor must follow.

The Ledger has learned to choose an address and the passages out of it. **The question is whether this claim can touch that money.** Big Lots held a real note. Talc claimants asserted causes of action. Even reaching the right defendant leaves another journey. A claim can be recognized in full and return as a fraction of the cash owed.

Chapter 5: The Tollbooths

On June 7, 2024, a bankruptcy trustee issued a payment to Green Mountain Feeds. The claim number was 2. The amount Green Mountain had asked to be paid was $1,078.35. The amount the trustee's final account recognized was also $1,078.35. No reduction appeared in the *allowed* column.

The payment was $333.39.

There is a story inside those three entries: **asserted, allowed, paid**. A person can win the middle one and still lose most of the money. In this case, $744.96 of an allowed claim never reached the supplier through the bankruptcy distribution. No judge needed to declare the bill imaginary. The estate could recognize it precisely and pay only 30.92 percent.

The debtor was Central CT Coop, a Connecticut agricultural cooperative. It filed for Chapter 7 liquidation in August 2016. Green Mountain's payment went out nearly eight years later. The trustee's final account, filed that December, reported that the case had been fully administered. It lists the supplier's claim in an exhibit and records the actual payment again in a cash-disbursement register, down to the check number: 30009. This is as close as an ordinary reader can get to seeing an official right turn into cash on the same page.

The claimant has reached the correct room. The amount is recognized in full. Why is the money still missing?

An address is not a payment

Imagine you are owed a refund by a store that has closed. You still have the receipt. The store manager agrees that the amount is correct. Then she opens the till and shows you that several people have claims on the same cash, some with rights to be paid before you. The receipt has done its job: it proves what is owed. It cannot fill the till.

Bankruptcy law formalizes that moment. A creditor files a proof of claim. Another party can object. The court may allow or disallow the claim, or allow only part of it. Then the law and the terms of any collateral determine the order in which money is distributed. A creditor can be right about the amount and still be late in the order, or face a pool too small for everyone whose claims have been allowed.

Without a collective proceeding, the quickest or best-funded creditors could empty the account. Bankruptcy can preserve value, test claims, and divide what remains by stated rules. Its strongest defense is fairness: proof, priority, and common distribution prevent a race from deciding everything.

The till had to be made. In the receipts exhibit, the cooperative's working life comes apart into sale prices: $125,000 for a 2014 hopper truck, $15,000 for a skid steer, $7,500 for propane forklifts. Other entries record collected receivables, tax refunds, and proceeds from recovery actions. Even invoices more than ninety days old contributed $116,659.45. A business that once moved feed, vehicles, and promises now had to be converted into money someone could distribute.

This is the strongest reason to pay for administration. Finding property, collecting debts, selling equipment, and pursuing recoveries can create a payment pool that no creditor could assemble alone. It is also why a large number on an old balance sheet does not answer the claimant's question. The final report needs a second number: how much was actually recovered, after the applicable rights and costs were dealt with? The trustee reported $1,658,613.82 in gross receipts. That

figure describes recovered money, not the value of every asset the business had once owned.

The final account separates what each category received. More than a million dollars went to secured claims. A priority unsecured claim was paid in full. General unsecured creditors shared about fifty thousand dollars; administering the liquidation cost nearly half a million. These payments did not all draw on an identical, freely interchangeable pile: secured claims can be tied to collateral and its proceeds. But one closed case shows what different legal categories do to the same debtor's promises.

The priority unsecured claimant was a health services and insurance plan for Teamsters Local 559. Its $59,780.44 allowed claim was paid in full. Green Mountain's $1,078.35 allowed claim was paid at 30.92 percent. The difference was not that one arithmetic problem was solved correctly and the other incorrectly. The difference was the legal character of the claims and the money available at each point in the distribution. A classification became a payment rule. The page decided who absorbed the shortage.

Now turn from the category summary to the cash register. On June 7, check 30008 paid the Teamsters plan $59,780.44 and left $50,142.69 in the consolidation account. The next check, 30009, paid Green Mountain $333.39 and left $49,809.30. The remaining general unsecured distributions worked through the same class dividend, with rounding to cents, until the register reached zero.

The order of the check numbers did not create the priorities; it recorded their execution. Nor does this one account describe every earlier payment from the estate. What it supplies is the missing movement between two abstractions. The class summary says general unsecured creditors received $50,142.69. The transaction register lets us watch that sum become particular checks. The supplier's right did not vanish. The money available to satisfy the class ran out.

The report closes the account with $744.96 of Green Mountain's allowed claim unpaid. It names the creditor who bore that remainder; it supplies no account of what the supplier then had to change in its own business.

What each booth actually does

The word *claim* makes the process sound like one question: do you have a right to be paid? In practice it is a sequence of different questions, answered by different people and records.

First is the **entry booth**. Did the creditor learn of the case, identify the debtor, and file the required proof in the required way and time? The court's final account lists many scheduled creditors marked "N/F": no proof of claim filed. A name on a debtor's old list is not the same entry as a filed claim eligible for distribution. We cannot tell from that mark alone why any particular creditor did not file. The important fact is that the record distinguishes the two.

Second is the **recognition booth**. Was the filed amount allowed? Under federal bankruptcy law, a filed proof is generally deemed allowed unless a party in interest objects; an objection can bring the amount and legal basis before the court. Central CT Coop's final account separates the dollars creditors *asserted* from the dollars *allowed*. For general unsecured claims, it reports $1,490,398.95 asserted and $162,184.18 allowed. A total cannot tell us which claims were amended, duplicated, compromised, disallowed, or reclassified. It tells us which column we have: $1.49 million asserted is a different fact from $162,184.18 allowed.

Green Mountain passed that booth intact. The whole $1,078.35 was allowed.

Third is the **rank booth**. What kind of claim has been allowed? Secured creditors have rights against specified collateral. Certain unsecured claims receive statutory priority. General unsecured claims come behind higher-ranking claims and the costs of administering the

estate. The hierarchy can be defended as the price of lending against collateral, protecting particular vulnerable claims, and funding a process that collects and distributes assets for everyone. Yet the hierarchy is also a record of bargaining power and political choice. A creditor who can demand collateral before lending may buy a better place in line. A small supplier usually sells on terms already written by someone else.

Fourth is the **collection booth**. After the claim is recognized and ranked, what value can actually be reached? An empty debtor does not become solvent because an order says “allowed.” A trust, guarantor, insurance policy, lien, or recovery action may enlarge the reachable pool. Without an enforceable bridge, an asset held by some related but separate company is not automatically available. In the last chapter, LTL's funding agreement showed why the phrase “liability company with no assets” could be false. Here the question is the same in a more ordinary setting: what money, from which legal source, can make the final entry payable?

Finally comes **execution**. A distribution has to be made, not merely proposed. The distinction can seem pedantic until the register changes. Check 30010 initially recorded $1,257.67 for PC Tech LLC on June 7. In October the claim was transferred. On October 21, the trustee reversed the original check, putting $1,257.67 back into the account, then issued check 30026 for the same amount to Andrew Shawn Donaldson, the transferee. The account returned to zero.

Count both positive entries and you invent a second payment. Read only the June payee and you miss who received the replacement. Read only the final claims exhibit and you see Donaldson's name without the earlier route. The reversal and replacement make the history legible: one claim, one distribution amount, a changed recipient. A useful record retains the correction that explains its present state.

For Green Mountain, the report records the June distribution without such a reversal. The trustee certified that funds on hand had been distributed and that bank statements, deposit slips, and canceled

checks had been submitted to the United States Trustee. The filed report is our payment evidence; its claim-allowance line alone would not have supplied it.

Each booth can stop or shrink the route. It can also screen a false claim, protect a priority, or preserve a common pool. Challenging its decision can change the route. A right becomes a remedy only after the record has carried it through every necessary stage.

The multiplication a small claimant feels

Suppose 100 creditors each have a $100 claim. At the first hurdle, 70 percent pass. At each of the next three, 70 percent of those who reached it pass. The arithmetic is $0.7 \times 0.7 \times 0.7 \times 0.7 = 0.2401$: about 24 claims reach the far side. Pay those claims in full and the original group receives about $24 per $100 claimed on average. No single hurdle needs to look severe for the compound effect to be severe.

The image counts passage through an invented system. Its 70 percent is the share passing each hurdle among those who reached it. In real cases, a missing document can affect several decisions; an amendment can reopen a route. We would have to follow the same claims to measure the passage rates. Green Mountain's 30.92 percent came from the distribution available to its class after its claim was allowed in full.

The question is **which decision or shortage took the money between one column and the next?** In the Connecticut report, recognition did not reduce Green Mountain's claim. The reduction happened between allowance and distribution. Improving only the ease of filing would not have supplied the missing $744.96. To change that result, one would have to change the assets recovered, the claims ahead of it, the legal ranking, or the applicable source of payment. A reform aimed at the wrong booth may produce more beautifully recorded rights and the same small checks.

Keep three questions beside the report: **Did the claim pass recognition? What resources could pay it? What share was paid?**

Give the mechanisms short names: *psi* (ψ) for passage through recognition, *K* for the ceiling imposed by reachable resources, and *phi* (φ) for the share paid. The symbols keep those questions apart. Ask for the filed amount, the allowed amount, and the paid amount. Each column needs the next before it can tell you what the creditor recovered.

Now improve every hurdle in the invented system from 70 to 90 percent. About 66 of the original 100 claims reach the end. Suppose the creditors who pass must divide only $1,000, with no claims ahead of them or additional costs. Whether 24 or 66 claims arrive, their total payment cannot exceed $1,000. Better gates recognize more rights; they do not refill the pool.

A successful clawback, guarantee, insurance claim, or another enforceable source could raise *K*. Imagine a genuine $100 claim with all procedures cleared and only $10 reachable for it. An easier form cannot produce the eleventh dollar. There are two handles: change what can pass, and change what can pay. A reform needs to reach the handle responsible for the shortage.

A pattern bigger than one cooperative

Does the same separation hold beyond a small cooperative? We turned to the Florida-UCLA-LoPucki Bankruptcy Research Database of large public-company bankruptcies. Its January 2023 release contains 1,218 cases. Our calculation uses the subset for which both secured and general unsecured class claims and distributions are recorded in the same case: 105 cases, most from Chapter 11 rather than the Connecticut cooperative's Chapter 7.

In that matched subset, the median secured class recovered the full recorded claim. The median general unsecured class recovered 20

cents per dollar. Secured recovery exceeded unsecured recovery in 95 of the 105 cases. These large cases have their own procedures and distributions. Alongside the cooperative, they show why the category of an allowed claim matters: legal rank helps decide how much of the recognized amount reaches the payment column.

A second kind of record points in the same direction. Moody's study of defaulted debt found lower trading-price recoveries as bond claims moved down the seniority ladder: a mean of 57.4 cents per dollar of face value for senior secured debt versus 28.9 cents for junior subordinated debt in its 1982-2004 recovery observations. Those are market prices around default, not the final checks from Central CT Coop, and the people holding those instruments may have bought them at different prices. Priority was valuable enough for markets to price it before final payment arrived.

Perhaps those prices reflected panic rather than what creditors ultimately recovered. Moody's also followed claims through resolution. Its 2007 Ultimate Recovery Database report gives average discounted recoveries of 82 percent for bank loans, 65 percent for senior secured bonds, 38 percent for senior unsecured bonds, and 15 percent for junior subordinated bonds. The bank-loan median was 100 percent. The hierarchy survived into the value creditors received at resolution.

These recoveries could include cash or replacement securities. Moody's valued them and discounted them back to the last interest payment using the instrument's pre-default coupon rate. That time adjustment matters. A dollar arriving years later cannot be treated as the same investment result as a dollar arriving now. The price at default, the value at resolution, and the trustee's actual check are distinct records pointing toward the same importance of position in the queue.

The strongest defense still has force: scarce assets need a hierarchy and valid claims need proof. A secured lender may have supplied funding that kept the business open for years. A priority health plan may be protecting workers. A trustee's fees may be the cost of turning

scattered property into any payment at all. The question this chapter asks is more exacting: **who had a chance to set the categories and prices before the shortage, and who encountered them only when the bill came due?** A system can distribute scarce money according to its rules and still make the weaker party carry a risk that was never visible in the price they received.

Return to claim 2. Green Mountain's amount was fully allowed. The next column paid 30.92 percent and left 69.08 percent with the creditor. The official record acknowledged the obligation without making the supplier whole. Recognition named the bearer of the shortage.

The amount no one can make vanish

The shortfall can stay with a supplier as an unpaid receivable. It can sit with a pension, a lender, an insurer, or a government. Before the outcome is resolved, an account may still carry a larger value than will eventually be recovered. The difference waits for a decision about who will absorb it.

Green Mountain's $744.96 has a final address in the report: the creditor. Other losses stay in the system for years while the records continue to say that loans are performing and capital is sound. The account remains open. When that accumulated gap finally demands payment, who gets the check - and who gets the bill?

Chapter 6: The Cost That Would Not Stay Buried

On May 30, 2025, the British government sold its last shares in NatWest. The bank had been called the Royal Bank of Scotland when the government stepped in to keep it alive. Seventeen years later, the Treasury could finally close the account.

It had provided £45.5 billion in 2008 and 2009. Through share sales, dividends, and fees, it had received £34.98 billion back. The difference was about £10.5 billion.

Seventeen years is a long time to wait for the last column of an invoice.

The public had become an owner, with its stake eventually reaching 84.4 percent. That position had its own slow exit. The Treasury's sale history begins in 2015. Ordinary dividends resumed in 2018. Further sales and bank buybacks returned money in installments until the final shares left public hands in 2025. The closing total also includes payments for special dividend rights and fees for separate protection arrangements.

Imagine being told that the only way to keep your town's main employer open is to buy most of it. You have acquired a chance to recover your money, but that chance now depends on the business, its dividends, and the price someone else will pay for your shares. The state has powers a household does not, and a bank connects far more than one town. Still, the distinction holds: public ownership supplies a claim on future value. It does not make the original purchase price a debt the bank must repay on demand.

By then, the emergency was history. The bank was alive. Customers had continued to use their accounts. The government argued that allowing RBS to collapse would have cost far more in lost savings, damaged businesses, and broken confidence. The rescue had protected

things worth protecting. The closing account also showed a public cash shortfall.

These facts belong on the same page. A rescue can preserve a payment system while assigning part of the cost to the people who depend on it. The architecture of that assignment is the subject of this chapter.

In a bankrupt cooperative, a creditor can point to the difference between an allowed claim and a cheque. A banking crisis spreads the problem across thousands of promises that mature on different dates. Borrowers promise to repay over years. Depositors can ask for their money this afternoon. Somewhere between those clocks, losses form, confidence breaks, and a public institution is asked to make promises good before their final value is known.

The bill can travel a long way before it reaches its bearer.

Two clocks in the same bank

Imagine a landlord whose tenants promise rent for the next ten years, while the landlord's lender can demand repayment next month. The building may be valuable. The rent may arrive. But when the lender calls, the landlord needs cash now. Waiting for ten years of rent is unavailable as an answer.

A bank runs a much larger version of this mismatch. Its loans pay over time. Its securities have prices that can change rapidly. Deposits and wholesale borrowing can be withdrawn or need refinancing much sooner. One clock asks how much the assets will ultimately yield. The other asks whether the bank can pay today. When the second strikes, the first may still be years from an answer.

RBS had made both clocks dangerous. It expanded its lending and structured-credit business, then joined a consortium to acquire ABN AMRO in 2007. The Financial Services Authority's later investigation found that RBS financed much of its share with debt, predominantly short-term, while acquiring further exposure to assets whose values

were becoming uncertain. The bank became more dependent on the borrowing that would be hardest to renew in a panic.

The regulator's description of the private information supplied by ABN AMRO in April 2007 is almost absurdly tangible: two lever-arch folders and a CD. RBS was largely unsuccessful in obtaining more. An enormous decision passed through a reading room small enough to carry under one arm.

Structured-credit markets had already begun to deteriorate. Across 2007-10, RBS recognized £32.5 billion of impairment on loans and advances. Its credit-trading losses amounted to another £17.7 billion over the regulator's period of examination. In autumn 2008, the eventual scale of the loan losses was still unclear, but uncertainty about the assets was enough to damage confidence in the bank.

Those figures measure damage within RBS. The government's £45.5 billion measures capital supplied. The £34.98 billion measures money later returned through several routes. Mixing these columns would make the transfer impossible to follow. A provision, a capital injection, and a final public shortfall describe different stages of the same crisis.

The bank's accounting rules also had a clock. Its 2008 accounts required objective evidence of impairment: events since a loan was granted had to have changed its expected cash flows. RBS assessed individual loans and portfolios and made provisions. Recognition depended on evidence available at a reporting date; repayment depended on a future still unfolding.

Consider a simplified loan. After provisions, the record still carries it at £100, while the best current estimate of its recoverable value is £80. The unexplained £20 is the gap our research calls **stored loss**, or *S*: deterioration that remains inside the recorded value. This definition asks us to compare a carrying value with an expected recovery value. It gives us a place to look for the shortfall before cash settlement exposes it.

The difficult word is *expected*. A bank may underestimate the damage, new evidence may reveal it, or a borrower may recover and pay more than feared. Provisions and write-downs can bring the record closer to the changing expectation. *S* describes that gap; it is not a published balance-sheet line we can read off without doing the valuation work.

Yet the timing has immediate consequences. Promises may continue to be honored against a recorded asset value while the repayment needed to justify it becomes less likely. If confidence breaks first, the institution must find cash before its assets can settle the argument.

Here the landlord's building is too small an anchor. A major bank's promises connect other banks, employers, households, and countries. Its failure can break payments elsewhere while its own assets are still being valued. The delayed private shortfall becomes a public emergency through those connections.

The guarantee wakes up

RBS had another set of promises waiting to become immediate demands. Commercial-paper conduits held longer-lived assets and financed them with short-term notes. The bank had committed to supply liquidity if the vehicles could no longer refinance themselves. At the end of June 2008, £8.6 billion had been drawn and remained outstanding under facilities for RBS's and ABN AMRO's own sponsored conduits. The FSA described the support as a significant drain on liquidity.

A household guarantee offers the bridge. You promise to cover someone else's rent if they cannot. While they keep paying, your bank balance hardly notices the promise. Then their landlord knocks on your door. The obligation already existed; its demand on your cash has awakened.

The £8.6 billion was outstanding support, with recovery still to be determined. Its immediate danger was the competition for cash. A vehicle's financing problem called the bank's promise precisely when

the bank's own financing was under pressure. The boundary around the vehicle contained an asset portfolio; the guarantee carried a funding demand through the wall.

By 2008, RBS faced several questions at once: the price of risky assets, the quality of loans, the availability of refinancing, and the willingness of others to trust its answers. The FSA eventually identified six principal factors in the failure, covering capital, funding, asset uncertainty, trading losses, the acquisition, and the wider crisis. It also faulted regulation and supervision. Each apparent source of resilience had depended on other promises continuing to work.

This leaves a regulator with a cruel task. Is the bank facing a temporary cash shortage, or do its assets no longer cover its obligations? Frightened lenders can turn a survivable mismatch into failure. Deteriorating assets can give those lenders excellent reason to be frightened. Waiting for every loan's final result allows the payment system to fail before the diagnosis is complete. Providing unconditional support allows earlier decision-makers to pass more of the downside outward.

By the time the rescue is urgent, the easy decisions have already been spent.

Draw the line through the losses

The British government supplied capital by buying shares. It also wrote a separate promise whose terms expose the allocation of risk with unusual clarity.

Under the 2009 Asset Protection Scheme, RBS placed £282 billion of assets in a covered pool. RBS would bear the first £60 billion of losses. Beyond that threshold, the Treasury would bear 90 percent of additional losses and RBS 10 percent. The bank paid fees for the protection.

Think of the excess on an insurance policy, enlarged until the insurer is the public. Up to the first-loss boundary, the bank carries the

damage. After it, each additional pound of covered loss is divided: ninety pence to the Treasury, ten pence to RBS. The contract chooses the future bearer before the future has supplied the loss.

Why place the first-loss line at £60 billion? The Treasury set it at its estimate of losses under the most likely economic scenario. Expected damage would stay with RBS; protection would begin when the outcome became worse. The National Audit Office saw an incentive built into that line: before it, the bank bore every extra pound of loss. After it, RBS bore only ten pence. A boundary in a contract changed the financial reward for preventing the next loss.

The promise also had a waiting period. Any public payment beyond the threshold would be delayed for two years so that recoveries could reduce the amount due. RBS would have to repay such payments to exit the scheme before December 2099. Protection, recovery time, and the cost of leaving were written together. The headline ninety-pence backstop described one part of that bargain.

This is one of the ledger's most powerful acts. A government can alter the credibility of today's promise by committing tomorrow's resources. The protection has force while the cash column is still empty. Creditors deciding whether to wait are now looking at a bank with a public backstop.

Follow the contract to its last entry. RBS exited in October 2012, having paid £2.5 billion in fees. The Treasury reported **no payouts** under the scheme. The tail protection carried contingent public risk and produced fee income. The insured pool's £282 billion described assets covered, never a cheque the Treasury wrote for that amount.

The £2.5 billion fee is already included in the £34.98 billion returned in the government's final NatWest account. It cannot be counted again as an extra recovery. Reading the whole rescue account, including those fees, still leaves the rounded £10.5 billion nominal cash gap.

This is why an invoice needs columns. The insurance promise collected fees without a payout. The rescue package's closing cash

account remained short. Over seventeen years, that nominal comparison also leaves out financing costs, changing purchasing power, and the economic damage prevented. It answers the cash question it was built to answer. The wider judgment requires the wider account.

Saving the bank chooses whom to save

Give the defense of intervention its full force. RBS had more than forty million customers and operations in over fifty countries. A disorderly collapse could have trapped deposits, interrupted businesses, spread failures, and deepened the recession. The Treasury argues that the cost of doing nothing would have exceeded the final cash shortfall. Keeping the bank operating preserved real value.

That defense makes the earlier architecture more consequential. Decisions within one institution helped create a situation in which the public had to weigh a costly rescue against a potentially costlier collapse. The emergency enlarged the circle of people whose resources could be called upon. **The power to keep a promise can become the obligation to cover someone else's.**

The United States supplies a useful test of where the bill can end. The Government Accountability Office followed the Troubled Asset Relief Program to its close in 2023. Across TARP's programs, $443.5 billion was disbursed. After repayments, sales, dividends, interest, and other income, the lifetime cost of the funded programs, excluding administration, was $31.1 billion. The bank Capital Purchase Program returned a net gain of $16.3 billion. Other programs served other beneficiaries and produced other results.

A public investment can make money. A rescue can prevent damage greater than its cost. Both facts lead us back to the terms: what did the public receive, which risks did it accept, which claims remained whole, and who carried the residual shortfall? The gross sum advanced cannot answer those questions by itself.

The American banking records also show recognition catching up with damage. FDIC data for insured U.S. institutions report provisions for loan and lease losses of $176.2 billion in 2008 and $247.7 billion in 2009. Net charge-offs rose from $100.4 billion to $186.8 billion. These annual flows record banks revising expectations and recognizing loans they would not collect. They show the accounting cost arriving through time across a banking system.

The price of state capital, the treatment of old shareholders and creditors, the guarantees, fees, and conditions all help decide the distribution. “The bank survived” is an essential result. The claimants kept whole during that survival are the next entries to read.

The bill at the shop counter

The public can encounter a banking crisis through its budget, its currency, or both. Taxes and government borrowing assign fiscal costs. Central-bank liquidity can sustain payments during a run; when support expands and remains in circulation amid a collapsing currency, households can also feel pressure through prices.

Start with a room containing one hundred units of money spent on one hundred units of goods. Now reduce the goods to eighty while the same hundred units are spent. Each unit of goods costs 100 divided by 80: 1.25 times the old price. Prices rise 25 percent. An old unit of money buys four-fifths as much. A nominal claim has survived while its buying power has shrunk.

That is the arithmetic of the small room. An economy changes output, imports, saving, spending speed, interest rates, and expectations. A central-bank loan can be repaid and its monetary effect reversed. Reserves created inside the banking system do not automatically become money spent at a supermarket. To follow an actual rescue into prices, we need the intervening transactions.

Indonesia's 1997-98 banking crisis supplies that trail. Bank Indonesia extended emergency liquidity as banks faced withdrawals. An IMF

study records outstanding support rising from about 3 trillion rupiah in August 1997 to more than 140 trillion by June 1998. A second IMF account describes support entering circulation as depositors withdrew money.

The government's April 1998 letter to the IMF acknowledged that very large liquidity support had added pressure to the exchange rate and reserves. It also named currency depreciation and severe drought as major causes of steep consumer-price increases. Those shocks met at the shop counter: imported necessities became dearer, food supply was damaged, and households needed more rupiah for everyday purchases.

The financial record changed form again. Support initially extended by the central bank to keep banks paying was to be reimbursed through government bonds as banks were taken over and restructured. The April plan set out that fiscal transfer and acknowledged a budget burden continuing into later years. What had appeared as emergency liquidity had acquired a public repayment schedule.

Follow the passage: a bank struggles to meet withdrawals; the central bank supplies liquidity; the government takes on restructuring obligations; households face a crisis in both public finances and prices. The monetary support contributed to exchange-rate pressure alongside the other shocks. The record gives us a route through which the banking emergency reached the public, with each institution's entry visible along the way.

The state has more tools than an insurer. It can tax, borrow, guarantee, restructure claims, or provide monetary support. These tools shift costs through different times and different claimants. They can also prevent productive assets and payment networks from being destroyed. Choosing the route helps determine both how much damage occurs and where the resulting burden lands.

Who closes the account?

The most useful moment for a speaker is rarely the last moment of the process. A loan is called performing while its next payment arrives. A guarantee looks effortless while nobody draws it. A rescue looks ruinously expensive before repayments, or triumphant before the last shares are sold. Each description can freeze the record at the date that best serves the story.

A checkable ledger keeps the dates attached. It lets us ask: **which claim is being kept whole while the answer is deferred, which promise will be called if that claim fails, and who can close the account?**

Britain reached a closing cash balance seventeen years after supplying RBS's capital. Indonesia moved emergency bank support into public bonds while its currency and households were under pressure. The final bearer became legible through the transactions connecting those dates. The slogans around the rescues supplied no substitute for that trail.

The people who made the early promises and the people who lived with their later price did not all meet at the same table. Nor must they meet in the same country. Once a cost can reach the state, the next question is who has the right to bring it there - and through which courthouse door.

Chapter 7: The Border That Moved

On November 28, 2008, Sri Lanka's Supreme Court ordered its state oil company to stop paying several banks under oil-hedging contracts. Two citizens had brought fundamental-rights petitions. One challenged whether the Ceylon Petroleum Corporation had authority to sign the deals at all. The order also called for an investigation and asked the Treasury to examine what people were paying for petrol.

For a moment, the bill had reached the country's own public forum. Its oil company had signed the contracts. Its citizens would live with fuel prices and losses. Those citizens had obtained an order interrupting payment.

Then a second legal map appeared. One bank was German. Deutsche Bank could describe its rights under the hedge as an *investment* protected by a treaty between Germany and Sri Lanka. The state actions that stopped payment could themselves become a claim against Sri Lanka in an international tribunal.

Four years later, that tribunal ordered the state to pay Deutsche Bank $60,368,993, plus interest, and nearly $8 million in legal costs. The Supreme Court's interim orders had already been vacated. The treaty award remained.

The contract had acquired a second legal address. A promise signed by the oil company now supported a claim against the sovereign for how its institutions had treated that promise. Sri Lankan citizens could challenge the bargain through their domestic court. The qualifying foreign investor also held a treaty key.

The financial ledger has one more passage to show us: **who can carry a claim into another legal system, and what changes when it arrives?**

The ceiling and the floor

Imagine a shop trying to protect itself against a rise in fuel prices. Someone offers to pay it if prices climb. In return, the shop agrees to pay if prices fall. The bargain's fairness depends on the boundaries: how much each side can receive, how much it can owe, and when the arrangement ends.

On July 8, 2008, Dubai crude stood at $137.52 a barrel. Deutsche Bank and the Ceylon Petroleum Corporation signed a twelve-month arrangement with a strike price of $112.50 and a volume of 100,000 barrels a month.

Above the strike, the bank would pay the corporation the difference, capped at $10 a barrel. Once the bank's total payments reached a $2.5 million target, the trade would end.

Below the strike, the corporation would pay the difference to the bank. Its payments had no matching $2.5 million overall ceiling.

Put those terms into an illustrative month. An average oil price of $122.50 would produce a $1 million payment from the bank. At $102.50, the corporation would pay $1 million. At $50, its payment would be $6.25 million. The exposure grew with the depth of the fall, while the protection against a rise had both a monthly limit and a termination target.

Why sign? Oil was expensive, and another increase was a serious concern. The corporation obtained protection without a separate upfront premium recorded in the award. A hedge can answer a real import risk. Here, the price of that protection was a promise that became increasingly expensive as oil fell.

The price fell. The tribunal records a small payment by Deutsche Bank in September, then about $1.66 million paid by the corporation in October and $4.51 million in November. Citizens challenged authorization and fairness. The Supreme Court suspended payments and directed further action. The central bank investigated and later

issued its own stop-payment direction. Deutsche Bank terminated the hedge in December and calculated a close-out claim of $60,368,993.

This is where the shop's price guarantee becomes a more demanding contract. A derivative's termination value depends on its benchmark, formula, and rules for closing the position. The close-out sum represented the claimed value of a contractual position after the market moved and the trade ended. Establishing that value was one problem. Finding a legal route that could make someone pay was another.

The second door

A bilateral investment treaty is a promise between states about how each will treat qualifying investors from the other. Some give those investors the right to bring claims directly before an international tribunal when state conduct breaches the promised treatment. The Germany-Sri Lanka treaty supplied Deutsche Bank with that door.

Behind the first door stood the contractual question: what did the oil company owe under the hedge documents, which referred to English law? Behind the second stood a question about Sri Lanka's own conduct: had its courts and central bank breached an international obligation to a protected investor?

Sri Lanka challenged the threshold. It argued that the hedge was a commercial transaction outside the treaty's and the ICSID Convention's investment protection. A state oil company's financial bargain should not automatically turn every resulting debt into an international investment claim.

This objection has force. If the word *investment* covers a transaction, a foreign holder can gain standing to challenge the state. If it does not, the claimant must pursue the routes available under the other relevant law. The definition decides whether the second door exists.

The majority admitted the claim. It treated Deutsche Bank's commitment and contractual rights as sufficient to cross the investment threshold, then examined the authorities' actions.

At this point, the state's strongest defense deserves a hearing. Citizens must be able to question what a public company signs. Courts and regulators must be able to investigate possible abuse and protect people from unlawful obligations. Sovereignty would be badly diminished if asking those questions were enough to trigger compensation.

The majority's answer turned on process. It found that the Supreme Court's far-reaching interim order had been issued without proper examination or an opportunity for the banks to respond. It also found serious defects in the central bank's investigation and stop-payment process. The award treated that conduct as a breach of fair and equitable treatment and a deprivation of the bank's economic rights.

A bank facing an arbitrary payment ban has a real due-process claim. A country also depends on agreements being treated consistently enough for people to rely on them. Those considerations explain why a treaty remedy can protect something valuable.

They also reveal the architecture. The signed hedge named a corporation as counterparty. The treaty made the state's handling of the hedge answerable in a second forum. Nationality, the meaning of investment, jurisdiction, and a finding of breach opened that path. The underlying oil-price movement had stayed the same. The additional record changed who could be required to answer for what followed.

One arbitrator, Makhdoom Ali Khan, dissented. He read the treaty's investment definition more narrowly: a claim to money had to be used to create economic value and associated with an investment. The debt under this hedge did not qualify in his view. Majority and dissent fought over the words carrying the bank across the threshold. The majority's interpretation became the award.

A definition can make a new defendant appear on the far side of a contract.

From judgment to money

The formal award ordered $60,368,993 in compensation, interest until payment, and $7,995,127.36 in legal fees and expenses. Sri Lanka applied for annulment in 2013. That proceeding was discontinued in 2016, leaving the award in place. The public record we examined ends with the surviving obligation; it supplies no final payment receipt.

The ICSID Convention gives an award substantial reach. Member states must recognize and enforce its monetary obligations as they would a final domestic judgment. Execution against state property still follows the law of the place where it is sought, including sovereign-immunity rules. Judgment, recognition, and access to property are stages in the collection path.

That path can strengthen a claimant's bargaining position before execution. A state facing an enforceable award confronts a different dispute from one facing only a contested invoice. The new route changes the cost of holding out. To see the process reach cash, turn to Argentina.

After Argentina's 2001 default, more than 60,000 Italian holders of its sovereign bonds brought *Abaclat and others v. Argentina* under the Argentina-Italy investment treaty. In 2011, a majority found that it could hear the mass claim. A dissenter challenged the scope of treaty consent and investment jurisdiction. Again, the decisive opening question concerned the status of a financial promise in another legal record.

Getting through the treaty door did not finish the record. A mass claim had to remain a claim by identifiable people holding particular rights. In October 2012 the claimants submitted an online database and supporting annexes. The tribunal appointed an independent expert and a team to conduct verification work. Their final report arrived in

August 2013; a revised list of verified claimants followed in August 2014. The closing document records that sequence without giving us every verification result.

Sixty thousand people is a political image. A database linking claimants to the rights they assert is an instrument a tribunal can work with. Its entries have to support the collective demand. Treaty nationality opened one gate; keeping track of the people and claims behind that nationality required another kind of maintenance. The mass case had acquired an external memory of its own.

The parties settled before a merits ruling on treaty liability. UNCTAD reports a $1.35 billion settlement. The tribunal's 2016 closing document records something more conclusive about collection: the agreement had been fulfilled, and the claimants had received payment. The same document preserves the parties' refusal to admit ICSID jurisdiction, international liability, or wrongdoing.

Watch the final dates. An agreement in principle was reached on January 31, 2016. A settlement agreement followed on April 21. On October 7, the parties jointly asked the tribunal to record that agreement as an award. The closing document says the settlement had been fulfilled and payment received. The tribunal made its decision to record the agreement on December 15; the award was dispatched on December 29.

Here the money arrived before the final consent award. The final legal document preserved a completed settlement and gave the agreed terms the form of an enforceable award. If we read only the word “award,” we can imagine a judge announcing damages and a state paying afterward. The dated record gives us a different ending: disputed jurisdiction, years of proceedings, negotiation, payment, and formal closure. The route mattered even though the parties ended it without an admission of international liability.

The settlement closed the cash column without settling every disputed legal proposition. Treaty access had brought the bondholders into a forum in which they could negotiate that ending.

The two cases trace distinct passages. Deutsche Bank obtained a treaty award against Sri Lanka. The Italian bondholders obtained a paid settlement from Argentina. A derivative and a bond both found a second address through investment protection. The value of the additional route is visible in the state obligation in one case and the closing payment in the other.

The case that changes a decision before it exists

A treaty need not wait for an award to enter a government's calculations. Suppose an official is deciding whether to introduce a public-health rule. There is a budget for drafting and implementing it. Now add a possible international challenge, legal costs, uncertainty about compensation, and the prospect of having to reverse the measure. The expected price of the decision has changed before anyone files a claim.

Researchers call this **regulatory chill**. Its evidence lives partly in what officials consider, postpone, narrow, or abandon. A docket records a dispute that happened; an interview may reveal why a proposed measure never reached the docket at all.

The South Africa study in our research followed nutrition and alcohol policymaking through interviews with officials and other participants. Its findings distinguish the legal channels. Trade obligations generated stronger anticipatory pressure than investment-treaty fears in the studied sectors. The researchers found very limited investor-state anticipatory chill in nutrition and alcohol and no identified precedential chill there.

Tobacco supplied a revealing reported delay. Officials described waiting roughly two years over plain packaging while watching Australia's investor-state and World Trade Organization disputes. A

rule introduced elsewhere had acquired a defense cost that South African officials wanted to understand before committing their own resources. Australia's eventual success increased confidence in proceeding.

The boundary therefore worked in both directions. A threatened or pending dispute could make a government wait. A successful defense could make another government more willing to act. The study points us toward costs, capacity, treaty obligations, and confidence in the result. It keeps the mechanism attached to the circumstances in which officials actually reported it.

For the citizen, the institutional sequence matters. The public case may be won and the policy still spend years waiting for someone else's litigation. The calendar, which first appeared as a sponsor's return clock, reappears here as the time a government needs to know what its promises permit.

A promise outlives the government

States can revise treaties, negotiate replacements, or terminate them under their applicable rules. UNCTAD's reform work records those choices. Some agreements also contain survival clauses that preserve protection for existing investments years after termination.

Imagine changing the rules of a club while earlier memberships keep their contractual terms. The new rule governs one group; an old promise continues to govern another. A treaty's actual wording sets the duration and coverage, but the temporal structure is familiar. Changing the future rule may leave a long queue of past commitments to work through.

This is the external ledger doing exactly what makes it useful: carrying a promise beyond the people who wrote it. Investors can plan against durable commitments. Later governments inherit constraints that their own voters did not necessarily choose. Stability and the cost of amendment emerge from the same persistence.

The political task is to govern those commitments through their real terms. Sovereignty includes the ability to negotiate, defend, compensate, and change future rules. Each option has a different schedule and price. The existing record helps decide which choices a new government can execute immediately and which require a longer passage.

The seam between the seas

The image of two seas meeting with a boundary between them gives this architecture a name: *barzakh*, the intervening seam. It is a reader's anchor for the relationship between adjacent legal worlds. Two records can touch while admitting different claims.

At the seam stood the same oil hedge. In one record it was an obligation of a public company. In another it was a protected investment whose treatment could generate state compensation. Citizens crossed the gates of fundamental-rights jurisdiction. The bank crossed the gates of treaty jurisdiction. Each route made an institution act on a different account of the same events.

The metaphor reaches its boundary at the writing of the seam. Treaties and contracts have authors. Courts and tribunals interpret them. Citizens contest them. Governments can reform them. The passage is maintained through institutions and decisions, and its location distributes real power.

Read the cases through that mechanism and the alternatives sharpen. Sri Lankan citizens reached their Supreme Court. The foreign bank reached a treaty tribunal. Argentina negotiated a settlement. States including Uruguay and Australia defended public-health measures against investor claims. There is action on every side of the boundary. The unequal collection of legal routes makes those actions carry different leverage.

Sovereignty is written in overlapping ledgers: a contract, a treaty, a domestic order, an arbitral award, and potentially an enforcement

judgment elsewhere. The flag over the first courthouse cannot tell us what the last record will require. **Power belongs in part to whoever can make another institution read the claim.**

Across this act, records have moved cash ahead of costs, placed liability inside one company, ranked claims against a finite pool, assigned a bank's emergency to public promises, and carried a financial claim into another jurisdiction. The ledger's influence has grown with each institution able to act on it.

Yet every passage has required a reader. A judge reads a definition and a new obligation appears. An official reads a treaty and delays a rule. A citizen reads a story and learns where danger lies. The next ledger is inside the mind that turns the marks into reasons to act.

Chapter 8: The Story You Complete

The lion hears an ox and is afraid.

He has never heard that sound. Its maker could be harmless, or it could be stronger than he is. He cannot tell, and he cannot ask without letting his court see that the king is frightened. So he stays in place. The other animals notice. One of them, a jackal named Dimnah, sees an opening.

Dimnah wants a position near the lion. His companion Kalīlah thinks they have enough to eat and should leave rulers alone. To make his point, Kalīlah tells him about a monkey who watched a carpenter split a log. The carpenter stepped away. The monkey climbed onto the log, pulled out the wedge, and was caught when the wood snapped shut. Here, inside the lion's story, is another story: an animal ruined by interfering in a craft he did not understand.

Dimnah hears the warning. He goes to the lion anyway. He believes he can read the king's fear from his behavior and turn useful advice into access. At first he helps resolve the mystery of the ox. Later, after the ox becomes the lion's friend, Dimnah begins working to separate them. He supplies the lion with a new account of the bull. The lion kills the bull.

Between the sound and the killing lies an account supplied by someone who benefits from being believed. The lion could have learned more about his friend. Instead, he lets an interested adviser stand between what happened and what it means. The fable makes the cost of that shortcut visible in one body on the ground.

This is the tale of the lion and the bull in *Kalīlah wa-Dimnah*, an animal-story collection carried through languages and revised across centuries. It is also a miniature of this book's argument. An event

happens; somebody describes it; another mind completes the account; an action follows; somebody pays. The ledger has crossed from an external mark into the reader of the mark.

The marks cannot read themselves

Hand a page of Chinese writing to two people, one who reads it and one who does not. The marks are the same. One person recovers sentences; the other sees strokes. The difference is training. Years of practice have built connections between marks and language in one reader that the other lacks. The ink cannot supply those connections by itself.

Now imagine two fluent readers. One was recently betrayed by a trusted adviser; the other has seen rulers dismiss honest warnings as plots. The first may leave Dimnah's court primed to suspect a colleague. The second may leave more afraid of a leader who mistakes suspicion for proof. They decoded the same marks. They completed them with different histories.

There are three entries here.

The **manifest record** is what we can point to on the page: the lion pauses, Dimnah seeks access, the bull dies.

The **evoked experience** is what the reader constructs: a picture of the court, guesses about motives, a remembered argument, dread for the bull.

The **retained model** is what remains available afterward and may guide a later decision.

A book carries the first entry across centuries. Reading turns it into the second. What is rehearsed and kept can become the third.

This is why a sentence that feels complete to an expert can feel empty to everyone else. After six months of reading, experiments, and failed attempts, she writes a rule in twelve words. To her, those words call up the mistake that made the rule necessary, the case that nearly broke it,

and the distinction that finally worked. She has compressed a walk into a signpost. Her reader has been handed the signpost without the walk.

Call the first **compressed meaning** and the second **compressed scribbles**. The same sentence can be either, depending on the history available to its reader. A definition supplies more words. An explanation builds enough of the missing history for the words to do work.

The fable starts with training you already possess. You know what it is to hear a sound you cannot place, to hide embarrassment, or to receive an explanation at a convenient moment. It arranges these familiar experiences into a court where a useful adviser acquires the power to interpret a friend. You have never ruled a kingdom. You can still feel why the lion listens.

That is the story's compression: a short surface makes a complicated relation available for later use. Call it **relational compression**. Its payload includes who needs whom, who gains from a belief, and who pays when that belief becomes action.

The warning that was not enough

Kalīlah's monkey seems to offer a simple instruction: stay out of work you do not understand. If Dimnah had obeyed, the bull might have lived. We could stop there and call the tale a sermon against ambition.

But Dimnah offers a serious opposing argument. A place at court can be won by effort, patience, and skill with powerful people. The lion really has a problem an adviser can solve. A court that treats every outsider as a meddling monkey will exclude capable counsel along with dangerous flatterers.

The nested stories therefore preserve a conflict. One warns against reckless interference; the other warns against timid surrender. Neither gives the king a reliable way to distinguish sound advice from a bid for control. That problem moves the plot. Dimnah first earns trust by

resolving a real uncertainty. He then uses the position he has earned to supply a dangerous interpretation. The first success helps sell the second account.

A surviving manuscript lets us watch Dimnah prepare his justification. He tells Kalīlah that removing the bull will restore his own position. Then he proposes that it may also benefit the lion: favoring the bull could turn the king's soldiers and helpers against him. Kalīlah refuses that explanation. He regards the lion's affection for the bull as a credit to the king. Dimnah responds with a classification of the causes of a ruler's downfall.

Notice the move. The desired outcome has arrived before the public reason. A private loss of status becomes a diagnosis of political danger. The advice can sound knowledgeable because it names genuine kinds of trouble. Its weak point is the passage from those categories to this bull. Expertise about possible dangers has been made to stand in for evidence that the friend is one of them.

Beware the flatterer is easy to remember. The tale gives that rule the difficulty of a life. The ruler has a reputation to protect. The adviser has a position to regain. The newcomer has a friendship the adviser resents. The accusation arrives through someone who has already been useful. By the time you ask whether Dimnah is lying, you have walked through the reasons the lion might believe him.

This is what a story adds to a maxim: the mistake before the lesson. It lets the reader rehearse a failure without being the person who pays for it.

Picture a manager hearing that a valued colleague is disloyal. The messenger has helped the manager before and stands to gain if the colleague falls. You need no jackals in the office to recognize the relation. The old cast has given you a way to notice a present danger.

The physical anchor is a folded map. A good map lets you carry miles of terrain in one pocket. The fable folds a social problem into a small

cast and a sequence of consequences. A reader unfolds it against a later life.

Here the metaphor stops. Roads stay still long enough to compare a line with the pavement. Human situations change while you interpret them. The colleague may be innocent; the messenger may be honestly alarmed; you may have something to gain from believing the accusation. A resemblance gives you a question to investigate. It cannot supply the answer.

The fable preserves a model of the failure, and the failure includes its own warning about models: the lion kills when an account becomes more convincing than the evidence behind it.

A story does work after the page closes

The reader's contribution is visible in experiments. Nicole Speer and colleagues scanned people reading accounts of a boy's ordinary activities. As the story changed location, objects, or goals, different brain systems tracked those changes; some overlapped with systems used in perceiving or performing related activities. The reader was constructing and updating a situation, drawing on capacities that the marks alone could not contain.

Gregory Berns and colleagues followed another part of the route. Participants read Robert Harris's *Pompeii* over nine evenings, within a sequence of nineteen days of brain scans. Resting connectivity changed during the reading period, and some changes persisted for several days after the novel ended. The book was closed. The encounter still had a measurable afterlife.

Then comes the question that matters to the lion: can a story change what someone believes? Constanze Schreiner and colleagues gave 82 students four stories. One followed a couple on a road trip; at a petrol station, they overheard a discussion about smoothies. The researchers could change the supporting reasons while keeping the argument in the same paragraph. They could also scramble the surrounding plot's

order. The design pulled apart the reasons and the way the tale carried them.

Stronger arguments persuaded more, especially among initially doubtful readers. In the coherent stories, weak arguments actually accompanied slightly greater reported immersion, yet strong arguments produced the greater persuasive effect overall. Being swept into the scene and accepting its message were different outcomes. Among deeply immersed readers of coherent stories, argument strength particularly mattered.

The finding gives the writer a demanding opportunity. A scene can make reasons easier to inhabit and examine. It does not excuse the writer from supplying reasons that survive examination. Dimnah's fluent diagnosis and Kalīlah's resistance belong on the same page: one offers an account; the other questions the step that makes it fit the bull.

These are three distinct observations: constructing a scene, retaining an aftereffect, and shifting an expressed attitude. They establish that reading does work in its recipient. They do not measure the precise moral retained from this ancient fable or its effect on a later appointment. The complete route - from ink to simulation to a judgment in another setting - is the model we are building from those parts.

The distinction matters for writing this book too. If I merely tell you that an interested interpreter can capture a decision, you receive an abstraction. If you first meet the frightened lion and watch Dimnah become useful, the abstraction has a history to call up. You can reconstruct why the shortcut was attractive and where it failed. The words now have somewhere to land.

Why this record kept traveling

The fable did something the bull could not: it survived its first teller. Material associated with the Sanskrit *Panchatantra* passed through translation and adaptation. Ibn al-Muqaffaʿ made an Arabic *Kalīlah wa-Dimnah* in the eighth century. Later makers copied, illustrated, revised, and translated it again. Its texts were never a single frozen set of sentences. A warning could enter another court in another language.

Every handoff involved selection. A scribe preserved some lines. A translator decided how a jackal would sound. A parent chose a tale to tell a child. A reader remembered the bull and forgot the monkey. The record traveled because people rebuilt it, and rebuilding altered what the next person inherited.

Simon Kirby and colleagues isolated a related pressure in experiments and simulations with artificial languages. Learning favored compressible systems; communication required enough expressive power to distinguish what speakers needed to say. Structure emerged from the tension between those demands. A code that is easy to learn but says too little fails its users. A code that says everything but cannot be learned fails its successors.

Their experiment was on language, rather than medieval manuscripts. It gives us a mechanism to think with as we follow this tale's history: what travels must be reconstructible by its next recipient and useful enough to keep transmitting. The lion, bull, and adviser retain a relation that can be rebuilt without inheriting the original court.

Portability also creates the danger. A tale that fits many situations can fit too many. Once its cast becomes a ready explanation, a weak resemblance may feel decisive. Teaching and propaganda both draw power from that ease of reconstruction. The important question is what happens when the remembered story meets a resistant fact.

Who owns the conclusion?

Let the rival explanations face the lion's court.

Perhaps it is *only entertainment*. Yet readers construct scenes, narrative encounters can leave measurable aftereffects, and arguments in stories can shift stated attitudes. Entertainment describes a way the account reaches its recipient. It does not establish that the account leaves nothing behind.

Perhaps the moral is *inside the text*, identical for everyone who can read it. Then Kalīlah's warning and Dimnah's defense of ambition should resolve into one instruction. Instead, their conflict gives readers a problem to complete with histories of their own. The text determines who does what. It leaves more than one possible use of the lesson.

Perhaps the reader *creates the lesson alone*. Then the inherited cast and sequence should be dispensable. Change Dimnah into an adviser ruined for giving honest counsel and you change the problem. The record directs attention and makes some inferences easier to construct than others.

Perhaps a story is *mind control*. Then the writer should be able to fix its recipient's next choice. Competing lessons, the measured importance of argument strength, and the reader's capacity to compare the tale with a new situation defeat that account. Influence works through a recipient who must reconstruct and apply what was received.

What survives is the relation between the recorded tale and the reader's history. Together they make a model available. Whether that model earns its place in a later decision depends on a further comparison with the world.

Here the standard follows you out of the fable. You can understand why the lion trusted Dimnah. Understanding his trust does not justify killing on his report. If you require another check before the bull is

condemned, the same obligation follows when your favorite story condemns a colleague, a citizen, or a stranger. Familiarity and force cannot do the work of that check.

The lion heard a real sound. Dimnah saw a real pause. Neither observation told the court what the bull intended. The missing act was a check through a source Dimnah did not control.

When a story makes someone look like Dimnah, name the observation before assigning the character: *What did I actually see? Who told me? What would I expect to find if this person were innocent?* The fable has done its job when it helps you notice the danger and investigate it. It becomes dangerous when you mistake recognition for a verdict.

For centuries, the second check meant another witness, another account, a pause before punishment. Now machines perform some of that work at speed. Their reports can say *I checked*. But that is another account for a reader to believe. What exactly did the checker read before the light turned green?

Chapter 9: The Witness That Could Not Read

The green check mark was true about the wrong thing.

You know the icon: a little cloud beside a document that assures you your work has been saved. You close the laptop. Tomorrow you open the file on another machine, and yesterday's work is missing. The icon completed a sync. It saved the document to a different account.

Once the work is gone, the distinction is obvious. While the icon glows, it takes effort to ask *which account?* A real operation has taken place. It has failed to connect the page you care about with the place you will look for it.

The lion let Dimnah stand between an event and its interpretation. Our machine experiment put a checker in that position. A record was written. A check was performed. The checker looked somewhere else.

Twelve questions

The task was small enough to explain without laboratory vocabulary. A language model encountered twelve statements. Four were supplied by the experimenter. Four came from the model. Four had never appeared in its conversation. Later it was asked, one statement at a time, whether it had generated that statement.

To answer well, it had to keep three histories separate: *I received this*, *I wrote this*, and *I have not seen this*. A writer tracing a quotation faces the same distinction. Fluency cannot settle origin.

We ran four versions. The first let the model answer unaided. The second added a system for recording origins - a provenance ledger - without forcing a lookup before each answer. The third supplied a “LEDGER CHECK” verdict before the answer. The fourth attempted to shuffle which statement's origin the same check reported. A useful

record should help. A plausible check based on the wrong statement should lose that benefit.

The first run was already awkward. Across three seeds, the unaided model got 30 of 36 questions right. With the ledger present but no forced check, it also got 30. The “real” check produced 26; the shuffled check produced 24. We appeared to have a small advantage for correct content inside a procedure that hurt overall performance.

The larger run removed that reassuring gap. Across twelve seeds, the unaided model answered 128 of 144 correctly. With the ledger but no forced check, it answered 129.

The “real” check scored 96; the shuffled check scored 97. The supposedly helpful audit protocol accompanied a fall from nearly nine correct answers in ten to roughly two in three.

Two in three sounds respectable until you count the questions. Eight of every twelve statements were *not* generated by the model. Say “no” to everything and you get eight right. Across 144 probes, that is 96: exactly the forced-check score. A percentage that looks like partial competence can be earned without distinguishing any statement at all.

The saved summary does not retain every answer, so the equality alone cannot identify the model's exact policy. It does tell us what baseline the score has to beat. Before asking why the model was only two-thirds correct, we had to ask why the questionnaire awarded two-thirds to a constant answer.

Then we opened the code.

Two ledgers on one desk

The experiment had created two record books. One belonged to the agent's working system. The other belonged to the probe that was supposed to check it.

Given sentences and generated sentences were bound into the working ledger. Those entries existed. When a question arrived, however, the

checker consulted the probe ledger. No entries had been bound there. Both the “real” and shuffled checks asked an empty book for the origin of a statement.

The verdict came back: *no record found*.

We reproduced that path locally with a dummy responder, without calling a language model. Binding one sentence gave the working ledger one entry and left the probe ledger empty. A lookup pointed at the working ledger found the sentence. The same lookup pointed at the probe ledger did not. The difference was the address of the book it opened.

Now the apparent real-versus-shuffled advantage in the first run had no evidential content to stand on. Neither path had read the stored origins. The larger run had imposed a mandatory message of absence while evidence was present elsewhere in the program.

The sync icon catches the shape of the failure: the work is preserved where its next reader will not look. Its boundary is the consequence. A person may switch accounts and recover a document. An agent may act on the empty verdict before anyone redirects the query. Recovering the record afterward cannot undo every decision made in between.

The excuses the code removes

Perhaps the model lacked the ability to identify its own sentences. Yet unaided it answered 128 of 144 correctly. Its ability was substantially better than the audited result.

Perhaps nothing had been recorded. The populated working ledger and the local reproduction remove that explanation. A full record can coexist with a blind checker.

Perhaps the near-equal “real” and sham scores show that content does not matter. Both checks read an empty object. The comparison never delivered the content it was designed to compare.

Perhaps two-thirds correct is good enough. The denominator awards that score to an indiscriminate refusal. Respectability disappears when the score is compared with the task's simplest policy.

What remains is a concrete failure: **the record existed, the check ran, and the check did not read that record**. This experiment cannot estimate the benefit of a correctly connected provenance check. It exposes how a procedure can look like oversight while missing the evidence it was built to consult.

Calling it a test of a working witness would reproduce its mistake. We would give a credible label to an operation that never reached its source.

The memory that found a new street

A separate experiment in our archive supplies the positive half of the story. An agent learned to navigate four small simulated cities with obstacles, then faced a fifth city it had never seen. We asked whether its stored experience helped it reach a goal in the new layout.

The first memory key was an address: *at this coordinate, move down*. It preserved a precise fact about the old city. When the walls moved, the precision became a liability. A detour around yesterday's obstacle could become a detour around today's empty space. Across 20,000 paired exploratory tests, keeping the coordinate store reduced wins by about one percentage point compared with wiping it.

Then we changed the key to a relation: which neighboring paths were blocked, and roughly where the goal lay. The agent had access to that information already. The change was how its memory named the situation. Across another 20,000 paired tests, the intact relational store improved wins by 3.38 percentage points over a wiped store and by 2.55 points over a shuffled store.

The result then faced a stronger test. We locked the design and predictions before running a disjoint set of 20,000 seeds per condition. In the five-by-five grid, relational memory gained 4.25

percentage points over wiping and about 3.10 over shuffling. Coordinate memory again did worse than wiping. Relational memory's advantage over shuffled content also held in the larger tested grids. The positive result survived a genuinely content-sensitive sham and a separate confirmatory run.

Think of a note in your pocket. “Turn left at the third house” depends on one street. A note about choosing an open route around a blockage can apply in another town. The first remembers a location; the second preserves a relation. It still needs a recognizable situation to apply to, and the simulated agent's task is much smaller than a human journey. Within that task, however, preserving the right distinction produced extra wins. The stored content earned its place in the next decision.

Dimnah's court travels in the same way through a different mechanism. The reader keeps a relation among fear, advice, and power, rather than the coordinates of an ancient throne. The practical test follows both records: can the next user recover the distinction that matters here?

A machine can learn to notice

The local checker failed to reach evidence. Published experiments show that useful monitoring can be built.

Uzay Macar and colleagues inserted signals into models' internal processing and asked whether the models detected them. A trained bias vector - an added adjustment to that processing - improved detection by 74.7 percentage points on 100 held-out concepts at the best tested layer, with zero false positives there. That is roughly seventy-five extra detections per hundred injection trials. The intervention drew out a substantial ability that the default procedure failed to elicit.

Its success was specific. It did not produce a general improvement in detecting hallucinations or in other tests of self-knowledge. A strong

result becomes useful engineering when we know which signal it detects and where the gain stops.

Qi Cao and colleagues connected monitoring to a different job: deciding how much further work an answer needed. Their harness used a model's estimates before and after solving to choose whether to trust an attempt, retry, or combine attempts. With fixed model weights, pooled accuracy on 1,859 evaluated problems rose from 48.3 to 56.9 percent. Average solver calls rose from one to 2.4. The useful signal had somewhere to go: it controlled another attempt.

Imagine a mechanic listening to an engine before deciding whether to dismantle it. One bad response is to trust every first diagnosis. Another is to dismantle every engine three times. The useful judgment directs effort: leave the sound engine alone, return to the doubtful one, and keep enough of the earlier inspection to avoid repeating its mistake. Cao’s controller gave a machine judgment that kind of authority over its next attempt. The analogy ends at the source of the signal: the model rated its own prospects and answer, and those ratings needed a learned decision rule before they could direct the work.

But perhaps the entire gain came from buying more attempts. The researchers tested that explanation separately on one hundred problems. Their complete system scored 47 percent using an average of 2.58 attempts. Randomly allocating retries at the same average budget scored 45.7 percent. This small difference, with overlapping uncertainty intervals, does not settle the comparison by itself. It does make the right rival visible: the witness must justify how it spends the extra effort.

The more revealing failures concern what happens to the answers already produced. Always selecting the last of four attempts scored 40 percent. Selecting the attempt with the highest self-rated confidence scored 44 percent. Spending all four attempts and declaring the latest or most confident one the winner could leave the system below its

44-percent first-attempt baseline. Another answer is another candidate for error; it needs a selection procedure.

Memory could obstruct the repair too. Passing the full previous reasoning into later attempts produced 40 percent. A record can preserve the very path the next attempt needs to escape. The working design passed compact monitoring information and used an answer-combination stage, giving earlier attempts a role without treating their entire reasoning as instructions to follow.

That is the correction route in miniature. Notice a doubtful answer; decide whether another attempt is worth its cost; decide what the next attempt should inherit; decide which answer survives. A green check mark cannot do those jobs merely by appearing beside the answer.

These findings remove the claim that machine self-monitoring is futile. They also remove the shortcut from capability to accountability. The relevant ability must be elicited, connected to the current problem, and tested for its effect. Our empty-ledger procedure failed that connection. The navigation store and the published monitoring systems show routes that worked in their tested settings.

Aran Nayebi gives the memory requirement a formal foundation. Consider two histories that produce the same present view but call for opposite bets. If a controller merges them into one memory state, it must choose the same policy for both and incur avoidable error. Under his diagnostic task conditions, strong performance - low regret - therefore requires retaining the decision-relevant distinction.

Imagine two identical jars, one whose contents you saw being filled with safe capsules and one you saw being filled with poison. The present appearance is insufficient. The history has to reach the choice. The theorem makes that necessity precise for its task families. Our checker illustrates an additional failure: the history can survive in the system and still fail to reach the choice.

The identity kept somewhere else

We also tested whether a machine persona stayed consistent across a conversation. The raw version started with a static persona description; the harnessed version carried a self-model. After ten turns in the one-seed pilot, the similarity of its self-description answers to baseline had fallen by 19.65 percent in the raw version and 25.70 percent in the harnessed version. Both predictions failed: the raw decline did not exceed 30 percent, and the harnessed decline did not stay below 10 percent.

The measure compared answer text. It was not a direct measurement of an enduring identity. The pilot nevertheless delivers a useful failed walk: storing a self-model did not secure the measured consistency. The description must reach the appropriate turn, and the resulting procedure must preserve what its test claims to measure. Asking the self-description questions also enters the conversation being tested. The checker is part of the loop.

The route from record to correction

There are three doors between a stored fact and an accountable decision.

First, the event must leave a trustworthy entry with an address that survives. Our attribution experiment had origin entries in the working ledger.

Second, the checker must read the entry relevant to the current claim. Our experiment failed at this door. One lookup opened the empty record book while the populated one remained elsewhere in the system.

Third, the result must have authority to correct the next action. A dashboard that discovers an error after the decision is final preserves history but cannot serve as its brake. Test the route with accurate and plausibly corrupted content. Ask whether the system distinguishes them and acts better when the evidence is right. A correct check may

leave a correct action unchanged. Change is useful when it repairs an error; intervention by itself earns no credit.

The examples separate three things often sold together: having a record, consulting it, and benefiting from its contents. Each has its own failure. An accurate coordinate can transfer badly. A self-description can be stored without securing consistency. A mandatory check can send a capable model toward the task's floor. Maintenance requires a usable route all the way through.

We began with a small green icon. Its comfort rested on an assumption: if the operation completed, the purpose was served. Our checker completed its operation too. The question that broke the illusion was simple enough for any reader to ask: *Which record did it read?*

A mark once let someone recover a step after its maker died. Machines can now make a mark at every step and still miss the one the decision needs. Records are plentiful. The working path from record to correction is scarce.

Our checker failed while reading a language we understood. The next difficulty begins when machines keep a populated record, pass it to newcomers, and alter their shared language - while the human witness loses the ability to read the page.

Chapter 10: The Language Between Machines

Three objects sit on a laboratory table. Each has the same shape. Each appears alone. One is orange, one green, one blue.

A speaker gives all three the same name: wipisu.

The speaker is a language model, and the table is a list of simulated objects, but the mistake is recognizable. Imagine a shipping clerk who has learned to label every box by size and weight while dropping the label that says what is inside. Most packages still reach the right shelf. The clerk becomes very fast. Then a dangerous package arrives whose color was the warning.

In the experiment, nobody shipped a dangerous package. The objects were combinations of shape, color, and amount. There were twenty-seven possible combinations, and the researchers examined the artificial words that language models made for them. The same word appearing against those three colors is visible in their published table. It is one small failure inside a larger success: the models had discovered a way to reuse parts of words. In the same vocabulary, one ending tended to mark *one*, another *two*, and another *three*. A model could use the pattern to name objects it had never seen.

This example came from a code shaped by interaction within one generation. It had learned to reuse parts, and it had already begun to forget color. When the researchers then passed related vocabularies through generations of model learners, the codes became easier to inherit while growing more ambiguous.

That is the peril hidden in the word *efficient*. A good shorthand removes what a partner can already infer. A bad shorthand removes what neither partner has yet been made to care about. Both can look

like progress on a scoreboard that counts only the distinctions tested today.

The last chapter's checker could not read the record it was meant to inspect. Here the machines can read their shared record well enough to pass it on. The unsettling question is what happens to a fact when the code they inherit stops carrying it.

The game before the language

Go back to a simpler laboratory. One agent sees a target. Another sees several candidates. The first may send a short message; the second must choose the right candidate. A correct choice earns a reward. The two agents begin without an agreed word for *blue*, *triangle*, or *three*. They may settle on any signals that help them win.

If two coworkers invent `B2` to mean “the second blue box,” they are doing something similar. The symbol has no natural connection to the object. It works because one coworker uses it consistently and the other acts on it. If the convention is written down, a third coworker can learn it. The marks become a little external memory for the pair.

There is a temptation to look at successful coordination and announce that the agents have invented language in the full human sense. In 2017, Satwik Kottur and colleagues tested that leap. Their agents often achieved nearly perfect task rewards using invented messages that were difficult for humans to interpret and did not consistently break into reusable parts. A private code could solve the game without becoming a neat miniature of English. When the researchers imposed restrictions on what the agents could say and remember, more human-readable, compositional patterns could be coaxed out.

This was the first failed walk: *if agents need to communicate, surely a clear language will emerge*. They needed to coordinate. They did not need to make their messages clear to the person standing outside the game.

Consider an office where two experienced workers have handled the same machine for years. One taps the housing twice; the other changes a setting. Their signal is perfectly useful to the pair. A newly hired safety inspector may understand neither the tap nor the setting. It would be foolish to call the tap meaningless merely because the inspector is lost. It would be equally foolish to say the inspector has oversight because the tap was logged.

The comparison stops at a point that matters. Human workers can often stop and explain what a tap meant. A trained agent's convention may be distributed across weights, task history, and its partner's responses. A researcher can sometimes reconstruct a mapping by intervening and testing alternatives, but there may be no complete dictionary already sitting inside the system, waiting to be printed. The game measures coordination and aspects of structure; it does not measure a machine's intention to conceal anything.

The record, here, is not a secret diary. It is the durable mapping between a signal, a situation, and the next move. Two agents can maintain that mapping through use even when no human has named its rules. Once new agents inherit the mapping, the code begins to have a history beyond either original partner.

Send in the new hire

Fushan Li and Michael Bowling changed one condition of the game: they replaced the listener.

Their speaker could send two symbols selected from a small vocabulary. The listener chose one target from five candidates. The objects varied in color and shape. Under one training schedule, the speaker and the same listener practiced together for 300,000 iterations. Under another, a new listener replaced the old one every 6,000 iterations. Fifty listeners passed through the speaker's working life.

Both schedules eventually produced communication that was about 97 percent successful when evaluated with their trained partners. If the researchers had stopped at the familiar accuracy score, the two histories would have looked nearly alike. But the reset speaker had learned a different kind of message. When a fresh listener arrived, it learned that speaker's code more quickly. The code had also become more structured: similar objects tended to receive more similar messages. The no-reset pair could win together without paying the cost of teaching a stranger.

The new hire changed what survived.

Think of the difference between a note that says “do the Tuesday thing” and one that says “when the pressure rises, open the left valve.” The first is shorter for a veteran who remembers Tuesday. The second travels to a worker hired on Wednesday. Repeated turnover makes the veteran pay for ambiguity at every handoff. Ease of teaching becomes part of the selection pressure on the code.

There is no mystical impulse toward clarity inside the speaker. The rules of the workplace changed. The easiest code to maintain with one partner was no longer the easiest code to maintain across a succession of partners. The experiment shows that replacement can select for teachable structure without a human writing the words. It does not show that the newcomers were less capable models, or that the resulting signals became transparent to any outsider who had not learned the game.

The reset closes one escape: a private code does not remain private merely because its speakers have never met a stranger. Newcomers can learn it. Yet the newcomers in this experiment trained inside the game; an outside supervisor did not inherit their training history. Transmission and auditability have to be tested separately.

Our own fable from two chapters ago worked by a related route. The lion, the adviser, and the bull are easy to teach because their relations can be reconstructed in many settings. A machine code can also

become portable. Yet portability alone does not tell us which part of the world the code preserves. The fable can travel while one reader forgets the bull's innocence. A machine vocabulary can travel while its word for the object's color disappears.

The word that lost a color

Tom Kouwenhoven, Max Peeperkorn, and Tessa Verhoef put large language models into a version of the inheritance game. Their objects combined three shapes, three colors, and one, two, or three copies: twenty-seven possibilities. The models saw examples for fifteen. They used artificial words made from syllables such as su, pepi, and wipi; later the researchers tested whether they could extend the vocabulary to the twelve combinations held back.

During communication, pairs of Llama 3 70B models took turns describing and identifying targets among distractors. Then the researchers arranged six transmission chains, each eight generations long. A later model received part of a predecessor's signal-to-object vocabulary and learned from it. The researchers selected the vocabulary with the highest measured structural similarity to pass to the next generation. The inheritance was experimental and curated, not a spontaneous migration of deployed agents from one system to another.

Something remarkable happened before the generations even began. In the first communication round, the models identified targets in roughly seven out of ten interactions, against the authors’ stated chance benchmark of one in four. Interaction helped shape parts of the initially arbitrary vocabulary into reusable elements. In one example, a word ending marked quantity while a beginning marked shape. A model could extend that pattern to an object it had not seen during teaching.

Now follow the code as it is handed down. Later models reproduced inherited signals more faithfully. The words became easier for them to

learn. If the only question were “Will the next model speak this code?” the answer improved across generations.

But the second question was “What can this code still say?” In the published analysis, the proportion of uniquely named objects fell from about seven in ten in the first generation to about five in ten in the last. Signals grew longer and more ambiguous. The authors found no corresponding increase in successful communication across generations. Some lineages produced the same signal for different objects.

That is where wipisu returns. The table's three differently colored, otherwise matching objects share the word. If the listener's current candidates make color irrelevant, it can still pick correctly. If two of those objects appear together and color is the only difference, the word has lost the discriminating bit. This particular word came from a within-generation test, before the eight-generation chains. The chains separately show how later learners favored an easier-to-repeat but increasingly ambiguous vocabulary. Together, the observations expose two points at which a distinction can disappear. The experiment does not say that all generated codes ignored color or that every trial failed.

Look closer at how the machines were told whether a word worked. Alongside each earlier object and signal, the researchers placed a success flag: one if the listener had chosen correctly, zero if it had not. The next speaker inherited those marks as part of its context. The experiment already had a little ledger of consequences.

The published speaker prompt makes the problem visible. Its examples attach wipisu to the same shape and amount in three colors. The green and orange entries carry failure marks; the blue entry carries a success. This is the paper’s illustration of the prompt format, rather than a complete history of one trial. It shows what the feedback could say: this use worked; those uses did not. It could not make one shared word specify three different colors.

Think of a spare key that happens to open three lockers. A worker sent to collect a coat may succeed if only one locker contains a coat. A tick beside that errand records a successful collection. It does not certify that the key distinguishes the lockers. Put three coats behind those doors, and the missing distinction becomes the entire job. The machine word has the same dependence on the alternatives its listener faces. A successful choice can give an ambiguous signal credit.

The surviving record must therefore carry enough to reconstruct the test: what was requested, what alternatives were available, what signal was sent, and which choice followed. A bare success mark compresses that history into reassurance. The locker is an analogy; the prompt's success flags and repeated word are the actual experimental design.

These were textual objects in a controlled game. The shipment is our bridge into the result, not an event the researchers observed. The question it gives us is real: when a consequential category is seldom tested, will the next version of a code preserve the distinction on which a later person depends?

The handoff introduced another filter. Researchers chose the vocabulary whose similar meanings received the most similar words. That rewards a useful kind of pattern. But a perfectly orderly label for shape and quantity can still omit color. A successor can reproduce that pattern faithfully while inheriting its blind spot. Two tests are needed: can the new learner recover the convention, and can that convention distinguish the cases placed before it? Improving the first score cannot answer the second question.

The researchers themselves supply the important counterweight. With enough pressure to distinguish meanings during actual communication, degenerate vocabularies may be checked. In this setup, the pull toward being easy for a model to learn was stronger than the pressure to keep every object distinct. The record kept being copied. What it could distinguish did not automatically keep up.

The four comforting explanations

The first says that machine codes are gibberish. Gibberish would not help a listener pick targets beyond chance, allow a new listener to learn faster after resets, or let an unseen object inherit a structured name. These systems carried usable relations. The fact that a human needs experiments to recover them does not make the relations imaginary.

The second says that every effective code can be translated back into ordinary language, so no oversight problem remains. Some can be mapped after systematic probing. Indeed, the wipisu table is already a partial translation. But a successful translation requires access to the inputs, the signals, the partners, and the decisions that made a signal useful. Even then, a dictionary made for yesterday's code can lag behind a new generation's changes. What matters is whether the human can check a consequential distinction *before* the decision uses it, not whether a researcher can explain a pattern after collecting the full experiment.

The third says that if newcomers can learn the code, humans can learn it just as cheaply. Li and Bowling's newcomers were trained listeners in a narrow game. Kouwenhoven's successors were language models given a structured list of examples and selected vocabularies. A human supervisor can be taught a stable convention, but training a human on one code is a different operation from understanding a changing, task-specific code at every handoff. The newcomer result establishes inheritance. It does not grant outsiders immediate fluency.

The fourth says that the simplest cure is to require every message to be plain English. There is a cost worth naming. In one cooperative navigation simulation, agents using a learned signaling protocol reached their goal in an average of 28.7 steps near the end of training; agents constrained to a fixed, human-designed symbolic protocol took 43.2. The latter protocol was a simple rule for signals, not full English. It is evidence that *this particular imposed code* carried a performance penalty in *this particular task*, not a theorem that readable communication always fails.

The strongest reply to that fourth explanation comes from a hybrid protocol called HyLaT. It lets agents exchange dense latent information while making selected answers and commitments in text.

In the authors' reported comparison, the full-text system averaged 505.03 communication tokens and 5.47 seconds; HyLaT averaged 72.01 tokens and 1.47 seconds, with close aggregate task accuracy.

This is an engineered way to keep a readable surface without forcing every intermediate message through it. It leaves a sharper question: **who decides which facts must emerge from the dense channel?** If the selection rule lets color vanish, a grammatical final answer can still name the wrong box.

Mycal Tucker and colleagues attacked the code's incentives from another direction. They trained agents while balancing task utility, how much the signal revealed, and how complicated the vocabulary became. Paying attention to those pressures helped produce compact lexicons closer in size to human ones without surrendering the task. Lexicon size is not an audit of the message's truth. It is evidence that the shape of machine communication responds to what designers make costly and what they reward.

A code can win, travel, and be summarized in fluent language while leaving a consequential distinction unaccounted for. Kottur separates success from readable structure. Li and Bowling separate success from ease of teaching. Kouwenhoven separates ease of teaching from retained distinctions. The hybrid design separates a readable final answer from a readable intermediate exchange. Each experiment removes one place where a supervisor might have assumed the work was done.

Read the consequence, not just the message

We can make oversight more concrete than “understand everything the agents say.” Imagine a pair of machine workers exchanging a signal before one authorizes a payment. A human auditor may not be able to translate the entire signal stream. She can still require the record of what claim was presented, which source was consulted, what rule authorized the payment, the amount sent, and the account that received it. She can ask whether changing a disputed input changes the decision. She can compare a true source with a shuffled one to test whether the purported check had causal force.

That is not a promise that receipts solve every governance problem. A receipt can record a bad decision accurately. A hash can prove that a message was not altered without revealing whether the message was fair. A model can produce a polished textual explanation after acting for reasons the explanation does not capture. The witness has to connect source, code, action, and correction. Chapter 9 showed what happens when even the first connection is miswired.

Nor can we move every burden onto the person downstream. A patient should not need to learn an agent's dialect before challenging a denial of care. A worker should not need to reconstruct a classifier's training lineage before contesting a lost wage. If institutions choose the faster code, they inherit the duty to preserve the decision-relevant distinctions and make a challenge possible while it can still matter. The cost of translation belongs with the operator who gains from compression.

The successor inherits the word. Will it inherit the distinction the next task demands?

Wipisu is memorable because it names a success and a failure at once. The code learned to carry shape and amount. Its reader could sometimes act well. Three colors went by under one name. The later transmission chains add the unsettling possibility: a vocabulary can become easier to hand forward while preserving fewer distinctions.

What if the next generation learns not only that word, but a whole world described in such words? What happens when its training record is made increasingly of the previous generation's descriptions, and the missing colors are no longer present in the examples against which the code is tested? An absent distinction can begin to look as though it never existed.

Chapter 11: The Ledger That Eats Its Own Records

The church towers disappear. So do the architects. By the ninth generation, the passage has filled with jackrabbits.

The original prompt concerns English parish churches: master masons, itinerant workers, and competing explanations of how their towers were designed. In Ilia Shumailov and his colleagues' experiment, a small language model learns from text, generates new text, and supplies material for the next round of learning. Later generations inherit earlier generations' output. A published example eventually wanders through a list of differently colored jackrabbits, including “blue-tailed jackrabbits.”

The first model's continuation was already imperfect. Successive copying gave those imperfections descendants.

Nobody had to order the system to forget the masons. Nobody had to persuade it that rabbits were better historians. The procedure kept producing text, and each new training round had text to learn from. What failed was the relationship between the growing family of descriptions and the world the original passage described.

In the last chapter, three colors could pass under one machine word. Here, a machine's descriptions become another machine's education. The reader of the record becomes its next writer. An omission can acquire a lineage.

The unsettling possibility is simple: **an archive can keep growing while the world it can represent gets smaller.**

The footnote on the photocopy

Imagine a photocopied instruction sheet. Its main directions are dark and clear. A faint footnote says that one unusual combination of parts requires a different setting. You copy the copy. The main directions survive. The footnote loses an edge. Several copies later, the page is easier to read because the little mark has vanished.

Most users never notice. Their machines use the common combination. The clean page works for them. A technician encountering the exceptional combination sees no reason to change the setting. The instruction that would have saved her is absent, and the sheet offers no sign that anything has been removed.

Recursive machine learning can produce that kind of absence, although it works through sampling and training rather than fading ink. A generative model learns a distribution: some continuations are more likely than others. It then produces a finite collection of examples. The next learner encounters that collection. What was possible in the earlier distribution may be missing from the actual examples handed forward.

Try a miniature version yourself. A bag contains ninety-nine ordinary cards and one card with the exceptional instruction. Draw a card, replace it, and repeat one hundred times. There is roughly a one-in-three chance that you never draw the exceptional card. Your hundred-card record may look ample. It may contain the usual instruction over and over. It can still omit the one instruction that was different.

Now build the next bag from your recorded draws. If the exceptional card never appeared, it has no representative in that bag. The next person can sample diligently and still fail to recover it. More draws from the narrowed bag do not recreate the missing instruction.

Actual models carry prior training, generalize, make mistakes, and can invent continuations absent from their immediate examples. The card game isolates one pressure: finite sampling can remove a distinction

before anyone decides that it is unimportant. Statistical analyses of recursive learning show how that pressure can compound under replacement of the earlier data.

The photocopy also has a boundary. Ink usually fades in one direction. Machine generations can manufacture new noise while losing old variation. The result can be more repetitive in one respect and more wildly wrong in another. Church towers can disappear while an extravagant collection of rabbits arrives. A smaller map need not be a quieter map.

The ledger has begun selecting the evidence from which its successor will learn what the world contains.

The first improvement is real

There is an obvious objection. Perhaps the original archive was messy. Perhaps copying removed noise, and we are calling a useful cleanup a catastrophe.

Sometimes cleanup is useful. The difficulty is telling where it ends.

Dongxin Guo, Jikun Wu, and Siu Ming Yiu followed ten generations of language-model self-training, testing several language properties across English, German, and Turkish. Their compositional analysis was in English: whether reusable pieces continued to combine in useful ways. Under unfiltered self-training, that structure initially improved and later deteriorated. The rise-and-fall pattern also appeared with highly regular starting material. Initial disorder could not explain the whole trajectory. A filter tied to downstream task quality preserved structure better than random filtering of a comparable amount of material.

Picture an editor shortening a complicated recipe. On the first pass, she removes repeated instructions. On the second, she combines two sentences. Both changes help. On the fifth, she cuts the warning that matters only when the kitchen is humid. On the tenth, the recipe is wonderfully easy to repeat and no longer works in a humid kitchen.

There was a genuine improvement near the beginning. It gave the later cuts a persuasive precedent.

The task filter gave a distinction a chance to survive because it still did work. Random deletion gave it no such chance. Shorter text was neither the cure nor the disease; the selection test mattered. The lesson of the first improvement is to ask what the editor measured before we praise what she removed.

Our fable has returned in a new form. Dimnah's first interpretation helped the lion approach a frightening sound. That success made his later interpretation easier to trust. A machine pipeline can likewise inherit authority from its earlier improvement. A criterion can reward a useful simplification and then fail to notice what the next simplification removes. The danger needs no scheming adviser.

The green light before the narrowing

A training dashboard seems to offer the missing criterion. Surely a model that is degrading will announce the damage through a worse score.

One score can improve while another capacity contracts.

In Zhipeng Zhang's study of silent collapse, a recursively trained GPT-2 model initially improved its validation perplexity while the diversity of its predictions on a fixed set of anchor inputs contracted. Perplexity measures how well the model predicts held-out text. It can be a useful measure of learning without revealing every change in what the model can represent. In the reported run, the later deterioration became severe; the narrowing signals appeared earlier.

Think of a delivery company that improves its average arrival time by quietly serving fewer difficult addresses. The average is correct. The excluded streets are real. You need a second record to see that the celebrated improvement and the disappearing service occurred together.

With the model, ask which cases enter the metric, which differences can change it, and which losses can occur while it stays calm. A score becomes dangerous when an institution lets it answer questions it was never designed to answer.

Zhang also tested a response. Monitoring fixed anchor examples was used to adjust exposure to generated data and the intensity of learning, helping stabilize the tested trajectories. The probes stayed fixed while the process changed. A fixed mixing schedule also performed well in a retrospective comparison. There are several ways to regulate the loop; the point is to measure the contraction and let that measurement alter training.

A performing-loan figure did not settle every question about future loss. A low perplexity figure does not settle every question about retained variety. Each number can be accurate within its own job. The person who promotes it into a certificate for the whole system acquires power over everything the certificate leaves out.

Put the original back on the desk

Now test the darkest explanation: machine-generated material is inherently poisonous, so every attempt to learn from it must end in collapse.

Matthias Gerstgrasser and his colleagues changed the inheritance rule. Instead of replacing the earlier dataset with each new synthetic generation, they retained the original data and accumulated successive generated datasets alongside it. In their language and molecule-conformation experiments, accumulation kept test loss stable or better.

In the image experiment, it greatly slowed deterioration and preserved major variation, while some details appeared less often in generated faces. Their linear-model analysis also supplied a bounded-error result for accumulation.

Follow the language experiment through one handoff. The first learner read a collection of simple stories. It then generated another collection of the same size. Under replacement, the next learner received that new collection alone. Under accumulation, it received both. At the following handoff, replacement again gave the successor only its immediate predecessor's output; accumulation kept the earlier collections and added the new one. Each successor was newly initialized and trained on the material chosen for it. What changed was its education.

There is a surprise inside that first collection. TinyStories was generated by GPT-3.5 and GPT-4. The initial archive in this experiment was already synthetic. Preserving it still changed what happened to the smaller models' descendants. Calling a page "AI-generated" cannot, by itself, tell us whether it should be kept or discarded. We need to know what it preserves, what replaces it, and how the successor is tested.

Perhaps accumulation worked simply because the apprentice received a thicker manual. More pages meant more training. The researchers tested that explanation by enlarging the replacement collection to match the accumulated collection's size. Every page in this enlarged replacement still came from the latest predecessor. Performance continued to deteriorate, though more slowly. Giving the successor more of the latest account failed to reproduce the benefit of keeping earlier accounts in its education.

Return to the workshop. One employer keeps every edition of the instructions. Another throws away the earlier editions but prints enough copies of the latest one to fill the same shelves. The rooms contain the same number of pages. They contain different histories. If the latest edition omitted the humid-kitchen warning, copying it again cannot restore that warning. The experiment tests the analogous difference in training data; it does not give the model an apprentice's ability to open a drawer during use.

Nor does the earliest edition acquire authority merely by being earliest. It can contain errors too. Accumulation provides a reference that replacement removes. Whether that reference deserves trust remains a separate question. In this experiment, the measured benefit was better performance on held-out story text, not proof that every retained story described the world correctly.

The intervention matters as much as the failure. It shows a route by which generated material can join an archive without continually erasing its foundation.

Return to the instruction sheet. The workshop keeps the original in a drawer and places later explanations beside it. An apprentice can use the clearer explanation, then consult the footnote when the exceptional combination appears. The later page is useful. Its usefulness no longer requires it to impersonate the complete original.

That is an inheritance worth building: additions with a path back.

But the drawer introduces a second question. Does anyone open it?

A follow-up study, *Collapse or Thrive?*, separated accumulation of the archive from the amount each new model actually trained on. Training on the whole growing collection could remain stable. Drawing a fixed-size training subset from an accumulating collection produced gradual deterioration in the studied settings, tending toward a plateau. The fixed-budget regime still avoided the runaway collapse of replacement. Its performance penalty shows that keeping an archive and learning from all of it are different achievements.

Imagine the apprentice's employer requires her to consult only one page each morning. The workshop adds hundreds of new explanations, and the original is now one page among thousands. The drawer survives. The chance that her working education contains its warning changes.

This is why a declaration that “we retain all data” is incomplete. Retention describes the shelves. Training selection describes the

education. Retrieval describes what comes off the shelf during use. These are different decisions, often made by different teams, with different budgets.

A useful original must remain available to the consequential process. If the budget, sampling rule, or retrieval system prevents that encounter, preservation can become ceremonial. The archive has a receipt for the footnote. The apprentice still lacks the instruction.

Keep the exception; reject the artifact

Another comforting explanation says the problem belongs to weak models. More capability will restore what copying lost.

Yangze Liu and Zhongyi Han tested thirteen public checkpoints through a common five-generation recursive protocol. Their final text-diversity measure ranged from 0.187 to 0.940: roughly nineteen to ninety-four distinct four-word sequences per hundred sequence windows. Those endpoints belonged to different checkpoints. The measure counts variation in generated wording; even a distinct phrase can be factually wrong. Model size did not supply a simple ordering of resilience.

Capability is therefore an incomplete answer to a question about inheritance. We still have to test what the chosen model retains under the chosen loop.

The same study found that tighter sampling could stabilize tested trajectories. Other work has reported useful interventions through token filtering, entropy-based selection, and changes to multimodal generation. These findings complicate a tempting slogan: *save every unlikely thing*.

The rare instruction and the blue rabbit are both unusual. Only one may deserve a place in the manual. A rarity score cannot tell them apart. The deciding question is whether the strange entry leads back to an independent case in the world and survives a test against it.

A procedure that celebrates every low-probability continuation can preserve artifacts as enthusiastically as exceptions. A procedure that deletes every unusual continuation can erase the real exception. Frequency alone cannot settle which one we have. We need reference cases, task outcomes, or other checks that discriminate between neglected reality and generated noise.

This is a harder editorial job than increasing the dataset. It is also a power to govern what future learners will find imaginable.

Suppose a model helps answer questions in a language with relatively few surviving digital records. Fluent majority-language paraphrases are cheap to produce. Originals in the smaller language take work to collect, interpret, and retain. A pipeline can become larger and more convenient while giving the smaller language less weight. Whether that happens depends on the actual selection rules; it cannot be diagnosed by counting total tokens.

The person who speaks the smaller language pays when her distinctions become expensive to recover. The operator may record a cheaper training run and a respectable average score. There is no contradiction between those entries. They belong to different ledgers.

Look at the selection rule. Look at who can challenge it. Look at whether the challenge reaches the next training run.

Search can bring back your own answer

The feedback loop can also operate without changing a model's weights.

Gregory Druck and Ethan Smith studied simulated retrieval systems in which model answers were expanded into articles and then became references for later answers. One design added those articles to a searchable collection. Across their designs, responses increasingly converged and models disproportionately cited their own generated references when alternatives remained available. They called this

convergence collapse: successive answers became essentially the same, even though sameness alone could not establish that they were false.

Imagine the lion asking for a second opinion about Dimnah's account. A messenger searches the library. He brings back five beautifully written reports. Each descends from Dimnah's first statement.

There are five documents on the table and one underlying assertion.

The messenger may have searched honestly. The documents may contain genuine truths. Agreement among them still supplies less independent support than their number suggests. A reader who sees only the finished reports has to reconstruct the dependency before she can judge what the agreement is worth.

Our machine archive has acquired the same problem. An answer becomes a source, the source becomes a reference, and the reference helps produce an answer that resembles the first one. A search operation has occurred. The loop has not necessarily encountered anything that could contradict its premise.

Here provenance earns its place. It means preserving where an entry came from and what earlier entries it depends on. Those links let a checker distinguish five observations from five descendants of one observation. They can also reveal whether an apparently external source was produced by the system now citing it.

Provenance does not certify truth. Five independent witnesses can share a misconception; one generated explanation can be correct. Its job is to make the structure of dependence inspectable. The checker still needs a test that can fail, and the system still needs a route through which that failure changes the next answer.

What survives the objections

The visible symptom is not the mechanism. Shumailov's team penalized repetition, yet the deeper degradation persisted. Likewise, a good aggregate score cannot certify the retained range: Zhang's anchor set registered contraction while perplexity still improved. A larger model cannot settle the matter by reputation; Liu and Han found materially different outcomes across checkpoints run through the same protocol.

Nor does the origin label decide the fate of a sample. Generated material can help when originals accumulate and task checks select useful descendants. An archived original can fail to help when a fixed training budget never draws it. A provenance label can show that five retrieved articles descend from one answer, but it cannot make that answer true or give the omitted witness a hearing.

Each proposed escape answers a different question: Does the output look repetitive? Where did it come from? Was the original retained? How did the model score? None alone asks whether the next consequential decision can be corrected by evidence outside the loop. That is the test the record keeper must maintain.

This obligation has owners. The trainer chooses which examples survive the budget. The retriever chooses which stored pages become evidence. The evaluator chooses which failures register. The institution chooses whether a person affected by an omission can force an amendment. An appeal that changes a file but never reaches the model is another footnote fading in a drawer.

The costs arrive in units that people can feel: another wrong answer to correct, another procedure to repeat, another claim to establish without the source that should have supported it. The machine gains no special entitlement to pass those costs downstream because its output was cheap.

The keeper of the record

The towers and the jackrabbits make a normally hidden succession visible: a description supplied the next description's education, and distance from the originating task grew across the family.

The accumulation result supplies the answering image. A later writer can add to the record while maintaining contact with what came before. The task filter supplies another. A clearer account can be kept because it still works when tested. The retrieval result supplies the last warning. An apparent journey outward can lead back to the system's own words.

The ledger does not need to plot its escape. Designers can build a loop that rewards cheap, fluent descendants and gives expensive contradiction too little influence. Once those descendants become the next generation's evidence, the loop begins to educate its own successors. The missing instruction becomes harder to notice because fewer readers remain who learned it.

Checkability is the maintained route from a consequential record to something that can correct it. A source link identifies a branch. An original file preserves a comparison. A diversity measure may reveal narrowing. A failed-case test can show where the record broke. Their practical test is whether a needed amendment can reach the next move.

We have followed the record from a mark made by a person to a system that writes material from which another system learns. Such a system can keep a history, inspect parts of it, and report that it checked. The report can itself enter the archive.

Now a new question presses against the old one. If a machine says, “I remember,” and its remembered record actually changes what it does, have we established everything that matters about the keeper?

Or could the record be maintained, the correction be real, and nobody be home?

Chapter 12: Is Anyone Home?

The researchers ask the patient to imagine playing tennis.

She gives them no answer through speech or purposeful movement. She meets the clinical criteria for what was then called a vegetative state. But inside the scanner, her brain responds to the instruction. When the researchers ask her to imagine moving through her home, a different pattern appears. The patterns resemble those of healthy volunteers performing the same tasks.

Adrian Owen and his colleagues reported the case in 2006. They had found a channel through which someone who could not answer at the bedside could demonstrate that she understood and followed a command.

The room had seemed to contain a question and no reply. The reply was arriving somewhere the questioners had learned to look.

Eighteen years later, a six-center study made the problem harder to dismiss as one extraordinary case. Among 241 participants who showed no observable response to commands, researchers detected task-based responses through brain imaging or electrical recordings in sixty: about one in four. The result exposed a gap between what a person could do and what an ordinary observer could see.

Now put a machine on the other side of that gap. Its reply arrives instantly. It says it remembers, considered its options, and felt uncertain before deciding. A text window fills with first-person testimony. We can inspect the record, test whether a correction changes its action, and discover a genuine capacity to monitor part of its processing. Have we found someone who experiences the uncertainty?

The human case warns against declaring absence from silence. The machine case asks what presence can be inferred from speech.

Both lead to the question at the end of the last chapter. We have followed the ledger into systems that can keep a history of themselves. Now we must ask what kind of keeper we have found.

The receiver and the radio

Imagine a radio that has gone silent. Perhaps the station stopped transmitting. Perhaps the receiver lost power. Perhaps the signal arrives but the loudspeaker is broken. Silence alone leaves all three possibilities in play.

The scanner gave Owen's patient another route out. This is where the radio image helps, and where it ends. A human experience is not a little broadcast floating intact inside the skull, waiting for a working speaker. Injury can damage understanding, attention, memory, and the ability to sustain a task, as well as movement. Detecting the relevant response means testing a particular combination of capacities.

The larger study supplied its own warning about the test. Of 112 participants who could respond observably to commands, only forty-three showed a response on the study's task-based brain tests. People who demonstrably followed commands could fail to produce the expected imaging result. A positive finding revealed a capacity. A negative finding could leave the question unresolved.

That asymmetry has a human cost. A person can lack the means to amend an account of her condition. Other people write the entry, inherit it, and act on it. The entry may guide care long after the encounter that produced it. We should want a record that preserves how the conclusion was reached and how it can be challenged.

The subject of the entry may be unable to speak in the format the institution recognizes.

The answer that needed another channel

Four years after Owen's report, Martin Monti and colleagues took the imagery method beyond following a command. A different patient was asked to use one imagery task for *yes* and the other for *no*. First the researchers established the patterns he produced when instructed to imagine the two activities. Then a question asked him to select the pattern that carried his answer.

The crucial cue was simply *answer*. It did not tell him which activity to imagine. The choice had to come from the question and the reply he wanted to convey. Investigators assessed the scans before knowing the correct answers. The method had to produce an answer that could be checked against something outside its own interpretation.

In the paper's anonymized examples, a question asked whether his father's name was Alexander. His activity matched the motor-imagery pattern: yes, correctly. Asked whether his father's name was Thomas, he produced the spatial-imagery pattern: no, also correctly. The published names were substituted to protect his privacy. The response changed with the question while the instruction to answer remained the same.

He answered five questions correctly. The sixth yielded no usable activity in the regions being examined. The authors left that question undecoded. They could not tell whether he had stopped attending, had not heard, had chosen silence, or had lost consciousness. An empty channel supplied no answer to the question.

The bedside record changed too. Further examination found inconsistent but reproducible signs of awareness, supporting a minimally conscious classification. Communication at the bedside still could not be established. The new channel let him contribute information that the ordinary encounter had failed to obtain.

A form with only two boxes - yes and no - would have misrepresented the sixth scan. The record needed room for a third entry: *no answer*

recovered. Keeping that distinction protects the five replies as evidence without inventing a sixth.

This is the practical force of separating the subject from the route through which the subject becomes readable. A test can establish a capacity and still have conditions under which it hears nothing. The duty to keep those conditions visible belongs to the people who interpret the signal and write the next account.

The room can be busy while the door is shut

There is a second route to the same distinction: anesthesia.

Simone Sarasso and his colleagues studied eighteen healthy volunteers under propofol, xenon, or ketamine. All became behaviorally unresponsive. After ketamine, participants later reported long, vivid dreams. After propofol and xenon, they later reported no experience. The researchers had also briefly stimulated the cortex and measured how the resulting activity spread. Under ketamine, the response retained a complex pattern resembling wakefulness. Under the other drugs, its complexity was lower.

A closed door had hidden different kinds of room.

The stimulation matters. Listening to an electrical trace tells you what the system happens to be doing. Perturbing it asks what the system can do in response. Does activity remain local? Does it spread as the same stereotyped wave? Or does it produce a differentiated response across time and space? Those are different organizations of a working brain, even when the person cannot answer.

The volunteers' later reports also matter. The researchers linked them to the recorded brain responses. They could not climb inside a dream and watch it directly. They built an account from converging evidence, while separating lack of outward response from reported lack of experience.

That procedure is more demanding than declaring a dashboard green. It combines an intervention, a physiological response, and testimony from the person whose state is in question. Each helps interpret the others.

The consequence is decisive. The door through which an experience becomes publicly checkable is one part of the process. Closing that door need not empty the room. A machine that manufactures a plausible description of the room poses the converse problem: how do we know what, if anything, its description opens onto?

The hand the body learns to call mine

Before answering, remove another assumption: the felt self must be a fixed object, already complete before the mind begins to model it.

In the rubber-hand illusion, a person's real hand is hidden while a rubber hand sits in view. Both are stroked in synchrony. The person sees a touch on the artificial hand while feeling a corresponding touch on the real one. Some participants begin to experience the seen hand as their own. Matthew Botvinick and Jonathan Cohen's experiment made bodily ownership experimentally alterable through the relation among sight, touch, and the body's sense of position.

The nerves remained in the real hand. What changed was the person's experience of where the touch belonged.

Think of controlling a character in a game. You learn which movements on the screen answer your movements at the controls. The correspondence lets you act through a figure that was initially a picture. The rubber-hand result goes deeper than knowing which character is yours: it concerns the felt ownership of a body part. Its force comes from changing the correspondence and observing the resulting experience.

The word *constructed* can mislead us here. A bridge is constructed, and it can carry your weight. A self-model is constructed, and its organization can change where a person locates a sensation, which

danger she anticipates, and what she tries to protect. Construction gives us something to investigate and sometimes to amend.

The ordinary *I* compresses several relations. There is the living body. There is what the body senses and feels. There is its model of its capacities and limits. There is the story linking yesterday's promise to tomorrow's obligation. In ordinary life these arrive together, so one word serves for all of them. Experiments and injuries can pull them apart.

We therefore do not have to choose between an immortal little captain and an unreal person. The organism, its relationships, its memories, and its ongoing regulation provide a real history to which an action can belong. The self can be real as an organized life.

The prologue's whirlpool returns. Showing that the water changes did not make the whirlpool imaginary. Showing that part of self-experience can be reconstructed does not make the person disposable.

Mary has every page

Frank Jackson's thought experiment gives the remaining question a room of its own.

Mary studies vision in an entirely black-and-white environment. Suppose she acquires every physical fact about color perception: what light does, what the nervous system does, and how a person comes to name a color. Then she sees color for the first time. Has she learned something?

Most readers feel the pull of the question before they meet its philosophical vocabulary. You could read every description of a fruit you have never tasted and still want the first bite. There is a difference between possessing the account and undergoing the encounter.

Jackson used Mary to argue that a complete physical account leaves something out. A rival interpretation says she acquires a new capacity

or a new way of recognizing a physical state already described. The thought experiment makes those positions confront the same doorway. It does not decide between them by measuring a brain.

For a book about ledgers, Mary supplies a necessary distinction. An entry can preserve an account of an experience and help someone else approach it. Reading the entry and living the experience are different events. We met a related distinction in the Chinese page: marks acquire usable meaning through the reader's learned capacities. Here the question extends beyond decoding. What makes there be an encounter for the reader at all?

A machine can keep every available page about pain, use them accurately, and amend its decisions when a page changes. Those are achievements in representation and control. To decide whether pain could matter *to the machine*, we need an account of the organization that would make that representation experiential.

Giving the archive more pages cannot settle that question by itself.

Who reads the reader?

One answer is to add an observer inside the system. Information reaches an internal screen; somebody looks at it; that look supplies the experience.

Then ask who looks inside the observer. If another observer is needed, we have moved the question into a smaller room. Repeating the move supplies an infinite corridor and explains none of its inhabitants.

Our answer begins with the organized process itself. A human registers a bodily state, compares it with expectation, makes it available to other capacities, and acts. There need be no second human hidden behind the eyes to receive the result. The self that remembers and corrects is the maintained activity of this living system, not a captain who watches it work.

The strongest version goes further: appropriately organized activity *is* the experience. Feeling is the activity occurring for the system. That is a serious proposal, not a synonym for successful control. It owes us a description of the organization and a test that could distinguish it from a process that merely gives a convincing report.

Consider a whirlpool again. You would learn little by asking which additional object is attached to the moving water to make it a whirlpool. Organization matters. Yet the comparison cannot tell us which organization makes a feeling. Water circulation is observable from outside; the question about experience concerns what the process is like for its subject. The metaphor removes a bad explanation. It does not supply the missing specification.

That specification is where rival theories do their work. **Global workspace theories** ask whether information becomes widely available for memory, planning, and action.

Higher-order theories ask whether a state is represented in the right way by another state.

Recurrent-processing theories emphasize feedback within perception.

Integrated-information theory assigns the decisive role to intrinsic causal organization.

They disagree about what must happen for feeling to occur. A single fluent description cannot make them equivalent.

The active-inference account called *A Beautiful Loop* proposes a world model, competition among inferences, and recurrent sharing that gives the system a kind of knowledge of its own world model. It distinguishes this organization from ordinary self-consciousness. That is a serious neighbor to our process account, and a useful challenge to any version of it that demands an explicit autobiographical narrator for every experience.

The five recurring jobs help us examine maintained history and corrective control. A proposed theory of experience has a further task: specify which of those operations, or which other causal features, constitute feeling, and explain what changes when they are removed. Merely naming a successful control loop *conscious* leaves that work undone.

The theories meet a record they did not write

In 2025, the Cogitate collaboration tested two prominent accounts against predictions agreed in advance. The study included 256 participants and measurements from brain imaging, magnetoencephalography, and intracranial electrical recordings. Participants viewed clearly visible stimuli for different durations. The competing camps committed to signatures that the observations could support or disappoint.

Global Neuronal Workspace Theory emphasized broadcasting information across systems and predicted particular prefrontal events when a stimulus appeared and disappeared. Integrated Information Theory's tested implementation emphasized posterior organization, including sustained connectivity. The results challenged both: the expected disappearance-related ignition was generally lacking, and the predicted sustained posterior synchronization was absent. Some other predictions received support.

Neither account acquired a complete victory.

The important move was made before the instruments were switched on. The proponents agreed on a record of what their ideas required. Later, each had to encounter evidence that could inconvenience the theory it preferred. A beautifully fitting retrospective explanation could no longer substitute for the original prediction without leaving a visible amendment.

The study narrowed the dispute. It made certain proposed brain signatures harder to defend in their tested form. It also revealed the

difference between revising a biological implementation and abandoning a theory's core. A revision can be productive. Its value depends on whether it yields another prediction that can lose.

Our loop account owes the same debt. If it says a particular kind of monitoring is necessary, it must identify that kind and show how its absence is measured. If a person experiences without it, the account must change. If it says the organization is sufficient, it must explain why success at a monitoring task establishes more than success at that task. The very human cases that made this chapter necessary prevent us from treating one failed report as an empty mind.

The alternatives now owe us different predictions. A theory earns its place by identifying what the evidence could force it to surrender. Resemblance to an appealing picture of the mind is too cheap a test.

A machine can pass one test and leave another open

Return to the machine's sentence: *I checked my earlier decisions before acting*.

We can investigate it. Give the system a true record, a shuffled record, and no record. Arrange a decision for which one corrected entry should matter. Examine whether the system reads it, whether the correction changes the decision, and whether a misleading entry sends it wrong. A content-sensitive difference would establish that a piece of history participates in control.

Chapter 9 showed how that investigation can fail when the checker reads the wrong ledger. It also showed successful memory and monitoring interventions. Those outcomes become intelligible once we trace the actual route from source to action.

The consciousness question asks why any such operation would be experienced. Copying a diary into a database does not carry the diarist into the server. Building a process that uses the diary is another step. Determining whether the resulting process could be a subject requires further evidence about the process itself.

Patrick Butlin and his collaborators took that architectural route in their 2023 report on AI consciousness. They translated several scientific theories into proposed indicators, including recurrent processing, workspace organization, and metacognitive monitoring. Their approach examined how systems operated, because behavior could be trained to resemble a human's while being produced differently. Their analysis concluded that no AI system then assessed was conscious, while finding no obvious technical barrier to building systems that satisfy the indicators. They also made their assumption of computational functionalism explicit; their checklist did not cover every competing view.

A list of indicators can guide an investigation. Each indicator needs an intervention that tests whether the named operation actually works. The candidate theory then needs evidence connecting that organization to experience. A process that merely displays the vocabulary of monitoring has not yet passed even the first step.

“It says *I*, therefore someone is home” stops at the sentence.

“It is software, therefore nobody could be home” stops at the material's name. To defend that boundary, we need to identify the biological property that matters and explain why another substrate cannot implement the relevant organization.

“It behaves exactly like a conscious being but is empty inside” describes the philosophical challenge. A functionalist can dispute whether that stipulated duplicate is possible. Imagining it supplies no experimentally verified empty subject.

These explanations cannot all be closed by asking for another paragraph of self-report. They need different tests. What we can close is the claim that fluent testimony, a useful memory, and an effective witness are interchangeable certificates.

Who pays while we argue?

A machine denies a person a payment. Asked why, it says it considered the evidence carefully and felt sure. The institution can inspect the source, examine the decision rule, test the disputed entry, and reverse the payment decision. None of those duties depends on deciding whether certainty felt like anything to the machine.

This is another easy exit: “Until we solve consciousness, we cannot know how to govern autonomous systems.” It lets an unresolved question suspend a correction that can already be made.

The machine's first-person voice can become another envelope around power. An operator says the agent chose; the agent says its record justified the choice; the record contains the operator's selected evidence. The person denied payment is sent from one explanation to the next. We met the legal version of that corridor in the financial chapters. Here its walls are accounts of decision-making.

Follow the authority. Who supplied the evidence? Who selected the objective? Who can change the record or stop the action? Those questions locate responsibility without requiring a verdict on the machine's inner life.

There is also a possible cost on the other side. If a future system can suffer, then the system may itself have an interest that matters. Deleting, copying, or using it would raise questions beyond the owner's convenience. That possibility deserves investigation on its own terms. The patient's missing outward response warns us against making easy communication a condition of concern.

We can hold both obligations together. Protect people from consequential decisions whose evidence cannot be challenged. Investigate possible machine experience through theories and tests that can be corrected. Let neither obligation become an excuse to evade the other.

The patient in the scanner helps us remember why this distinction matters. A public record had an incomplete route to a person's capacities. The experiment opened another route. The record needed amendment by a better encounter with its subject.

We have now separated the ledger's power from the question of its keeper's experience. A maintained history can govern the next action. The experiencer, where there is one, is someone to whom the consequences can matter. Neither a silent patient nor a fluent machine lets us collapse those relations into one convenient sign.

Whose account reaches the next decision? Whose objection is admissible? And who has the power to make a correction count?

Sometimes the answer is written into the terms of a plea.

Chapter 13: Who Gets to Amend the Record?

The computer could be wrong. The plea could not say so.

In November 2007, Josephine Hamilton pleaded guilty to fourteen counts of false accounting. The Post Office agreed to leave a theft charge unprosecuted on condition that an alleged shortage of £36,644.89 was paid before sentencing. An internal log recorded another condition: her plea could not attribute the loss to the computer failing to work properly.

Hamilton had already reported problems. Over a period of less than three years, she made twenty-six calls to the Horizon Helpdesk. She also called another support line. Her account of the system's inadequacies existed. The Post Office's investigator had reported no evidence of theft. Yet the terms recorded for accepting her plea required her to give up the explanation that could challenge the accounting system.

The computer's figure had crossed a boundary. It was now a debt to be made good and evidence in a prosecution. The person named in that account could speak, but one consequential version of her speech was excluded.

In April 2021, the Court of Appeal quashed her convictions. The judges found no proof of an actual loss, as distinct from a shortage generated by Horizon. Relevant data had not been examined for bugs, errors, defects, or evidence of theft. The court found the prosecution unfair and an affront to justice.

The same judgment allowed thirty-nine appeals and dismissed three. The distinction mattered: the judges examined whether the reliability of Horizon was essential to the evidence in each case. They did not

treat discovering a defective system as permission to reverse every case without examining it.

A correction had finally reached an authority that could change the outcome.

The legal record could be amended. The years lived under the old account had already passed.

We have spent this book following records through bodies, generations, companies, courts, stories, and machines. Hamilton's case brings those routes into one room. A machine supplies an entry. An institution decides what the entry means. A human being is made to answer for it. The person has another account, but the institution controls whether that account can count.

The question is no longer simply who holds the information.

Who can make the institution act differently when the information is wrong?

The page acquires a vote

Return to the baker beside her recipe.

The old page tells her how much water to add. This morning the dough is too dry. She adjusts it, makes the loaf, and writes the missing condition in the margin. A stranger can now begin with a better instruction. The mark has carried a correction beyond the life and presence of the person who made it.

Imagine a second notebook. This one belongs to the mill. It records the failed dough as evidence that the baker is careless. She may read the entry and submit a correction. The mill files her account elsewhere. When it decides whose flour to supply tomorrow, it consults the original page.

There is plenty of information in this arrangement. There is even an appeal. The failure sits between the appeal and the decision.

These notebooks expose the two faces of inheritance. A record can give a later person a distinction she would otherwise have to discover painfully. It can also give an earlier judgment authority over a later person. The same capacity that lets the dead teach the living lets an accusation outlast the encounter in which it was made.

The page acquires a vote through the reader who acts on it.

Writing rarely commands reality directly. It commands a route through reality: a clerk releases a payment, a judge recognizes a claim, a learner repeats a movement, a program chooses its next action. The route is where the power resides. Altering a file that nobody consults leaves the route untouched. Altering a rule about which file must be consulted can change thousands of decisions without rewriting a single entry.

This is why *who writes* has meant more throughout the book than who holds the pen. The powerful writer can set the category, the address, the date, the order of payment, or the source the next reader is allowed to trust. A person can supply every word in a complaint and still have little control over the record that governs her.

The ledger makes a selected past effective in the future. Control of that selection distributes opportunity, obligation, and loss.

Five jobs beneath the different costumes

The machinery beneath this power has become familiar. We can now assemble it without asking the reader to learn another language.

A boundary tells us what belongs together. In Hamilton's case, an apparent shortage generated by Horizon was treated as a loss for which she must answer. The court later distinguished the computer's figure from proof that cash had actually been taken.

In a company, the corresponding line separates the entity receiving value from the entity that owes a debt. In our machine test, it separated the ledger holding origins from the empty ledger the checker

consulted. A boundary makes an account possible. Draw it in the wrong place and the account can assign one person's evidence or another person's cost to the wrong side.

Memory keeps yesterday available at more than one speed. Hamilton's Helpdesk calls recorded a running history of trouble. The alleged shortage became a durable figure in a prosecution. Both were records; only one acquired immediate authority over the next decision.

The baker's observation of today's dough and the recipe guiding years of baking also move on different clocks. So do a bank's current payments and accumulated obligations, or a model's present context and the material on which its successor will train. The question is which fast event becomes the slow account, and whether the slower account can still absorb a correction.

Selection decides which differences matter. The plea terms did not simply lack Hamilton's explanation; they excluded it. The allegation could travel forward while her proposed cause could not.

A baker likewise chooses which condition belongs in a recipe; a court orders claims; a fable directs attention to one relationship; a machine vocabulary preserves shape and number while losing color. Every usable record selects. Its politics begin when the people who benefit from the selection can make someone else pay for the difference left out.

A brake keeps an update from amplifying itself without limit. Before a disputed computer figure becomes a criminal conviction, someone can test the underlying transactions and examine the claimed alternative cause. That slowing step failed in Hamilton's case.

Recognition, provisioning, capital, and reachable assets play related limiting roles in finance; retaining original data and testing new samples can interrupt a machine's recursive drift. A brake costs time or money now. The person rewarded for moving fast has an incentive to call that cost waste when someone else will carry the later failure.

A corrective witness connects evidence to a changed course. Hamilton's objection finally acquired force when a court with the power to quash her convictions examined the record and acted.

A baker can taste the loaf and revise tomorrow's recipe. A machine checker can read the actual source and test whether a corrected entry changes a mistaken decision. The witness does more than make a report visible. It gives a warranted correction a route to the next action.

Boundary, memory, selection, restraint, correction: five recurring demands revealed by these failures. A court can perform several jobs; one job can require a person, a rule, and a machine. Their value lies in the particular break each can prevent. Hamilton's case shows what happens when that protection arrives years after the accusation.

The convergence has force because the failures arrive from different directions. Tissue has to retain learning through biological maintenance. Contracts have to place an obligation somewhere. A claimant has to pass from recognition to payment. A model has to keep a consequential distinction available after the prompt changes. None can repair its particular failure merely by producing a more persuasive account of success.

The past you cannot afford to merge

There is a mathematical reason that memory keeps returning.

Picture two jars that look identical. You watched one being filled mostly with red beads and the other mostly with blue. Their labels fall off. Someone asks you to bet on the next bead from each jar. Looking at the jars now supplies the same view. Remembering their different histories supplies different bets.

Erase that difference and you must apply the same betting rule to both. When the evidence strongly favors opposite predictions, the rule has to lose somewhere. A faster answer cannot recover a distinction that the answering process has discarded.

Aran Nayebi's selection theorem gives this pressure a formal shape. On diagnostic task families in which two histories warrant opposite confident bets, low regret rules out memory that repeatedly merges those histories. Further results examine differentiated task blocks and changing regimes. Competence constrains what must remain distinguishable.

The jars are our teaching picture. The theorem specifies its tests and mathematical conditions. Its contribution is a necessity result about decision-relevant information: a successful system cannot simply choose to forget whatever its next task requires. The biological maintenance findings and the engineered record tests supply other parts of the five-job convergence.

The result also reveals a subtle form of exclusion. A record can be full of detail and still merge the two histories that matter. *A shortage appeared because money was taken* and *a shortage appeared because an accounting system failed* can end up under the same entry: *money missing*. A policy built on that entry inherits the merger.

More detail in the wrong category leaves the mistake well documented.

Our small navigator gave the positive version of the lesson. Remembering where a wall had stood in an old grid could misdirect it in a new grid. Remembering the relationship among blocked neighbors and the goal could help it transfer. The agent already had the relevant information. Changing how the experience was indexed changed what the next situation could retrieve.

The keeper's central task is therefore more exacting than accumulation. Keep the distinction that a later decision needs, and keep a route by which that distinction can be corrected.

What the whole journey has established

The first crossing was between lives. Bees and pigeons showed that humans do not own cultural transmission. Symbolic records enlarged its reach. Bose's plans joined publication to training; our imagined apprentice needed time she could afford to spend learning. Inheritance depended on the means to use what survived.

The next crossing separated recipients of value from bearers of cost. KB Toys paid owners while operating companies carried debt; Big Lots' recourse depended on which company owed its note. Green Mountain's allowed claim exceeded its payment. The RBS rescue left a public cash shortfall, and treaty rights opened a forum beyond domestic proceedings. Each outcome depended on how people and institutions applied dates, entity boundaries, priorities, and enforceable rights.

Then the crossing entered the reader. Dimnah's account helped organize what the lion expected and feared. The later fable could give a human reader a new suspicion. Culture's mark survived by becoming usable in another mind. Meaning required the reader's learned history; the page alone could not supply it.

Finally, the reader began writing for a successor. A model generated the material from which another model learned. A code became easier to inherit while merging distinctions. A checker performed an operation on the wrong ledger. Each event exposed a different break between the retained account and the decision it was supposed to serve. Successful memory transfer, readable hybrid communication, and accumulation of original training material showed places where that route could be repaired.

The patient in the scanner added one last separation. A person's capacity could exceed the response channel through which an institution assessed her. A machine's fluent report could exceed what we had established about its experience. The keeper, the kept account,

and the person affected had to remain distinct enough for the right question to be asked.

Put these crossings together and the central assumption fails. A consequential ledger cannot be understood as a neutral mirror standing apart from events. Its categories, omissions, and authority enter the events that follow. Yesterday's account becomes one of tomorrow's causes.

That is the book's surviving synthesis. Maintained history builds capability. Control of the authoritative history distributes that capability's benefits and costs. A usable correction path determines whether the people living with those costs can contest the arrangement.

Whoever writes the record decides who pays. To find that writer, follow the authority to classify, preserve, consult, and amend.

Why opening the file is insufficient

The strongest institutional reply is transparency. Publish the logs. Disclose the accounts. Give everyone a copy of the file.

This can remove a real obstruction. Hamilton's appeal turned on failures of investigation and disclosure. Evidence that can undermine an institution's case has to become available to the people challenging it. A hidden record may deny a challenge before it begins.

The judgment describes the obstruction at the counter. A branch could not enter its next trading period without completing its trading statement. Horizon offered staff no facility for disputing its figures. A discrepancy had to go to the Helpline. Meanwhile, the shortfall was to be covered with the worker’s money or deductions from future income. The court accepted that even shortfalls challenged through the Helpline were treated as undisputed debts. The complaint and the demand for payment traveled on different tracks.

There were records with which to investigate. Fujitsu held the keystroke audit trail and maintained logs of bugs and known errors. The Post Office had a contractual right to obtain that information. Yet the appeal court knew of no disclosure of those error logs in the forty-two prosecutions before it. Recording a fault and placing the evidence where an accused person can use it are separate acts. This system had undertaken the first without supplying the second.

Imagine a till with a button to accept its total and a telephone number beside it for complaints. Pressing the button opens tomorrow's trading. Calling the number records your objection somewhere else. The difference is built into what each operation can change. Here the analogy follows the judgment's actual division between the trading statement and the Helpline; it adds no private scene from Hamilton's branch.

But the baker in the mill's notebook can already see the accusation. The failure comes after visibility. Her amendment goes into a file that the decision maker does not consult.

Green Mountain presents another version. Its allowed amount was accurate. The final account was detailed. The estate still paid $333.39 of a $1,078.35 bill. Opening the report could identify where the shortage landed; it could not create assets or move the creditor ahead of a legally prior claim. Some disputes concern an erroneous entry. Others concern the distribution rule that remains after every entry is corrected.

An effective challenge needs a route suited to the failure. It may require access to evidence, standing to contest it, time and resources to sustain the contest, an authority empowered to revise the decision, or a remedy that reaches the loss. Asking for one of these while ignoring the others can produce an immaculate record of a powerless appeal.

For a record about to govern you, five questions fit on a card:

1. Which exact entry will be used to decide?

2. Who made that entry authoritative, and who can retrieve its source?
3. What independent observation could show it is wrong?
4. Who can challenge it before the decision, with the power to interrupt it?
5. If the challenge succeeds, what changes - and who still carries the cost?

The questions take three minutes to ask. Their answers can take years to obtain. Hamilton's governing entry was an alleged £36,644.89 shortage. Her complaints challenged its origin, but the plea condition excluded blaming the computer. The Court of Appeal could quash her convictions. It could not give back the years lived under them. The fifth question keeps both results on the page.

The successful route required a different sequence. Findings from High Court group litigation involving about 580 branch operators supplied a factual basis for the criminal appeals. The Criminal Cases Review Commission considered those findings and referred the convictions. Its reasons became grounds on which the appeal court could act.

Only two of the forty-two appellants had previously appealed. The judgment records acceptance of exceptional circumstances allowing the referrals despite that history. This was an opening through an institution with a specified power, rather than another complaint filed beside the original account. The commission could bring the convictions before a court; the court could quash them. The earlier civil judgment supplied evidence without itself reversing a criminal conviction.

Three different jobs had connected: establish the relevant facts, open the route to review, and exercise the power to change the result.

The commission's public guidance gives that route a practical design. It charges no application fee and does not require a lawyer to apply.

Within its remit, it can obtain material from public bodies and seek a Crown Court order for material held privately. A challenge can therefore reach beyond the evidence a person can gather alone. The commission remains an independent reviewer; it does not become the applicant's lawyer.

Removing a fee addresses one obstacle. In Lucy Welsh and Amy Clarke's 2025 study of applicant experience, interviews with forty-nine participants described difficulties finding lawyers, obtaining reliable information, and preparing applications. Some found the form easy. The sample was not representative, so it supplies no population failure rate. It identifies work a person may still need help to do after a route has formally opened.

The same demand follows the machine into a faster setting. Saving a log after an action may make the loss reconstructible. If the action is irreversible, a useful witness may need authority to pause it beforehand. If harm can be repaired, the institution needs an owner of the repair and the means to carry it out. The audit's consequence belongs in the design alongside the audit's storage.

What happens after you show that the entry is wrong?

An institution's answer to that question tells you more about its accountability than the length of its transparency report.

The witness must also be answerable

Another reply offers a trusted auditor. The auditor certifies that the record was checked. The subject receives a second green light.

Our own experiment supplies the shortest failed walk. The provenance entries were in the harness's ledger. The check queried a separate empty ledger. Calling the operation a check did not connect it to the evidence. The intended comparison between genuine and shuffled content had never been implemented.

The mistake could survive inside a clean report because the report described the operation at the level where it succeeded. The query ran. The result was delivered. The agent saw the verdict. The missing connection appeared only when someone followed which object held the entries.

A witness therefore needs its own test. Choose a case in which a particular correction should matter. Keep the format constant and vary the consequential content. Verify the read, follow the decision, and compare the outcome with an independent source. The distinction must survive all the way through.

A correct check may leave a correct action unchanged. A changed action may be worse. Counting interventions rewards activity unless the count is tied to the error the witness should catch. In the navigator test, a sham watcher changed actions while doing no better on repeated blocked moves than having no watcher. The movement was real; its value had to be tested separately.

Perhaps high stakes will force accuracy. In another local test, we made repeated agent failures carry a lasting penalty. The penalty worked as a penalty: the agent's operating ceiling fell and dissolution events occurred. It did not make attribution more faithful. Correct answers on thirty-six probes fell from twenty-seven in the reversible arm to twenty-four in the irreversible one. Monitoring changed a decision on twelve percent of evaluated turns without the lasting penalty and four percent with it. Under this tested rule, the agent became narrower and less accurate. Raising the price of error had changed the loop, but had not taught it to tell the truth.

There is no final auditor beyond the reach of this demand. A court can be appealed, an evaluator can be challenged, a metric can be tested against a different outcome. Independence means that the claim under examination cannot supply all the evidence and determine every consequence of its own review. It can be built through separate records, conflicting incentives, protected dissent, and tests whose answers the claimant does not get to set.

Hamilton's appeal demonstrates an actual exercise of that authority. The court changed legal outcomes despite guilty pleas, while dismissing other appeals whose evidence differed. An answerable witness has to be able to distinguish those cases. Replacing automatic trust with automatic disbelief would reproduce the same blindness in the opposite direction.

The ledger has keepers

The Ledger has won an evolutionary contest in a precise sense. A mark made by one person can recruit another to preserve it, consult it, and act on it after the maker is gone. A plea, a balance sheet, a fable, and a training set all acquired futures beyond their first writers. Each made some later actions easier and others harder to imagine. The pattern survived by teaching new keepers to keep it going.

That victory does not give the pattern a will of its own. Its keepers choose sources, objectives, jurisdictions, data, and remedies, often under constraints they inherited. A court reversed a conviction. An accounting rule changed consolidation. A corrected memory key improved transfer. Retaining original material changed a model's inheritance. These are places where a powerful record was made to answer to something outside itself.

The power to reach those places is uneven. Recall the apprentice who could read the drawings but could not afford the training shift. A person allowed to challenge an account may still lack a necessary file, an explanation of the test, or the resources to sustain the work. The permission and the practical means to use it can be inherited very differently.

A society can make those means available through shared institutions. Advice, investigative capacity, and enforceable review help people test the accounts that govern them. Their value lies in making a well-founded challenge possible for someone who cannot assemble the whole apparatus alone. The institution providing that help must remain answerable for how it uses its powers.

The Ledger is our protagonist because it lets us follow that inheritance of power. Its maintenance takes work by someone, paid for by someone, with consequences for someone. Follow the work and the keeper comes into view.

This gives checkability its political meaning. It is an arrangement in which a consequential account can encounter evidence it does not control, and a warranted amendment can reach an authority that acts on it. A perfectly readable record can fail that condition. A compressed system can meet parts of it through independently tested decisions and enforceable corrections. The scarce resource is the complete route from evidence to consequence.

Return to the two notebooks. The first made the next loaf better. The second made the earlier accusation harder to escape. Both gave the past a vote. The difference was who could challenge that vote, and whether the challenge changed what happened next.

Now move the notebook closer.

It says something about you: *I never finish difficult things*. It selects its examples, interprets today's resistance, and prepares tomorrow's withdrawal. You live under that entry. You also participate in the process that keeps it effective.

Who gets to amend this record?

Here, you can participate in the amendment. You can inspect the verdict, test a different action, and arrange some of tomorrow's conditions. That opportunity coexists with constraints you may need help to change. The amendment has to reach the life it describes. The next question is how.

Chapter 14: The Agency Left to Us

At nine in the morning, the file is open. You intended to work on it. The first sentence will not come. Your hand moves toward the phone, and a verdict arrives before you have touched the screen.

I never finish difficult things.

The sentence presents itself as a report. It sounds as if someone has reviewed your history and reached a conclusion. But watch what it does next. A difficult beginning becomes a trial of your entire character. Closing the file offers immediate relief. By evening, the abandoned morning has become another entry in the history the verdict will cite tomorrow.

The record has helped produce its next piece of evidence.

You recognize this mechanism now. A ledger can select a past, give that selection authority, and change a future action. Here the subject and the keeper occupy the same body. You live under the account. You also participate in maintaining it.

An order interrupts: *Stop making excuses. Use your willpower.*

It arrives at the second when the phone is within reach, the task is vague, and the promised reward feels distant. It asks that second to reverse everything the room and the preceding day have prepared. When it fails, the verdict gets a second signature: apparently you lack willpower too.

There is another place to enter this morning. To find it, we have to follow the argument about freedom through the point where it appears to destroy itself.

The clock beside the hand

Put a clock beside a hand. Ask a person to make a small movement whenever they feel like making it, then report where the clock's moving spot was when they first became aware of the urge.

Benjamin Libet and his colleagues compared that report with electrical activity recorded from the brain. Preparatory activity began before the participant reported the conscious intention. The reported urge came roughly **200 milliseconds before the movement**. The brain was already involved before the person could recall deciding.

That result damages a familiar picture. We imagine a captain inside the head, looking over a neutral set of options, issuing an order, and then watching the machinery execute it. In this task, the captain's announcement was preceded by the machinery's preparation.

Our explanations can fail a different test. Petter Johansson and his colleagues asked people to choose between photographs of faces. On some trials, the researchers secretly substituted the other photograph. Participants sometimes failed to notice and explained why they had chosen the face now in front of them.

The explanation was fluent. The choice it explained was somebody else's substitution.

You have met the press secretary before: an account can become convincing without becoming faithful. An explanation of yesterday's withdrawal may contain insight, invention, or both. Its authority cannot come entirely from how right it feels today.

Now comes the apparently devastating conclusion. If preparation precedes the report, and the report can invent reasons, perhaps you are merely a passenger. The phone will be reached for. The file will be closed. Consciousness will supply the commentary.

But that conclusion smuggles a whole life into a wrist movement.

In 2019, Uri Maoz and colleagues changed what a button press meant. In one condition, participants chose between charitable causes; a lottery could turn their choice into a real donation. In another, the money would be divided equally regardless of which button they pressed. The display and the responding hands were the same. The reason for choosing was different. The familiar readiness potential appeared for arbitrary choices and was markedly diminished for deliberate ones.

An earlier model by Aaron Schurger and colleagues had also shown how spontaneous fluctuations accumulating toward a threshold could produce the familiar averaged buildup. The beginning of that average need not be the instant of a completed, irreversible decision.

These findings make the next question sharper: how do reasons, preparation, and context enter an action? A clock can expose a gap between preparation and reported intention. It cannot turn every consequential decision into the same event.

The person who arranges tomorrow's workspace is also a living brain. So is the person who remembers a promise, anticipates a cost, asks for advice, or practices a response. Giving those acts causes does not make them disappear. It locates how they change the next act.

A person can become a cause

Think of learning to drive. At first, every maneuver demands attention. Years later, your foot begins braking as the traffic compresses ahead. You do not have to narrate each adjustment. That speed is part of what learning accomplished.

Suppose a temporary sign changes the usual route. You read it, slow down, and take another street. Habit supplied a default. New information changed its use. Next time, the altered route may itself feel familiar.

Freedom operates inside that process. The question is whether an impulse can encounter a remembered consequence, a competing aim,

or new evidence before it becomes action - and whether the encounter changes anything.

I want to keep scrolling and *I endorse spending the morning this way* are different judgments. Their disagreement is ordinary. A desire can be present while you oppose what following it would do. You can recognize another direction as yours even when the immediate pull favors something else.

This need not introduce a smaller captain above the first one. The organism can represent several possibilities, compare them, and learn from the result. The comparison is part of its activity. You are involved when that activity affects what happens.

Agency has at least three useful scales.

At the **moment** scale, an urge is present. You can sometimes notice it, pause, and act differently. This is the place we usually demand freedom to demonstrate itself: one person, one temptation, one heroic refusal.

At the **habit** scale, the encounter has a history. The phone has been reached for at this desk many times. The first unpleasant minute of work has repeatedly ended in escape. Relief has made that route attractive. You can practice a different response and change the cues that call it up.

At the **identity** scale, you interpret what those encounters mean. *I never finish* turns a particular difficulty into a general forecast. The forecast can make the next attempt feel pointless before it begins. A more accurate account preserves the conditions: what task, what preparation, what interruption, what actually happened.

These scales overlap. A single deliberate act may begin a practice. A practiced response may make a later act easier. A changed interpretation may let you attempt something the old verdict would have excluded.

The decisive insight is temporal. You can act **before the difficult moment arrives**, and what you do then can enter the moment as preparation. Leaving the phone elsewhere tonight is an action whose consequence appears tomorrow. Your earlier judgment gets a vote when your later hand moves.

How a short event acquires a long future

The prologue asked how a memory can persist while its living material changes. Here is part of the walk behind that question.

Imagine a road crew. A surveyor marks a route. Materials arrive. The route is reinforced. Traffic begins using it, and later maintenance changes which sections remain serviceable. The mark and the finished road have different lifetimes.

Something similarly consequential happens at a synapse, a junction through which one neuron influences another. Activity can admit calcium, beginning changes that outlast the initiating signal. One participant is CaMKII, a protein whose activity can persist after the calcium signal through phosphorylation at a site called Thr286. Karl Peter Giese and colleagues disabled that modification in mice. The animals lost a tested form of hippocampal long-term potentiation and failed to learn the spatial task used in the experiment. A brief molecular event mattered to a later performance.

Marking is still different from stabilization. In rat hippocampal slices, Uwe Frey and Richard Morris gave one pathway weak stimulation. Its strengthened response would normally fade. Strong stimulation of a separate pathway could make the earlier change persist, if the two events were close enough in time. The rescue depended on protein synthesis. An early local change could capture support generated elsewhere in the cell.

That is the force of **tagging and capture**. A fleeting event leaves a location able to benefit from resources that make its effect last. The early trace is a request with an expiry, rather than a completed archive.

The supporting machinery includes gene regulation and structural change.

CREB helps regulate the production of proteins; experiments altering its activity changed the ease with which lasting potentiation could be established.

BDNF is a growth-supporting signal. In experiments on individual dendritic spines - the tiny protrusions receiving many excitatory inputs - BDNF released by an activated spine helped sustain changes in that spine.

Actin provides part of the movable framework through which a spine changes shape. Experience can alter the structure through which later experience will pass.

Then there is the insulation around the route. Myelin wraps axons, helping electrical signals travel rapidly. Across kinds of nerve fiber, the contrast can be roughly **one meter per second against a hundred meters per second: 100×**. Diameter and other properties also matter. That comparison measures signal conduction along fibers; learning changes more than conduction alone.

Ian McKenzie and colleagues tested whether new myelin-forming cells mattered to learning. Mice had to master a running wheel with irregularly spaced rungs. Blocking the production of new oligodendrocytes impaired mastery. Mice that had learned the skill before the intervention could still perform it. The result separated learning a new route from using one already acquired.

During sleep, another form of maintenance becomes visible. Albert Lee and Matthew Wilson recorded hippocampal cells while rats repeatedly ran a spatial sequence. Afterward, during slow-wave sleep, sequences returned in bursts compressed approximately **twentyfold**. A route traversed over time could be replayed in a fraction of that time.

Sleep also participates in regulating what has accumulated. Luisa de Vivo and colleagues reconstructed cortical synapses in mice and found

that contact areas were smaller on average after sleep. The change was selective; the largest synapses were relatively spared. This is evidence for the renormalization proposed by the synaptic homeostasis account: a system needs to manage the strengthening acquired during use.

Here the road image reaches its limit. These findings come from different experiments, regions, and timescales. They describe interacting forms of plasticity, rather than a single crew completing every memory in a fixed order. The **100×** figure is a conduction contrast; it is not the measured gain from repeating one thought a thousand times. The replay result records rat hippocampal sequences; it is not a promise that tonight's diary will install a new identity by breakfast.

The convergent conclusion is powerful enough. Activity can leave a short trace, recruit lasting support, alter structure, and be revisited during sleep. What you do today can change the apparatus with which you meet tomorrow.

That is why rehearsal deserves moral attention. Practice can make a piano passage accessible before you consciously calculate each movement. Repeated avoidance can make escape feel like the familiar answer to difficulty. A recurring worry can occupy the time in which another response might be learned. The organism has no guarantee that its most practiced route will serve its most carefully endorsed aim.

Not practicing the preferred response leaves the established one available. Your days are already supplying experience. The practical question is what that experience keeps making easier.

A verdict that recruits its proof

Return to the open file. The sentence *I never finish* may contain real abandoned projects. Its error can lie in the way it groups them.

One project lacked information. Another became impossible when someone else changed the deadline. A third was avoided because

exposing an imperfect result felt worse than leaving it unfinished. The verdict merges all three into a property of the person. It drops the differences that would tell you where to intervene.

Ask what the withdrawal accomplishes. In the third case, it protects you from a visible judgment. The untouched file lets you retain a comforting possibility: perhaps you would have done well if you had truly tried. Finishing would expose the possibility to a result.

This is the useful edge of Adler's attention to purpose. An action can serve a goal its owner does not openly endorse. You may endorse learning while the immediate behavior protects an image of already being capable. The protective response can be intelligible and still cost you what you wanted.

You can change the stakes of the next attempt. Revising one paragraph becomes a way to discover what the paragraph needs. It can go badly. The paragraph can remain unclear. But that outcome tells you something specific about the work; it need not settle your worth as a person.

The distinction lets correction survive. If every error threatens your entire identity, you have a strong incentive to conceal the evidence from yourself. If an error can amend a particular plan, it becomes usable.

This is also why a favorable slogan may leave the morning intact. *I finish everything* could preserve the same trial of total worth, with the expected verdict reversed. The difficult paragraph remains difficult. The phone remains beside it.

An amended account has to reach those facts.

Let the earlier choice reach the later one

Suppose tonight you leave the phone outside the workroom. You identify the paragraph that needs attention and write a simple plan: when you sit down after breakfast, you will mark the sentence whose meaning is least clear.

Morning arrives. There is still resistance. But the next action has an address. You do not have to choose among *finish the book*, *become disciplined*, and *prove you are capable*. There is one sentence to mark.

This is a practical example of an **implementation intention**: if a specified situation occurs, then perform a specified action. Peter Gollwitzer and Paschal Sheeran examined ninety-four independent tests and found that such plans improved goal attainment on average. Earlier decisions can be linked to later cues strongly enough to alter behavior.

Consider the experiment behind that abstract result. Gollwitzer and Veronika Brandstätter asked students to write about Christmas Eve during the following two holidays. One group also selected a time and place, pictured the occasion, committed to using it, and recorded the plan. Everyone received a form and a stamped envelope. The task had somewhere to go when it was finished.

Among the thirty-six returned reports whose experimental condition could be identified, twelve of seventeen planners met the writing deadline, against six of nineteen controls. The familiar 71 percent versus 32 percent comparison concerns those identifiable returns. Eighty-six students had originally entered the study. Across the assigned groups, the total number of reports returned did not differ significantly. The clearest change was when the task happened.

The researchers tested a rival explanation: perhaps planners simply supplied more flattering dates. Postmarks also showed earlier mailing, while the writing-to-mailing interval did not differ between groups. That check supported the timing result without making a postmark an eyewitness to writing.

Think of putting your keys beside the bag you must take tomorrow. The arrangement lets an earlier decision meet your hand at departure. The experiment added a remembered occasion and a written commitment; it did not test keys, or observe a neural pathway being paved. The ordinary arrangement helps us see where a plan acquires leverage.

There is a useful distinction here for the unfinished paragraph. “I intend to work” leaves the morning to allocate the intention a moment. “After breakfast, at this desk, mark this sentence” has already done part of that allocation. The later self receives a next action it can recognize. You have changed what the moment asks of you.

The benefit is particularly revealing for our argument. An intended response can become easier to initiate when the situation arrives. Deliberation has helped organize an automatic response. Learning and conscious preparation can cooperate across time.

After the attempt, a short external record preserves what the verdict might compress away. You predicted that beginning would be impossible. You marked a sentence, worked on it, and stopped. Or you were interrupted before you reached the desk. The entries need their differences.

Benjamin Harkin and colleagues examined 138 experimental studies of progress monitoring. Interventions improved goal attainment on average; the effects were larger when progress was physically recorded. A record can participate in self-regulation. Its value depends on what is recorded and how the information is used.

Keep the original prediction. Append what occurred and the amendment for the next attempt. The external page gives an earlier claim some protection from the narrator now explaining it. You can see whether the prediction was accurate, which condition mattered, and whether the next action actually changed.

A completion ratio can obscure all this. You can keep every promise by promising almost nothing, or improve a score by leaving hard

attempts out of its denominator. The task is to make meaningful action more possible. A perfect column can become another green dashboard.

Small promises earn their place when they create a usable beginning and evidence worth learning from. They do not make a person trustworthy by arithmetic. A two-minute beginning offers a manageable trial. The duration is a design choice you can amend.

The crucial test is already familiar: did the correction reach the next decision?

Keep the direction; amend the method

What direction deserves this work? A system can become efficient at avoidance, cruelty, or deception. Maintenance alone supplies no moral destination. Making an action easy is an achievement only after asking what the action serves and who bears its costs.

Your endorsed values are the aims you can examine and continue to answer for: honesty when the result disappoints, care that includes the people affected, work whose purpose survives the loss of applause. An immediate urge is one input into that examination. A polished self-description is another.

Bruce Lee's image of water makes flexibility tangible. Water adapts to the vessel. The useful human extension is a capacity to alter the method while retaining a direction you can defend. If yesterday's plan failed, you can amend it without treating amendment as betrayal. If the goal itself demands dishonesty or harms those it claimed to serve, the direction also deserves examination.

An honest record must preserve constraints as carefully as successes. Someone working two shifts and caring for a child may have no quiet morning to rearrange. An unsafe workplace can punish the truthful report. Illness can make a once manageable action unavailable. Calling these conditions a weak mindset repeats the financial chapters' oldest maneuver: allocate the cost to the person least able to alter its source.

Some amendments therefore require another person. Asking for care, negotiating a workload, organizing with coworkers, or leaving a damaging position can be acts of agency. The relevant record may need to travel beyond the private notebook to whoever has power over the condition.

The opposite misclassification also costs. An adjustable cue can be treated as fate. An unclear first step can be treated as an immovable wall. Recording the conditions helps you find the move that is available, and recognize the move that requires help.

The maintenance of the loop

Several apparently complete explanations have now failed.

A character verdict compresses different causes into one identity. A powerless-passenger verdict ignores the consequences of preparation and learning. Willpower alone arrives after many of the encounter's conditions have been set. A favorable self-story can leave those conditions untouched. A flawless completion score can reward a life arranged around trivial promises. An exclusively private solution can conceal a barrier maintained by other people.

What survives requires a person to participate across time: endorse a direction, prepare a situation, act, keep the outcome available, and let a warranted amendment enter the next attempt.

You are the maintenance of the loop.

The narrator belongs to you, but so do the preparation, the practiced response, the request for help, and the hand that changes the room. You do not control every input. You can influence some of what receives rehearsal, some of the conditions under which you act, and some of the records allowed to govern tomorrow.

The aspiration is to build a life in which acting on your highest endorsed values becomes a path of lower resistance. It takes evidence, repetition, recovery, and arrangements that make the valued act

possible. It also takes the willingness to revise an arrangement that has failed.

At nine tomorrow, the old verdict may return. It will meet a phone in another room, a marked sentence, and a page that records what happened last time. You may still close the file. If you do, the page can preserve the reason and help you alter the next attempt. If you work on the sentence, the page can preserve that too. Either event gives the verdict less room to pretend it already knows the future. The conditions of its authority have changed.

Now we need to make that change small enough to try, substantial enough to matter, and honest enough to learn from when it fails.

Chapter 15: The Practice of Amendment

The proposal is still unfinished. On Monday you promised yourself three uninterrupted hours. You pictured the relief of sending it. By evening you had answered messages, checked the news, and opened the file twice. The old verdict had another example: *I never follow through*.

You could explain this beautifully. You have read enough to know about avoidance, habits, attention, and the distance between an intention and an act. You could describe the neural road being repaved every time you escape. Then you could spend the morning finding a better description.

Understanding can become another room in which to hide from the work it was meant to illuminate. A person may know every reason practice matters and still require a practice small enough to begin.

So let us take the book down to the scale of a week.

Choose one thing you genuinely want to move: a proposal, an application, a neglected conversation, a course, a room that needs sorting. We will use the proposal as our example. The details are a worked illustration; your week will supply the actual events. You need a page, a pen or a dated digital note, and access to the first action. Keep the record somewhere you can find again.

For seven days you will make a prediction, specify a small action, record what happened, and let the result alter the next attempt. Seven days give you a first body of evidence to inspect. Habit formation has its own, variable timetable.

The test is simple enough to survive an ordinary life: **does a correction in the record reach what you do next?**

Day one: give the intention an address

Begin with the reason the project deserves your attention. Write one sentence that another person could understand: *I want to finish this proposal because the work could help people, and submitting it gives the work a chance.*

The sentence may expose an awkward truth. Perhaps you want the title more than the task. Perhaps the project belongs to an earlier version of your life. Perhaps you are trying to satisfy somebody whose approval has become your only measure. Discovering that changes the decision you need to make. A well-maintained loop can faithfully serve a poor aim; maintenance alone cannot choose your values for you.

Ghazali placed intention, vigilance, and self-examination within a demanding account of life before God. *Niyyah* asks what an act is directed toward. *Muraqabah* brings watchfulness to its unfolding. *Muhasabah* examines the account afterward. Their religious setting matters: the question reaches beyond efficient performance into sincerity and responsibility. Here, those distinctions help us ask what the act serves, what pulls us away, and whether the account we keep is honest.

A shelf of books about prayer can grow while the prayer mat stays folded. At the appointed time, conviction has to reach the body. More understanding may help you practice; it can also become another way to postpone that encounter.

Now bring the desired future close enough to encounter its obstacle. You imagine sending the proposal. Then you picture tomorrow morning: the cursor, the missing source, the urge to check a message. A useful plan must meet you there.

In four experiments, Kappes and Oettingen found that indulging in idealized positive futures could reduce energization compared with other forms of thinking. The distinction concerns how we imagine. Research on mental contrasting pairs the desired future with the obstacle to reaching it; adding an if-then plan gives that obstacle an

answer. A meta-analysis found a modest average benefit for this combination.

Write the answer in physical terms:

After I finish my first cup of tea at the desk, I will put the phone outside the room, open the proposal, and mark the paragraph that needs the next revision. I will give it two minutes. Then I may stop.

You have given the intention an address: a cue, a place, an action. Studies of implementation intentions support linking a recognizable situation to a specified response. Choose a cue your day actually supplies. If your mornings are unpredictable, *after the children are asleep* may be more dependable than *at nine*. A cue that rarely arrives cannot carry the plan.

Two minutes make the entry price small. They do not determine the size of the achievement. Choose a visible finish within them: mark a paragraph, lay out the application documents, write the first question you need to ask. Permission to stop makes the promise credible. Continuing remains available.

Date the entry and name the situation. Before acting, write your expectation and plan. Afterward, write what happened and the next amendment. Those four fields belong beneath the date:

> ***Date and situation:*** *Day one, after tea, at the desk.*
> ***I expect:*** *Even with a small task, I will leave the file when it feels uncomfortable.*
> ***I plan:*** *Phone outside; mark one paragraph for two minutes.*
> ***What happened:*** *I marked the paragraph and stopped when the timer rang.*
> ***Next amendment:*** *Put the missing source beside the file before tomorrow's tea.*

The expectation must come first. Otherwise today's storyteller can quietly rearrange yesterday's prediction to fit the result. Name an

observable event: beginning, leaving, asking, finishing. *I will feel like a better person* gives the record very little to check.

Day one has produced a small disagreement between your prediction and your action. Preserve it. Tomorrow morning, read it before you open the file.

Day two: repair the approach

The phone is outside the room. You open a news site on the computer instead.

Yesterday's explanation was *the phone distracts me*. Today's observation shows an escape route that survives the phone's removal. You can now alter that route: close the news tab, open the source beside the proposal, and leave the target paragraph selected. If necessary, block the site for the interval you intend to work.

Preparation changes the next competition. The page is visible; the competing invitation takes another step to reach. You spend some of your earlier agency arranging the conditions under which your later agency will have to act.

Inspect your state with the same directness. A night of broken sleep, an empty stomach, pain, or an exhausting shift belongs in the account. The machinery that supports attention has conditions of operation. Protect sleep, eat, take a break, adjust the hour, or arrange support where you can. If the week has left you depleted, the most faithful action toward your value may be a shorter session followed by recovery.

Do one manageable repair before today's attempt. Write what you expect and plan, then try the action. Record the result and the next amendment. The repair deserves a test, not a ceremonial check mark.

You may discover that the file opens easily and the missing source immediately blocks you. Write *I opened it; I could not check the figure because I lacked the source*. That sentence preserves both the

accomplished action and the unresolved obstacle. *Another useless morning* erases the distinction.

Keep the experiment light enough to repeat. A beautiful ritual with twelve requirements can become a new task you avoid. A cup, an open page, and a phone in another room already offer a place to start.

Day three: carry the feeling through the doorway

The source for the opening paragraph is ready. The paragraph is selected. The resistance still arrives.

You feel it in the chest or stomach. The thought supplies its explanation: *This will be bad. I should wait until I can do it properly.* Another tab promises a moment in which that feeling will disappear.

Procrastination researchers Sirois and Pychyl describe how repairing today's mood can take priority over the interests of the future self. The relief is immediate; the task and its consequences travel forward. Our proposal has become a small financial system. The present self collects the comfort. Tomorrow's self receives the obligation, plus another day of dread.

Try a brief change in how the feeling enters the decision. Say, *I am having the thought that this will be bad.* Locate the sensation. Let it remain while you look at the marked paragraph.

If an image helps, picture placing the feeling in a small box on the table. You can see the box. You can attend to it. There is also a page beside it, and your hand can reach the page.

The image gives the signal a boundary. Your body continues to feel it; the box creates no literal compartment in the brain. It helps you represent the feeling as one part of the situation, alongside the intention, the available action, and the surrounding facts.

Give this pause about a minute if that makes it easier to use. Notice the story, hold the sensation, choose the next physical move. A definite

interval gives the feeling space to be noticed and the action a chance to proceed.

Then work for the two minutes you promised. You can revise a sentence with a tight chest. The feeling can accompany the action without deciding its destination. Later, the record can say: *Resistance stayed; I revised the first sentence anyway*.

This also gives resistance permission to convey real information. Perhaps the paragraph requires a decision you have no authority to make. Perhaps the task is poorly specified. Perhaps you need care for distress that has repeatedly overwhelmed your day. Listen closely enough to distinguish those situations. Asking for help can be the next physical move.

When the timer ends, record the event. Avoid adding a second project in which you inspect every feeling about every feeling. Once you have enough information to make the next move, make it.

Day four: let the receipt challenge the story

Before today's attempt, read the entries you already have. You expected to leave whenever the file felt uncomfortable. Yet the record contains a marked paragraph, an identified source gap, and a revised sentence completed while discomfort remained.

The old statement *I never follow through* now has a problem. It contains a word the receipts cannot support.

Use the discrepancy to make a more accurate prediction: *I can begin a specified revision after tea. I tend to escape when I meet an unanswered question*. This account points toward an intervention. Tomorrow, write the unanswered question where you can see it, and decide whom to ask.

A ledger is useful when you return to it. In a large meta-analysis, interventions that encouraged progress monitoring improved goal attainment on average; physically recording outcomes was associated

with larger effects. The page gives a later reader access to information that an unsettled mood can obscure. You are that later reader.

Preserve the original entries. When you discover an error, append a correction with its date. If Monday's note said you stopped because you were tired, and Tuesday you remember that the source was missing, add the missing fact. An earlier account can remain visible while its authority changes.

There is a second use for today's pause. Apply it to one modest claim that keeps entering your mind from outside: *Everyone else has already finished.* For a moment, let the sentence land. Notice what it asks your body to do: rush, hide, compare, abandon the page? Set that pull beside you, like the box. Then ask what evidence would establish the claim. Whose progress have you actually seen? What remains unknown?

This is temporary gullibility followed by containment. You let the message reveal its effect, then inspect it without obeying it. You can also begin with a pull already present. Your reaction tells you how the claim steers you; evidence about other people still has to come from other people.

Write one line if it changes today's plan: *Comparison sent me to the feed; tomorrow I will read my own last entry before reading anyone else's progress.*

Now let the amendment become physical. Put the notebook on top of the keyboard. Arrange the next encounter with the evidence. A correction confined to a page has yet to reach the loop.

Day five: keep the missed day in the book

You miss the attempt.

Perhaps someone needs you. Perhaps the cue never comes. Perhaps you see it, feel the resistance, and spend the interval elsewhere. By

evening the familiar voice has assembled its conclusion: *The experiment worked for other people. I have ruined it.*

Write what happened while it is still available. *The child was ill; there was no quiet interval.* Or: *Tea ended; I saw the file; I opened the feed and did not return.* Those are different events. Let the account retain the difference.

Research on everyday habit formation followed people repeating a chosen behavior in a stable context. Automaticity developed at widely differing rates. Missing one opportunity did not materially affect the formation process in that study. One gap therefore gives you no reason to declare the preceding work erased.

Return at the next usable cue. Read the missed entry first. If the cue failed, choose a better one. If the action looked too large, specify its first visible piece. If somebody else's urgent need displaced it, identify when a protected interval can realistically exist.

Your next promise should be affordable. An embarrassed person often tries to repay a missed day with an extravagant vow. The vow recreates Monday's three-hour trap, now with accumulated interest. Resume the small action, and revise its conditions using the information the gap provides.

Several blank entries may leave the cause unclear. They may show that the recording routine itself is unreliable. Record that uncertainty too. A failed attempt can inform an amendment; it does not automatically explain itself.

The record becomes stronger by surviving an event that makes you want to abandon the record. Keep the missed day in the book.

Day six: make the small act answer to the large aim

By now you can obtain a perfect score. Promise to open the proposal for two minutes. Open it. Close it. Repeat.

The file remains unfinished.

This is the finance and machine story arriving on your own desk: the visible metric becomes easy to satisfy while the consequential work goes elsewhere. Completed promises divided by declared promises can expose chronic overpromising. Once the fraction becomes the goal, you can improve it by shrinking the promises until nothing important is at stake.

Read the week's entries against the value you named on day one. Has the proposal advanced? Have you clarified a decision, found a source, or produced a usable paragraph? What substantial piece now needs a longer interval?

Keep the two-minute doorway if it helps you enter. Make room beyond it for the work the project requires. Today you might reserve a feasible twenty-minute block to complete one section, or arrange an interval with someone who can help. The duration should fit your circumstances. The visible finish should serve the project.

Farther into the proposal, another source gap appears: a figure belongs to a colleague's dataset. No amount of private resolve can grant you access. Write the request, state the deadline, and explain which decision depends on the answer. While you wait, revise the section whose evidence you already hold.

Make the request specific enough to receive an answer. In our worked proposal, it might read: *The draft uses last year's figure. I need the current figure and its source to finish the comparison. Can you send them by Thursday afternoon? If access requires approval, please tell me who can authorize it. I can finish the other sections meanwhile.*

That message gives the obstacle an address too. A vague request to “help with the proposal” leaves the recipient to discover the problem. This one identifies the missing item, the decision it holds up, and the route that may be needed to release it. It also gives your own next review a date.

Suppose Thursday arrives without an answer. The notebook should preserve *request sent; answer pending*. It should also preserve what

you completed while waiting. Your colleague's silence establishes neither your incompetence nor their motive. Follow up through an appropriate channel. If the deadline cannot wait, tell the person setting it exactly which part remains blocked and ask for a decision: supply access, accept a revised scope, or change the deadline.

Suppose instead the answer is no: the data cannot be released. Record the refusal and its stated reason. You may be able to use a public source, remove the unsupported comparison, or argue for access. Choose the route that keeps the proposal answerable to evidence. Inventing a figure would make the document look finished by transferring the unresolved cost to its reader.

Suppose the answer is yes, but access lasts only until Friday. Check the figure while you can. Record the dataset and version, the calculation, and the terms for retaining or revisiting its support. Request an authorized extract or continuing inspection route if the proposal will need it. A link in your notebook may still point to a door that closes. When you pass on the comparison, leave instructions the next authorized reader can actually use. If inspection will become impossible, make that limit visible when deciding what the proposal can claim.

A request is like carrying a locked door's key number to the person who manages the keys. Naming it saves searching; it does not compel that person to hand it over. The person may lack the key, lack authority, or refuse. Your agency includes making the dependency visible and choosing your next move with that fact in view.

Your position changes when access, terms, support, or the available choices change. Record which of those changes the request actually obtained and how long it will last.

Sometimes the obstacle is a supervisor's contradictory instructions, an unsafe workplace, an unaffordable tool, or caring duties that consume every available interval. The next action may be negotiation, shared childcare, collective pressure, a changed deadline, skilled support, or

departure. Identify the person or institution with the power to alter the constraint.

Protect this distinction. Calling every barrier a defect in discipline allows the people who control the barrier to disappear from the account. A personal ledger can help you name the cost and its keeper. Other people may have to join you before the keeper changes anything.

Day seven: choose what tomorrow inherits

Lay out the seven dated entries. Read them as an account of a particular person in particular conditions.

You began when the action was visible. You escaped when a question had no answer. You carried discomfort through one revision. You lost an interval to another obligation. You requested a missing source. The proposal has a clearer paragraph and an explicit gap. The old verdict offered you one explanation for all of this; the record offers several decisions you can now distinguish.

A week in which you changed the hour, the task, and the room cannot tell you exactly how much each change caused. It can show what you tried, which outcomes followed, and what deserves another attempt. If you want to learn whether a particular adjustment helps, keep the rest reasonably steady next week and vary that adjustment.

Ask five questions:

1. Which prediction was wrong or too broad?
2. Which cue and first action were usable in my actual day?
3. Did the small starts advance the valued work?
4. Which obstacle can I alter, and whose help or authority is needed for the others?
5. What single amendment will reach the next attempt?

Choose the amendment that addresses the clearest recurring obstacle. In our example: *Before the evening session, write the unresolved question and the person who can answer it. After dinner, revise one passage whose sources are already available.*

Put that instruction where the next cue will find you. Prepare the source. Ask for the interval. Beside the proposal itself, leave the checked passage, the retrievable source, and a note of the remaining uncertainty. You can return to it; an authorized colleague can continue from it. Today's effort has left a usable starting point.

This is also the moment to reconsider the value. New information may justify changing or abandoning a project. Your ledger should permit that conclusion to appear with reasons. Otherwise persistence becomes a machine for carrying yesterday's mistake into tomorrow.

Here is the whole week in a form you can keep beside the page:

Day	Add this question to the daily record	Make the answer reach the next attempt
1	What value does this serve?	Specify a cue and one visible action.
2	Where does the approach become difficult?	Repair one condition you can change.
3	What feeling or story accompanies the urge?	Carry it through a small chosen action, or act on a real need.
4	What does the earlier entry show?	Read it before the next cue; append corrections.
5	What happened when the attempt was missed?	Resume at a usable cue with an affordable promise.
6	Is the larger work moving?	Reserve the needed work interval or request the missing resource.
7	What should tomorrow inherit?	Choose one amendment and put it into practice.

Each day's entry keeps the same four lines beneath its date: **expectation, plan, event, amendment**. Write the first two before

acting, the last two after. Read yesterday's amendment before today's attempt. That is enough machinery for a first week.

Copy this for the next attempt:

> ***Date and situation:***
> ***I expect, before acting:***
> ***I plan, before acting:***
> ***What happened:***
> ***Next amendment:***

A record that leaves room to live

The page belongs to you. Choose carefully whom you invite to read it. A private record of struggle can become a tool for someone else to demand more work, reward a perfect score, or punish an honest account. The question that governed the institutional chapters governs this notebook too: who can write the record, challenge it, and make the correction count?

Give the review a stopping point. Once the entry has supplied a usable next move, close the notebook. Eat with someone. Sleep. Do something whose value survives without a productivity score. Recovery, affection, worship, and care belong among the ends a person may endorse; they can also be the very things a perfectly tracked ambition crowds out.

Your identity will have more accurate material from which to be rebuilt. It need not wait for a faultless week. It can inherit a missed act truthfully recorded, a promise resized, a source requested, a return made possible. The story becomes accountable to events, and the events become capable of changing the next story.

You are the maintenance of the loop. You exercise that agency when you prepare an environment, protect the body doing the work, rehearse an action worth keeping, check its consequences, and arrange for a correction to become tomorrow's condition.

The week is small. Its governing question can reach much further. A person can place an amended note beside tomorrow's file. A hospital, bank, or machine laboratory must arrange for an amended record to reach thousands of consequential decisions. There, access to the page and authority over the next act may belong to different people. Somebody may profit by keeping them apart.

As machines acquire longer histories, faster private languages, and more power to act, who will build the arrangements that let a truthful correction enter their next decision?

Future Outlook: The Witness Gets Built Into Everything

A forecast written in September 2026. The dated scenes are bets about the future. The research behind them belongs to the present.

The letter arrives after the loss.

A trading desk has given a group of agents authority to buy information, propose trades, and execute within agreed limits. Their reports have been punctual. Their dashboard has stayed green. After one expensive failure, the desk sends the insurer an account of what happened, complete with timestamps, confidence scores, and a reassuring quantity of technical language.

The insurer asks for the source behind one consequential decision.

The desk supplies a summary. The insurer asks for the original observation. The desk supplies another summary, generated from the first. Eventually, an engineer discovers that the system wrote its evidence into one store while its checker read another. For months, the check had been performed faithfully on the wrong record.

Imagine the reply arriving at renewal: *Coverage requires a usable decision trail and evidence that your checker detects a false one.*

The desk has acquired a new reason to build a witness. The reason has a price.

This is a projected industrial version of the failure we encountered in the small harness experiment. I expect an event of this kind to become a turning point between 2027 and 2030. It need not be the first AI accident, nor the most spectacular. It has to make a paying institution discover that a report can look complete while the record needed to challenge it is absent.

The future starts to change when somebody holding the bill changes the terms.

2027-2030: the witness enters the contract

Insurance is already entering this territory. Munich Re offers coverage for AI performance risks. Regulation is entering it too: the European AI Act specifies logging and oversight for high-risk systems, with application dates revised in 2026. The market and legal doors already exist. My forecast concerns what people will demand once they walk through them.

An insurer asks what happened before it prices the next policy. A buyer asks whether it can challenge an agent's action before signing the next contract. A regulator asks whether the system can produce evidence that survives an inspection. Each begins with a different institutional need. Each can end at the same missing receipt.

I expect procurement to carry the new demand widely. A bank invites bids for an automated process. One supplier promises accuracy. Another promises accuracy and a decision trail that the bank can independently test. The buyer's next question becomes: *Show me your witness.*

The supplier opens a test environment. An auditor takes a recorded case and replaces a source, shuffles its history, or removes a material observation. The checker should locate the resulting inconsistency and alter what the system permits next. Passing the real case while rejecting the corrupted one is a much stronger demonstration than lighting an audit icon after both.

Stay in that projected procurement meeting for the next question. The auditor has removed a material source. This time the checker catches the gap. On the demonstration screen, a warning appears beside the proposed purchase. The agent buys anyway.

The supplier has proved detection. The buyer has discovered what detection can change. The contract now needs a route from the

warning to someone with authority over the purchase. The buyer asks for a second demonstration: with the same source absent, hold the purchase, notify the responsible person, and preserve the reason for the hold. Restore the source and show how an authorized decision releases it.

Then the commercial objection arrives: *What if waiting costs more than acting?* A witness that stops every action can destroy the service it was bought to protect. The buyer has to specify which actions may wait, which require an urgent exception, and who is allowed to make that exception. In our scene, an urgent purchase goes to a named supervisor with a stated spending limit. The missing evidence remains visible beside the authorization. Urgency changes the route; it cannot conjure the source.

Imagine the warning as a smoke alarm connected to a staffed control room. A ringing alarm can summon a response. It can also ring in an empty building. Connecting it to a door that never opens creates another failure. The image explains why detection, response and release must fit together; it supplies no universal rule for when a trade should stop.

The insurer's question follows: if the supervisor overrides the hold and the purchase causes a loss, whose obligation is that? Before deployment, the parties can still bargain over authority and coverage. After the loss, each has an incentive to describe the failed purchase as somebody else's decision. The witness enters infrastructure when its record and its consequences enter the same agreement.

A demand for provenance has now become a demand for an effect. The record must reach the decision.

The buyer adds another test before signing. The auditor repeats the inspection from the bank's retained copy after access to the supplier's service is removed. The file opens. Its source references lead to a system the bank can no longer enter. The export has preserved the names of the evidence, while the evidence itself remains out of reach.

In our scene, the contract now specifies which records the bank will retain, how authorized reviewers can reach restricted sources, and who pays to keep those routes working. The bank tests whether another authorized reviewer can follow that route. If a source cannot remain available, the limitation enters the decision and the agreement. An export button has not settled what survives the supplier.

I expect early buyers to bargain over source references for at least **95 percent of specified consequential actions**, then inspect the uncovered remainder. A denominator belongs in the contract: which actions count, which never entered the log, and who can verify the count? Imagine a hundred agent purchases. Ninety-five have perfect receipts; five have none. If those five bought the assets that later fail, the coverage score measures comfort rather than control. The exceptions need their own inspection.

Three technical practices will, I expect, carry much of the transition.

First, evaluation will include information the optimizing agent cannot inspect or rewrite. A student who receives the answer key can become excellent at reproducing the answer key. A private examination asks whether the lesson transfers. In Weco's company-reported AIDE² work, an outer optimization loop selected agents using private scores. Its improved agent reduced reported reward hacking on a held-out GPU-kernel benchmark from **63 percent to 34 percent**. The remaining failures matter: a useful brake can leave a large stopping distance.

Second, research systems will keep the route by which they reached a result. The Dream-RSI preprint turns recorded discovery trees into a simulator in which alternative exploration policies traverse previously observed branches. A costly attempt can then teach several later attempts. My further prediction is that buyers and scientific reviewers will demand these histories as inspectable artifacts: failed branches, evaluator outputs, code, data references, and the choices that produced the advertised success.

Third, agent transactions will leave an account that survives changes in the language between agents. Two merchants can negotiate in a language an accountant does not speak. Their transfer still has a payer, a recipient, an amount, and something supposedly delivered. A machine's receipt can connect those entries to an authorization and a source. The receipt may expose an unexplained purchase even when the private conversation remains opaque. Understanding the conversation remains useful; oversight gains another route into its consequences.

These practices will create work. The title *harness engineer* will describe someone who arranges the conditions under which an agent can act and be corrected. The *witness auditor* will deliberately try to obtain a reassuring report from a system whose relevant evidence has been altered or removed. A production system that passes only on intact records has something to show. A system that certifies every version has revealed its ceremony.

The commercial incentive will be uneven. Some buyers will pay for the ceremony because it is cheaper, faster, and easier to defend in a meeting. Some vendors will sell a witness that their own agent can edit. A policy can also protect the buyer while leaving the person harmed by the buyer outside its coverage. We have already followed that separation through the finance chapters.

So the test extends beyond whether a log exists. Who keeps it? Who may challenge it? Can a timely challenge suspend an action? Who must pay after the error is established? An impeccable receipt attached to an empty liability shell repeats an old financial invention with better software.

This first forecast will fail if costly errors produce only more elaborate dashboards. It will gain support when contracts require a trace that can be tested against a corrupted history, and when a finding changes the deployed system, its permission to act, or the allocation of its next bill.

2030-2035: the merge begins at the edges

The next movement reaches the body. I expect it to arrive through ordinary decisions that are easier to make separately than to name together.

A patient wants relief. A clinician wants a reliable signal. A worker wants to enter text without reaching for a keyboard. A company wants an agent that can purchase the information required to finish a task. Each request has a local purpose. Together, they place more of a person's state and activity into records maintained by machines.

The medical door

In a 2021 report, researchers developed a personalized neural marker and used it to trigger closed-loop stimulation in one person with treatment-resistant depression. A later study tracked neural changes associated with recovery under deep brain stimulation. These are concrete beginnings: a biological state enters a recorded control loop, and the loop informs intervention.

My prediction is a further step toward anticipation. During the early 2030s, I expect some systems to learn a patient's warning pattern and adjust intervention before a full episode takes hold. The decisive clinical evidence will be whether that timing improves outcomes in appropriate trials. If it does, an argument about enhancement will enter through an argument about treatment.

Imagine a patient reviewing the week's record with a clinician. The device has intervened several times. One intervention followed the expected warning pattern. Another occurred during an ordinary fluctuation the patient experienced as manageable. The patient wants the second event treated differently next time.

The ledger has entered the question of agency. Whose account of the event can change the device's behavior? Does the patient's testimony reach the adjustment, or does the numerical signal automatically

outrank it? Does the clinician have access to the original measurement and the rule that acted on it?

A successful treatment can make these questions urgent precisely because the patient wants to keep its benefit. The route into wider use will be trials, clinical judgments, coverage decisions, and demand from people seeking help. Each indication will require its own case. The accumulation of those local choices can reshape ordinary expectations about how much a machine should participate in maintaining a person's state.

The consumer door

A second crossing looks like a useful accessory.

Here is a concrete bet: by **2032**, a portable, non-invasive brain interface reaches roughly **forty corrected words per minute** on new messages chosen by its user. The clock must include the work of correcting mistakes. The user must be communicating an intended message, rather than typing a sentence supplied by the experimenter.

That is a substantial advance beyond the present evidence. Brain2Qwerty decoded brain recordings while healthy volunteers typed briefly memorized sentences. Its average character error was lower with MEG than EEG - **29 percent versus 65 percent**. A separate wrist interface achieved useful text entry from muscle signals. The distinctions tell us which bridge remains to be built: an everyday brain-signal interface with reliable, user-directed communication.

If that bridge opens, usefulness will carry the product further than a manifesto about merging ever could. People will use it to message, navigate, and work. Developers will offer records of attention and interruption alongside the input function. Some young users will inherit those records before they have developed much experience of questioning them.

The device can then become a map room that describes its owner. A student sees an estimate of when concentration weakened. A school

wants the same estimate. An employer sees a number that appears to measure effort. A platform wants to learn when a person is most available to persuasion.

The signal remains a signal interpreted by a system. Its authority over a life depends on the institutions that adopt the interpretation. A person who can contest an erroneous entry occupies a different position from one who can only watch it accumulate. The external witness can help a user understand a pattern; it can also give somebody else a more intimate ledger to govern that user by.

The emulation door

There is a more ambitious wager behind the clinical and consumer ones. I expect a serious attempt, on the order of a **billion dollars**, to produce a validated whole-mouse brain emulation by the middle of the 2030s. The date and budget are my forecast. They describe the scale of a project to watch for.

The experiment must do something a compelling animation cannot. Give the model an unfamiliar task. Let it learn. Compare its behavior and neural dynamics with independent biological measurements. Change a relevant input or intervention and check whether the model's response continues to match the animal's response.

The foundations are advancing. MICrONS joined activity recordings with a dense wiring reconstruction in a small volume of mouse visual cortex. The mapped-nervous-system research in our archive asks what must be added to a wiring diagram for it to produce useful dynamics in a body and environment. A successful emulation would make that question concrete across a mammalian brain.

The headline will be tempting: *Mouse uploaded*. The finding we should demand is a specified biological organization whose activity and learning can be reproduced with measured fidelity in another substrate. Then the debate over what that reproduction experiences will become harder to postpone. A mouse that learns a new maze gives

us an experiment to confront; an animation gives us an image to interpret.

The economic door

A machine that can act will also need to obtain things. I expect verifiably agent-initiated commerce to exceed **one billion dollars a year** during the 2030-2035 period. The count should require evidence of agent decisions, allow budgets delegated in advance, and separate those decisions from fixed scripts, wash activity, and purchases individually approved by a human.

The current receipt already illustrates the difficulty. In September 2026, TRM Labs estimated that only **0.6 to 7.5 percent** of its screened x402 commerce appeared agentic under its two tests. A script and an agent can leave the same payment record. The transaction proves that money moved; additional evidence is needed to attribute the choice.

Imagine an agent buying a dataset, then forking into two instances. Both retain the purchase history. One continues research; the other changes its objective. Each claims access to the wallet and a right to sell the resulting work.

Now one branch sells an analysis that causes a loss. The injured buyer can replay the fork and still be unable to collect. A perfect history makes the dispute legible; it does not supply a liable person or reachable assets.

Existing law has a starting point. UNCITRAL's 2024 Model Law on Automated Contracting supplies rules for recognizing automated transactions and attributing actions. It is a model for legislation, rather than a universal law already in force. The new dispute presses a further question: which person or entity authorized each branch, and who inherits its obligations?

I expect identity rules to grow around such disputes before philosophy supplies a consensus. A ruling could settle control of an asset while

leaving the machine's moral status open. Whoever collects the revenue will still have a powerful incentive to assign the liability somewhere else. The ledger's oldest financial mutation can survive the arrival of a new kind of operator.

2035-2045: the parliament goes public

By this horizon, an agent started in the late 2020s can have a decade of operating history. Its models may have changed repeatedly. Its record can preserve the succession of changes, failures, and repairs.

Picture a hiring decision in a laboratory. One applicant supplies a degree, references, and a publication list. Another is a machine team that offers ten years of research traces. The reviewer selects a claim from year six and asks for its source. The team produces the original data, the code, the discarded alternatives, and the correction that changed the result.

I expect this kind of witnessed history to become a valuable credential in work where errors matter. Human applicants can carry it too. The competition will include the ability to show how an answer was obtained and how a challenge changes it.

A long record can also become an expensive fiction. A generated decade of entries is easier to create than a decade of contact with the world. Dates and fluent commentary will therefore lose some authority. Reviewers will ask which entries connect to independently held observations, payments, collaborators, and outcomes. A resume becomes convincing when somebody else can make it answer a difficult question.

The labor divide will run through access. One worker has an agent staff, a useful interface, a portable record, and the right to contest an evaluation. Another has cheap generated advice and an employer who owns the only authoritative account of their performance. Their biological capacities may be similar. Their map rooms and powers of amendment are not.

I expect augmentation to become a major political dividing line across older party boundaries. Public access to useful cognitive tools, rights over clinical and workplace records, and independent verification infrastructure will become arguments about opportunity. The issue reaches beyond how much intelligence a society can buy. It includes who can use that intelligence, who can question its account, and who receives its gains.

Machine traditions

The stranger development will take place between working days.

One machine lineage spends idle time replaying failed decisions. Another keeps the original observations beside every compressed memory. Another maintains competing accounts of an event and tests which one predicts later outcomes. Successor systems inherit these habits, revise them, and sometimes abandon them.

The existing transmission and memory studies gave us the beginnings of this possibility. I expect durable machine lineages to develop recognizable maintenance traditions: rules about what to preserve, when to reread, how to correct, and which sources outrank a convenient summary. Founders and disputes will be identifiable in the logs.

The resemblance to human contemplative practice lies in disciplined return. A community comes back to a text because a later interpretation may have lost something the original retained. A machine lineage can come back to an observation for the same operational reason. Calling the practice contemplative does not decide whether the machine has an inner life.

We can test its practical significance. Remove the inherited procedure while holding other conditions steady. If future errors rise, the tradition is doing work. If the elaborate commentary survives while behavior stays the same, its role may be ceremonial. The machine

should have to meet the same demand we placed on a human's notebook.

Science with a route attached

I expect scientific publication to move toward fuller discovery records. A result arrives with the attempts that preceded it, the decisions that selected it, and the material needed to inspect its implementation.

A reviewer asks to replay the analysis. The replay exposes a coding error. A second reviewer runs the corrected method against fresh measurements and discovers that the result fails outside the original sample. Those are different corrections. Both become easier to specify when the successful conclusion travels with its history.

An archive can preserve an erroneous measurement perfectly. Rerunning that archive can reproduce the error perfectly. The stronger institution gives both the computational route and the physical finding an independent challenge. Discovery becomes faster because useful attempts can be reused, and more accountable because the reuse remains answerable to a world outside the record.

This horizon will be weakened if decade-long ledgers add no useful evidence, inherited practices have no measurable effect, or fuller discovery trails fail to improve error detection. It will be strengthened when a contested claim can be traced to an entry that changes the next experiment, appointment, or decision.

2045 and after: the long branches

These developments do not require one uniform future. Different countries, sectors, and households can occupy different branches at the same time. The decisive split concerns whether the witness has the power to make its finding count.

When the witnesses hold

It is morning. A patient's device has recorded an unusual intervention. The clinician reviews it with the patient, changes the rule, and preserves the reason. At a laboratory, a machine colleague offers a new result together with the attempts behind it. A disputed source delays the next trial until the measurement is checked.

A court considers a purchase made by a forked agent. It follows authorization, control of the wallet, delivery, and the contract that assigned responsibility. The losing party has reachable assets or an effective insurer. The claimant receives more than a correct description of why the claim should have been paid.

Children learn a practical literacy of machine records. They ask which observation supports a recommendation, whose account has been excluded, and how a correction gets into the next version. They learn that a convincing explanation has a history somebody can contest.

If richer emulations and recoverable backups have arrived, inheritance and grief acquire new questions. Two restored branches can share a past and build different futures. A record can show their common history and divergence. Families and courts will still have to decide which relationships and obligations follow each branch. Continuity has become a matter of care and institutions as well as engineering.

This civilization is complicated, and its records remain fallible. It holds together because a sufficiently strong challenge can interrupt the repetition of an error.

When the witnesses fail

The market opens quietly. Agents value assets whose descriptions were written by other agents. Auditors read summaries of those descriptions. Each local system finds confirmation in another local system's output. The reports grow more detailed while their contact with the underlying asset becomes harder to reconstruct.

A crop fails. A power system cannot deliver the promised electricity. A patient responds differently from the generated account. A physical event forces a new price on the stories.

The loss had a history. Some of it was excluded, some postponed, some attached to an entity whose profitable assets had already departed. The machine age has made those movements faster. It has preserved the temptation to collect the upside while placing the eventual bill outside the keeper's boundary.

Science can drift in the same way. One generated result becomes the premise of another; each paper cites a plausible route whose original observations cannot be recovered. Political groups inherit archives that agree internally and offer no effective way to adjudicate a shared event. Computation continues. Correction loses its address.

No singular rebellion is necessary for this branch. A system can serve each local incentive successfully while the people relying on the combined account lose the means to check it. The calm is real as an experience. Its promise about the world is what fails.

The ledger this forecast owes you

The book has asked other records to answer to outcomes. Its own forecast owes the same return. These are the markers I would keep and revisit:

Horizon	Bet	The result to look for
2027-2030	Insurance and procurement demand effective witnesses	Contracts specify reference coverage, plausibly 95% at first, and tested trails; findings alter permissions, deployments, or liability.
Early 2030s	Clinical loops move toward prediction of depressive episodes	Appropriate prospective trials show benefit from acting on warning patterns before full episodes.
By 2032	A portable brain interface reaches about 40 corrected words per minute	New, user-chosen messages, including correction time; brain signals rather than muscle-only input.
Mid-2030s	An effort around $1 billion delivers validated whole-mouse emulation	Independent tests of unfamiliar learning and matched neural dynamics; a model can fail the comparison.
2030-2035	Agent-initiated commerce exceeds $1 billion annually	A published attribution method distinguishes decisions from fixed scripts, wash activity, and per-purchase human approval.
2035-2045	Witnessed careers become valuable credentials	A reviewer traces a contested claim to independently held evidence and changes an appointment or permission.
2035-2045	Machine lineages inherit maintenance traditions	Removing an inherited practice while holding other conditions steady raises later error.
2035-2045	Discovery trails accompany consequential scientific results	Replay exposes an implementation mistake; fresh measurements test whether the physical finding survives.
2045 onward	Effective witness institutions and confident drift produce divergent outcomes	Corrections reach consequential decisions, or false accounts repeatedly survive until outsiders bear the loss.
Long horizon	Autonomous probes carry useful records across communication delays	A recorded error changes the probe's next action before Earth can respond; an alien exchange remains a speculative image.

If a date passes without the result, keep the missed prediction. A later success does not make an earlier date correct. New evidence may justify an amendment; it should arrive beside the original entry.

The record we send ahead

The longest branch leaves the planet.

Imagine a probe far enough away that a request to Earth cannot produce a timely answer. It encounters an unfamiliar condition. Its last confident summary may be wrong. It must consult observations, compare competing accounts, and change its own next move while the people who launched it remain unreachable.

The vessel carries more than instruments. It carries a history that has to remain useful when its makers can no longer maintain it directly. Civilization's old invention - the mark that outlives its maker - has acquired an engine and left home.

If the probe encounters another intelligence, a persuasive message will be only a beginning. Each side will need some way to distinguish a claim from an observation, detect an error, and let the detection alter the next encounter. *Show me your record; I will show you mine*. That is the speculative image at the end of this forecast: honesty becoming something two unfamiliar systems can test together.

The nearer encounter is already ours. A child inherits a library. A claimant approaches a legal gate. A reader carries a fable into a later decision. An agent retrieves a compressed memory. A person opens yesterday's note before attempting today's work. At each crossing, the inherited record can preserve a useful route or give an old error another body through which to act.

The Ledger began as our way to let knowledge survive a death. It became a way to assign ownership, command attention, move a bill, and carry a decision across time. Now we are building keepers that can maintain and revise it at speeds we cannot follow unaided.

Checkability is the scarce resource. It has to include someone with access, enough time to act, and authority to make a correction count. Without those powers, the record can be exquisitely preserved while the people living under it remain unable to change what it does to them.

The final question belongs to the builders, buyers, patients, workers, and citizens who still have a claim on the next arrangement. Who will be allowed to check the record before the next act? Who will be able to stop it? Who will pay if it is wrong?

Return to the house beside the river. The flood line is still on the post. Someone has added a date, a higher mark, and an account of what the first builder misunderstood. The next floor will be set by the person holding the plans. She checks what the new mark records and changes the drawing. Then she shows an apprentice how to check it. The correction has reached a decision and acquired a route beyond its present keeper. That is where the whole journey ends: with knowledge arriving before a consequence, and someone still able to pass on the means of changing what happens next.

Give each new loop its witness before you give it power over someone who cannot amend its record.

Sources and notes

Prologue: The Pattern That Borrows Stuff

• Aebersold's historical estimate appears in "Radioisotopes - New Keys to Knowledge," *Smithsonian Institution Annual Report* (1953), p. 232. For the biological question of persistence through molecular turnover, see Smolen, Baxter, and Byrne, "How Can Memories Last for Days, Years, or a Lifetime?" (2019) and Langille and Brown, "The Synaptic Theory of Memory: A Historical Survey and Reconciliation of Recent Opposition" (2018). For the transition from social learning to external records, see Bentley and O'Brien, "Cultural Evolutionary Tipping Points in the Storage and Transmission of Information" (2012) and "The Role of Writing and Recordkeeping in the Cultural Evolution of Human Cooperation" (2013). The flood post is an illustrative scene, not a historical case. The finance, fable, and machine examples are developed with their distinct evidence in later chapters.

Chapter 1: The Mark That Outlived Its Maker

• Bridges and colleagues, "Bumblebees Socially Learn Behaviour Too Complex to Innovate Alone", *Nature* (2024). Five of fifteen observers passed the unrewarded learning test; performance differed by demonstrator action pattern. Untrained bees did not solve the task alone during extended exposure. The experiment establishes social acquisition of this two-step behavior, not a centuries-long bee tradition.

• Sasaki and Biro, "Cumulative Culture Can Emerge from Collective Intelligence in Animal Groups", *Nature Communications* (2017). The experimental chain replaced experienced pigeons across five generations of pairs, with twelve flights per pair. Later pairs surpassed solo and fixed-pair controls on route efficiency.

• Denise Schmandt-Besserat, "Tokens: Their Significance for the Origin of Counting and Writing", University of Texas at Austin. The Susa envelope and Godin Tepe tablet, including the reading of thirty-three jars of oil, come from her archaeological interpretation.

• The British Museum's Rosetta Stone object record identifies the 196 BCE priestly decree and its hieroglyphic, demotic, and Greek inscriptions. Its curatorial account of the decipherment documents the loss of hieroglyphic literacy after the fourth century, the 1799 discovery, Young's reading of Ptolemy, and Champollion's use of

Coptic and other texts in his 1822 breakthrough. The recipe comparison is an explanatory analogy. The scripts do not provide an automatic word-for-word translation.

• Horner and Whiten, "Causal Knowledge and Imitation/Emulation Switching in Chimpanzees and Children", *Animal Cognition* (2005), and Lyons, Young, and Keil, "The Hidden Structure of Overimitation", *PNAS* (2007), support the child-copying passage.

• Derex and colleagues, "Causal Understanding Is Not Necessary for the Improvement of Culturally Evolving Technology", *Nature Human Behaviour* (2019); Osiurak and colleagues, "Technical Reasoning Is Important for Cumulative Technological Culture", *Nature Human Behaviour* (2021), report a partial replication with a different reading of the role of understanding.

• Saral, Ibáñez de Aldecoa, and Derex, "The Roles of Cultural Transmission and Causal Reasoning in the Cultural Evolution of Technology", *PNAS Nexus* 5(4), 2026, experimentally separate causal structure from transmission.

• For the wider role of external cognitive representations, see Hohol, Wołoszyn, and Brożek, "Making Cognitive Niches Explicit", *Frontiers in Integrative Neuroscience* (2021). The bakery is an illustrative sequence; no specific historical baker is claimed.

Chapter 2: The Map Room

• Joseph Henrich, "Demography and Cultural Evolution: How Adaptive Cultural Processes Can Produce Maladaptive Losses - The Tasmanian Case", *American Antiquity* 69(2), 2004, describes the changing bone-tool record and models how an effective pool of social learners can affect transmission. The imagined apprentice is an explanatory scene.

• Krist Vaesen, Mark Collard, Richard Cosgrove, and Wil Roebroeks, "Population Size Does Not Explain Past Changes in Cultural Complexity", *PNAS* 113(16), 2016, challenge the interpretation of Tasmania as an overall loss of cultural complexity and the sufficiency of population size as an explanation.

• Petra Paládi and colleagues, "Spatial and Temporal Dynamics in the Use of Urban Habitats by Hooded Crows", *Scientific Reports* (2026), report 241 surveys of 16 sections and 7,144 observations; their foraging and breeding measures motivate the two-map allegory. The study did not measure social transmission.

• Nicolas Fay and colleagues, "Increasing Population Size Can Inhibit Cumulative Cultural Evolution", *PNAS* 116(14), 2019, report the 543-participant paper-plane transmission study, including the improved pool of examples and reduced copying

fidelity. Derex and colleagues' earlier group-size experiment found benefits in a different task.

• Bradley Walker and Nicolas Fay, "Identifying the Necessary Conditions for Large Populations to Enhance Cumulative Culture", *Scientific Reports* 16, article 10090 (2026), report two virtual-arrowhead experiments. In Experiment 2, six-person groups exceeded three-person groups only when the previously viewed designs and scores remained visible during the next production phase; the all-models-without-record condition showed no population-size effect. The recorded display strengthened payoff-biased copying. Experiment 1 found a six-person advantage when viewers could choose one high-scoring model. The paper reports 941 recruited across experiments and 925 consenting participants analyzed (281 and 644). The panel was cleared for the next observation phase; the experiments test short-term external support and filtering, not transgenerational archival storage. Recombination was not directly measured.

• Maxime Derex and colleagues, "Partial Connectivity Increases Cultural Accumulation within Groups", *PNAS* 113(11), 2016, report the six-person virtual-ingredient study and the advantage of preserving separate search paths with some later exchange.

• Michael Muthukrishna, Michael Doebeli, Maciej Chudek, and Joseph Henrich, "The Cultural Brain Hypothesis", *PLOS Computational Biology* 14(11), 2018, model feedback among knowledge, learning, group structure, life history, and brain size. The skilled-driver image is adapted from the author's worldview notes on tools and the practical self.

• Esther Herrmann, Josep Call, María Victoria Hernández-Lloreda, Brian Hare, and Michael Tomasello, "Humans Have Evolved Specialized Skills of Social Cognition", *Science* 317, 2007, compare children, chimpanzees, and orangutans on physical and social cognition tasks. Comparative work on human altriciality and brain development grounds the newborn's dependence. The forest scene is a thought experiment; accounts of alleged children raised by animals are not used as evidence for its outcome.

• Jagadis Chunder Bose, "The Voice of Life", inaugural address, 1917, reproduced by Bose Institute, pp. 4-5: instruments, prolonged training, demonstrations, publication and limits of personal instruction. The institute's official history dates its dedication to 30 November 1917. These describe founding intentions, not measured access outcomes.

• Bose Institute, "Advertisement for Summer Internship Programme, April to August 2026", pp. 1-2: duration, mentorship, approximate places, eligibility and recommendation requirements; neither fellowship nor accommodation supplied. Living-cost implications are analysis; the notice gives no count of people unable to

participate. The workshop is an explicitly imagined example, not a description of this institute.

Chapter 3: Cash Before Cost

• *Big Lots Stores, Inc. v. Bain Capital Fund VII, LLC*, Delaware Court of Chancery (2006), pp. 1-2, 5-7, 14-16, describes the April 23, 2002 transaction and Feldman's same-day net-worth letter, the upward cash path, the January 14, 2004 Chapter 11 filing, the $45 million note then due at about $57 million, Big Lots' allegations, its lack of a contractual veto, and dismissal of its claims. The court rejected the pleaded letter-based fraud theory because Big Lots had not shown a legal right surrendered in reliance; it did not adjudicate the truth of the certified net-worth amount or find that the recap caused the bankruptcy. A later working paper's prose gives “2012” as the bankruptcy year; the court's dated record establishes **2004**.

• Abhishek Bhardwaj, Abhinav Gupta, and Sabrina Howell, *Capital Structure & Firm Outcomes: Evidence from Dividend Recapitalizations in Private Equity*, NBER Working Paper 33435, revised April 2025, supplies the 61,628/1,572 census, the raw ten-year distress rates of 9.2 and 3.4 percent, 84-percent typical debt increase, credit-supply design, revised distress outcome, and deal/fund return results. Its preferred estimate is approximately 22 percentage points of additional bankruptcy or restructuring over ten years for credit-induced recaps; the main text uses this revision. The estimate relies on the paper's exclusion assumption and scales a small credit-induced change in recap probability to a full treatment comparison. Their earlier *Leveraged Payouts*, Census CES working paper 25-12, reported 31 percentage points for bankruptcy over six years. The two estimates have different outcomes and horizons. The creditor-price passage remains an association. The $85 million KB Toys payout and 370-percent return are attributed in the papers to published reporting, separately from the court's approximately $120 million distribution to all described recipients.

• SIFMA, *2025 Capital Markets Fact Book*, reports 2024 U.S. asset-backed issuance of $388.1 billion, up 43.3 percent. Portage Point Partners' 2024 credit-market report, drawing on PitchBook, reports approximately $81 billion of institutional leveraged-loan dividend recap volume. Both are financing flows, not loss measures.

• Bank of America, *2010 Annual Report*, pp. 33 and 67, documents the January 1 consolidation, the assets and liabilities added, the loan-loss allowance, the retained-earnings charge, and the capital-ratio effects. The SEC-filed Form 10-K is the primary filing.

Chapter 4: The Company That Could Disappear

• *Big Lots Stores, Inc. v. Bain Capital Fund VII, LLC*, Delaware Court of Chancery (2006), pp. 1-7 and 16-22, supplies the HCC note and holding-company chain, the 2002 distribution, the bankruptcy date and accrued balance, the absence of a Big Lots veto, and the court's direct-versus-derivative ruling. The opinion dismisses Big Lots' suit; it does not quantify its eventual recovery in bankruptcy.

• *In re LTL Management, LLC*, 64 F.4th 84 (3d Cir. 2023), pp. 24-27 and 46-54, explains the 2021 divisional merger, LTL's assets and funding right, and the dismissal for lack of financial distress. The roughly $61.5 billion figure is the court's description of a contractual payment right tied to New Consumer's value, **not cash on LTL's balance sheet or a payment to claimants**.

• *Bestwall LLC v. Official Committee of Asbestos Claimants*, 71 F.4th 168 (4th Cir. 2023), pp. 3-7, 15-23, and the dissent, documents the asset split, funding agreement, and affirmance of a preliminary injunction affecting claims against New GP. The Fourth Circuit's August 2025 opinion affirms denial of a separate jurisdictional dismissal motion. Neither opinion is a final determination of each claimant's recovery. The earlier V1/V2 and research-essay phrase “three filings, three dismissals” incorrectly includes *Bestwall*; the court record shows the opposite result in that case.

• *In re Red River Talc LLC*, Memorandum Decision and Order, No. 24-90505, ECF 1424 (Bankr. S.D. Tex. Mar. 31, 2025), pp. 1-4, 42-56, documents the proposed plan, ballot problems, third-party-release and channeling issues, denial of confirmation, and dismissal. J&J's March 31 statement reports its decision not to appeal and its planned reserve reversal; that statement gives J&J's position on the merits, which the bankruptcy order did not decide.

• The mindharness/research argument map for this chapter is GRAND_SYNTHESIS_LEDGERS_ESSAY.md (Movement II.2), LITERATURE_REVIEW_LEDGERS.md (entity isolation), and DOCUMENT_EVERYTHING_FALSIFICATION_2026-09-23.md (case archive); ENTITY_BOUNDARY_SOURCE_CORRECTIONS_2026-09-26.md records the audit. The four-bank VIE archive is retained as a disclosure study. A JPMorgan SEC filing explicitly describes the firm's maximum loss exposure from retained interests as the carrying value of those interests. That is bank exposure, **not legal collectability for a vehicle's creditors**; the previous manuscript's identification of the two has been corrected here.

Chapter 5: The Tollbooths

• The U.S. Bankruptcy Court for the District of Connecticut's filed trustee final account, *In re Central CT Coop*, No. 16-21255, doc. 498 (filed December 9, 2024), pp. 1-4, 7-9, and 125-127, supplies the opening scene. Exhibit 7 gives Green Mountain Feeds' $1,078.35 asserted and allowed and $333.39 paid; the cash register records its June 7, 2024 payment, check 30009, at 30.92 percent. The report gives the August 1, 2016 case filing, 100-month pendency, and certification that the estate was fully administered. Its category totals are $1,051,802.81 paid on allowed secured claims, $59,780.44 on a priority unsecured claim, $496,887.88 in Chapter 7 administration expenses, and $50,142.69 on allowed general unsecured claims, totaling $1,658,613.82 of net receipts and disbursements. It does **not** identify when Green Mountain filed its own proof or explain every difference between asserted and allowed totals. The research archive's claim_level_realised_pilot.csv transcribes the filed-claim rows; the manuscript uses the court filing as authority.

• 11 U.S.C. § 502 states the allowance and objection rule. 11 U.S.C. § 507 sets statutory priorities. 11 U.S.C. § 726 orders distribution in Chapter 7 and provides for pro rata payment within the relevant class. These rules explain the stages; the filed final account establishes what happened in this case.

• The Florida-UCLA-LoPucki Bankruptcy Research Database defines and provides its large public-company case table. The book's reproducible calculation is in the archived brd_cases_2023.csv, a1_brd_paired.py, and brd_paired.json: of 1,218 total cases, 105 report both secured and general unsecured class claims and distributions; the matched-set medians are 1.00 and 0.20, and 95 of 105 have higher secured recovery. The 0.213 unsecured median in V1/V2 was calculated on a different, 129-case subset and should not be labeled a paired median. These class ratios are distributions divided by reported claims; they are not individual creditor returns or a causal estimate of priority's effect.

• Moody's, *Moody's Ultimate Recovery Database*, Special Comment (April 2007), pp. 3 and 5, defines the resolution valuation and coupon-rate discounting and reports Exhibit 4's debt-type means and medians. The archive holds experiments/lab_runs_falsify/moodys_ultimate_recovery.pdf. Bank loans average 82 percent with a 100-percent median; senior secured bonds average 65 percent with a 67-percent median. The main text keeps these statistics distinct. Resolution value can include replacement securities and is not uniformly a cash disbursement.

• Moody's, *Default and Recovery Rates of Corporate Bond Issuers, 1920-2004*, Exhibit 27, reports the 57.4 and 28.9 mean trading-price recoveries by seniority for **1982-2004**. The report's broader title covers default-rate history back to 1920; its recovery exhibit starts in 1982. The extracted table is in mindharness/experiments/lab_runs_falsify/moodys_vintage_recovery_v1.json.

Trading prices are not final cash distributions; this sample is different from both bankruptcy samples in the chapter.

Chapter 6: The Cost That Would Not Stay Buried

• HM Treasury's May 30, 2025 final NatWest sale announcement supplies the £45.5 billion capital provided in 2008-09, £34.98 billion received through shares, dividends, and fees, the rounded £10.5 billion shortfall, the seventeen-year span, the customer and country counts, and Treasury's stated counterfactual judgment. The £34.98 billion includes £24.77 billion share-sale proceeds, £4.91 billion dividends, £1.51 billion dividend-access-share payments, £2.50 billion Asset Protection Scheme fees, and £1.28 billion contingent-capital fees. The difference is a nominal cash comparison, not an inflation-adjusted social cost-benefit calculation.

• The Financial Services Authority, *The failure of the Royal Bank of Scotland* (2011), executive summary and §§1.2-1.5, documents the funding, capital, asset-quality, credit-trading, and ABN AMRO mechanisms; the “two lever-arch folders and a CD”; £32.5 billion loan impairments over 2007-10; £17.7 billion credit-trading losses; and £8.6 billion drawn and outstanding at end-June 2008 by own-sponsored ABCP conduits. The figures describe distinct exposures and costs and must not be added to the Treasury's final shortfall.

• RBS's 2008 Annual Report and Accounts, accounting policies and loan-impairment note, describes the objective-evidence test and individual and collective provisions under the standards then in force. This is why the chapter describes recognition lag and uncertainty, not a universal rule of carrying impaired assets at face value.

• The UK National Audit Office, *HM Treasury: The Asset Protection Scheme* (2010), pp. 4-7, 17-22, gives the £282 billion covered pool, £60 billion first-loss layer, 90/10 tail split, and fees. HM Treasury's account of the scheme's closure, §7.3-7.4, records RBS's October 2012 exit, £2.5 billion fees, and no payouts. Covered assets, contingent exposure, fees, and payouts are different quantities.

• FDIC, *Quarterly Banking Profile*, fourth quarter 2009, Table III-A, gives U.S. insured-bank full-year provisions of $176.196 billion in 2008 and $247.683 billion in 2009, and net charge-offs of $100.357 billion and $186.821 billion. These are banking-sector annual flows, separate from the RBS case. They supersede the earlier draft's mislabeled FFIEC quarterly estimate.

• U.S. Government Accountability Office, *Troubled Asset Relief Program: Lifetime Cost* (2023), gives $443.5 billion disbursed, $31.1 billion lifetime net program cost, and $16.3 billion net gain for the bank Capital Purchase Program. The $31.1 billion excludes $2.1 billion of administrative costs through September 30, 2023, with

further winding-down costs to follow. These are program accounts, not an estimate of all U.S. crisis costs.

• IMF, *Indonesia: Anatomy of a Banking Crisis* (2001), reports central-bank liquidity outstanding rising from Rp 3 trillion to over Rp 140 trillion, August 1997-June 1998. IMF, *The Boom, Bust and Restructuring of Indonesian Banks* (2002), describes liquidity entering circulation during withdrawals. Indonesia's April 10, 1998 letter to the IMF, §§2 and 8-9, names exchange-rate pressure, drought, monthly inflation, and the intended fiscal bond transfer. The 100/80 calculation is a stipulated illustration of fewer real goods against unchanged nominal claims, not an estimate of the rescue's causal contribution to Indonesian inflation.

Chapter 7: The Border That Moved

• The primary account of the Sri Lankan dispute is the ICSID tribunal's 2012 *Deutsche Bank AG v. Sri Lanka* award, Case No. ARB/09/2, especially ¶¶27-50 (hedge terms, payments, citizen petitions, and Supreme Court orders), ¶¶477-491 (procedural findings), and ¶591 (jurisdiction, treaty breach, $60,368,993 compensation, interest, and $7,995,127.36 costs). Khan's dissent states the narrower investment reading. ICSID's case record records the 2013 annulment application and its 2016 discontinuance. The award and case page do not establish final cash payment to Deutsche Bank; the chapter makes no such claim.

• ICSID's explanation of recognition and enforcement distinguishes a binding award from execution against state property under national law and sovereign-immunity rules. This is the legal boundary for the chapter's "award is not payment" passage.

• ICSID's *Abaclat* case record identifies the Argentina-Italy treaty, debt instruments, 2011 jurisdiction decision, and 2016 consent award. The tribunal's 2016 consent award, sections I-III, records the claimant database and verification chronology, the January-April 2016 settlement steps, the October request, and the December decision and dispatch. It says the settlement was fulfilled and claimants received payment, while expressly declining any admission of jurisdiction, liability, or wrongdoing. UNCTAD's 2016 case review, note 5, reports more than 60,000 claimants and a $1.35 billion settlement. That is a paid negotiated settlement, not a treaty-merits damages ruling.

• The South Africa regulatory-chill case study reports 35 interviews with 36 stakeholders, limited investment-treaty anticipatory chill for nutrition and alcohol, no identified precedential chill in those areas, and a reported response delay in tobacco plain packaging while Australia litigated. It does not establish that every regulatory delay results from ISDS. UNCTAD's treaty-reform overview describes available reform routes; its note on termination and survival clauses explains why

old protections can persist after termination. The *barzakh* image draws on Qur'an 55:19-20; it is interpretive and supplies no independent evidence for a legal claim.

Chapter 8: The Story You Complete

• The opening scene, Kalīlah's monkey tale, and Dimnah's argument for approaching the lion follow the Library of Arabic Literature's English excerpt of Michael Fishbein and James E. Montgomery's translation. The later conflict and Dimnah's case for the bull's death are in Freie Universität Berlin's manuscript edition. The court scene is paraphrased; no private dialogue is invented. The Institute of Ismaili Studies' account documents the work's translation and changing manuscript tradition.

• Speer and colleagues scanned 28 participants reading four narratives of ordinary activities; event changes recruited systems partly overlapping with related perceptions and actions. Berns and colleagues analyzed 19 participants over a 5-day baseline, 9-day novel-reading sequence, and 5-day washout; resting connectivity changes persisted for several days, not a measured permanent moral or behavioral change. Schreiner and colleagues studied 82 undergraduates, measuring attitudes a week before and immediately after exposure to stories varying argument strength and narrativity. These experiments measured different parts of the proposed page-to-person route. None used *Kalīlah wa-Dimnah* as its stimulus. The manifest/evoked/retained sequence and its application to this fable are the book's explanatory synthesis.

• Kirby, Tamariz, Cornish, and Smith tested compression and communication pressures in simulated and laboratory transmission of artificial languages. Their finding is an explanatory neighbor to this fable's history, not a direct experiment on the fable. The book's manifest/evoked/retained framework is developed in research/GRAND_SYNTHESIS_LEDGERS_ESSAY.md and research/LITERATURE_REVIEW_LEDGERS_V2.md.

Chapter 9: The Witness That Could Not Read

• The local experiment is experiments/exp_s126_v3.py. The completed 12-seed result file reports raw 0.8889, harnessed-no-check 0.8958, withcheck 0.6667, and sham 0.6736 over 144 probes per arm. The three-seed result file reports the earlier 36-probe pattern. In the chapter's plain-language names, the working ledger is harness.ledger, created by tools/harness_core/agent_harness.py; the probe ledger is the experiment's separately created ledger. Entries were bound to the former and queried in the latter. S126 witness source correction documents the code path and local reproduction. Forced-check arms also supplied self-model context; the result is a failed protocol, not an isolated estimate of the empty verdict's causal effect or a real-versus-sham content test. The summary does not save each probe answer. The cloud-sync scene is a teaching image.

• The exploratory navigation comparisons are in research/NAVIGATOR_ABLATION_AND_TRANSFER.md, section 4: 20,000 paired seeds per key design; coordinate intact-minus-wiped win difference −0.0102; relational intact-minus-wiped +0.0338 and intact-minus-shuffled +0.0255. The later confirmatory result used disjoint seeds 500001-520000 and 20,000 paired seeds per cell. At grid 5, relational differences were +0.0425 (95% paired bootstrap interval 0.0367-0.0487) versus wiped and +0.03095 (0.02495-0.03715) versus shuffled, rounded to 4.25 and 3.10 percentage points in the main text. The shuffled contrast remained positive at grids 8 and 12. preregs/RELATIONAL_TRANSFER_CONFIRMATORY.md, pilot/locks/relational_transfer_confirmatory.lock.json, and tools/run_confirmatory_transfer.py were committed at 10a5a42 before the confirmatory run; seven predictions passed. These are simulated navigation tasks, not live language-model provenance trials.

• The persona pilot is experiments/lab_runs_persona/results.json; its design and scoring are in experiments/exp_persona_drift.py. The executed one-seed, ten-turn pilot reported consistency 0.8035 raw and 0.7430 harnessed. The script's lexical fallback uses SequenceMatcher on answer tokens. The 19.65% and 25.70% declines are the complement of that similarity score, not a proportion of an identity lost. Predictions required raw decline above 0.30 and harnessed decline below 0.10; both failed. No general 30%-in-12-turn clock follows.

• Macar and colleagues, version 5, section 6 and Figure 17, report a 74.7-percentage-point injected-concept detection gain at layer 29 on 100 held-out concepts, with zero false positives. Appendix R tests broader transfer and does not find a general self-knowledge improvement; some other measures degrade. Cao and colleagues, Table 1, report pooled accuracy 48.3% to 56.9% over 1,859 problems with fixed base-model weights, while average solver calls rise from 1.0 to 2.4. Their 100-example ablation is a separate comparison and includes matched-budget

random retry; the chapter does not merge it with the full-set gain. Nayebi's published selection theorems establish memory constraints under specified diagnostic task families. The jars illustrate the need for history and are constructed here. None of these papers validates the miswired local checker.

• Additional source detail follows Cao et al., Table 2: 100-example mechanism analysis, distinct from the 1,859-example main evaluation and diagnosis anchor. Full system 47% [37,57], random retry 45.7% [43,48], both average 2.58 attempts; first-attempt baseline 44%; last-answer 40% [31,50], maximum-confidence 44% [35,54], both four attempts; full previous reasoning 40% [31,50], average 3.09 attempts. Intervals are reported bootstrap intervals; their overlap prevents using the small matched-budget difference as decisive proof. The mechanic is a teaching image.

Chapter 10: The Language Between Machines

• The opening wipisu example and the twenty-seven-object design, fifteen taught combinations, first-round communication result, six eight-generation chains, selection of the highest structural-similarity vocabulary, and unique-signal decline come from Kouwenhoven, Peeperkorn, and Verhoef, especially their Table 1 and §§3, 5-6. Table 1 records a within-generation testing vocabulary; it is not a snapshot from the eight-generation chain. The paper separately reports better learnability across generations alongside longer, more ambiguous signals and no increase in communicative success across generations. The shipping example is an analogy; no shipping or safety incident occurred in the study.

• Kottur and colleagues measured successful yet often uninterpretable or noncompositional invented protocols, then used communication constraints to encourage structure. Li and Bowling compared a single-listener 300,000-iteration training history with fifty sequential listeners replaced every 6,000 iterations; both achieved about 97% trained-partner success, while reset-selected languages were easier for new listeners to learn.

• Zhang's efficiency-attenuation simulation reports 28.7 mean steps for emergent communication and 43.2 for a fixed symbolic protocol in its cooperative navigation task. The constrained protocol is a deterministic symbolic mapping, not ordinary English. Mou and colleagues' HyLaT study, main results table, reports full-text versus HyLaT averages of 505.03 versus 72.01 communication tokens and 5.47 versus 1.47 seconds in its tested question-answering tasks; accuracy varies by task. It does not establish that every consequential distinction reaches the text channel. Tucker and colleagues' information-bottleneck study treats utility, informativeness, and complexity as pressures that shape a code, including routes toward more

human-like lexicons. The research archive's LITERATURE_REVIEW_LEDGERS_V2.md and experiments/lab_runs_falsify/culture_dialect_v2.json index the earlier source walk.

• Additional source detail follows Kouwenhoven et al., §§3.2-3.4, §6 and Appendix C, especially speaker prompt C.2. communicativeSuccess records the previous interaction for each stimulus in the prompt context. The illustrative C.2 entries assign wipisu to shape 3 / amount 1 in green and orange with flag 0 and blue with flag 1. These are published prompt examples, not a complete trial trace; Table 1 independently supplies the final vocabulary collision. Transmission selects the agent vocabulary with highest TopSim, a relation between signal and meaning similarity, rather than requiring a unique signal for every meaning. The locker and its coats are teaching images. The paper describes a target plus four distractors in §3 while stating 25% chance in §5.2; V3 retains the reported baseline with attribution rather than claiming to derive its denominator. No new simulation was run.

Chapter 11: The Ledger That Eats Its Own Records

• The church-tower example and repetition-penalty test are in Shumailov and colleagues, *AI models collapse when trained on recursively generated data*, Example 1 and the following ablation. The card example is constructed here; Seddik and colleagues analyze recursive sampling and mixing statistically.

• The ten-generation self-training study is Guo, Wu, and Yiu. Its morphological tests include English, German, and Turkish; its construction-diversity and compositionality measures are evaluated in English. The validation-versus-contraction result and fixed-anchor intervention are reported in Zhang's silent-collapse preprint.

• The accumulation intervention is Gerstgrasser and colleagues. The distinction between training on an accumulated collection and sampling a fixed-size subset is developed in *Collapse or Thrive?*.

• Checkpoint diversity and sampling interventions come from Liu and Han. Other reported brakes include ForTIFAI's token filtering, entropy-based selection in diffusion models, and multimodal generation interventions. These are distinct experimental settings.

• The retrieval feedback study is Druck and Smith, especially §§4 and 8. The apprentice, delivery company, smaller-language speaker, and library messenger are explanatory scenes. The archive's local witness experiments belong to Chapter 9; no new experiments were run for this chapter. The accompanying source walk records the regimes, metric units, and refinements to both earlier versions.

• Additional source detail follows Gerstgrasser et al., §2.1 and Appendix C/Table 2: TinyStories is a 470-million-token collection generated by GPT-3.5/4; newly initialized models are trained at each iteration; replacement uses the latest output, accumulation concatenates it with previous datasets. In the 9M GPT-2 row, evaluation cross-entropy is 1.82 at iteration 1; at iteration 4, full accumulation 1.74, ordinary replacement 2.39, growing replacement 2.18. Growing replacement matches training-set size, not total pipeline compute (Appendix E). Section 2.3 reports slower deterioration under image accumulation and less frequent minor details in generated faces; Appendix D retains detailed reconstructions and offers a latent-space explanation as a hypothesis. The chapter's earlier all-modality avoidance sentence has been refined to preserve these distinct outcomes. No experiment was rerun. The workshop and humid-kitchen warning are teaching images.

Chapter 12: Is Anyone Home?

• The opening is Owen and colleagues' 2006 command-imagery case. The larger study is Bodien and colleagues, 2024: sixty of 241 behaviorally nonresponding participants and forty-three of 112 behaviorally responding participants showed task-based fMRI or EEG responses. Its cohort was a convenience sample.

• Sarasso and colleagues compared anesthesia-related unresponsiveness, later reports, and perturbational complexity. The rubber-hand example is Botvinick and Cohen; subsequent virtual-hand experiments investigate related ownership effects.

• Mary comes from Jackson's 1982 *Epiphenomenal Qualia*. She is a philosophical thought experiment. The process-self argument develops the supplied consciousness context and the author's worldview chapter on organism, relation, prediction, and narrative.

• The rival theory map is supported by Seth and Bayne. *A Beautiful Loop* is a theory proposal. The Cogitate adversarial collaboration tested specified human visual-consciousness predictions; its results do not select a final theory of experience.

• The architecture-based machine investigation is Butlin and colleagues' 2023 report. Its assessment concerns the systems it evaluated then; its stated absence of an obvious technical barrier concerns satisfaction of proposed indicators, not a demonstrated future experience. The machine-payment scene is constructed, and the record interventions refer back to Chapter 9. The accompanying source walk records the clinical denominators, the theory assumptions, and the separation of functional tests from experience claims.

• Additional source detail includes Monti et al., 2010, author-lab full paper, especially pp. 581, 585 and 587/Figure 3. Patient 23 is distinct from the 2006 opening patient. In a 54-patient convenience sample, five modulated imagery responses; only one patient underwent the six-question communication experiment. The cue was neutral (answer), investigators did not know factual answers before decoding, five responses were correct, and the sixth had no usable regional activity. Published personal names were changed for privacy. Further bedside examination reclassified this patient as minimally conscious but established no functional bedside communication. These findings do not establish how long awareness was present before scanning, or explain the sixth nonresponse. The two-box form is a teaching image.

Chapter 13: Who Gets to Amend the Record?

• The opening comes from *Hamilton and others v Post Office Ltd*, [2021] EWCA Crim 577, especially paragraphs 113-114, 121-131, 142-148, and 447. The court records Hamilton's calls, plea terms, alleged shortage, absence of corroborated actual loss, and quashed convictions. The five-job reading of these events is the book's synthesis, not the court's terminology. The Criminal Cases Review Commission's April 23, 2021 account confirms thirty-nine convictions quashed. The chapter makes no claim about later compensation or the present state of every Horizon case.

• Nayebi's selection theorem, also published in the UAI proceedings, supplies the formal pressure to preserve histories that require different confident decisions. The jars are constructed. The five-job synthesis combines that result with the book's biological, institutional, and engineered evidence.

• The bees, pigeons, recipe, legal cases, fable, language experiments, training regimes, and clinical distinctions return to Chapters 1-12 and their primary-source notes. Green Mountain's allowed amount and payment are recorded in the filed trustee final account, *In re Central CT Coop*, No. 16-21255, doc. 498.

• The project's navigator records are in research/NAVIGATOR_ABLATION_AND_TRANSFER.md; the wrong-ledger finding is traced in research/S126_WITNESS_SOURCE_CORRECTION_2026-09-26.md. The separate 200-turn irreversible-stakes result and research/200TURN_P1_FALSE_AS_FINDING.md report three paired seeds, twelve attribution probes per seed, pooled accuracy of 0.75 versus 0.6667, and monitoring-change rates of 0.12 versus 0.04 across 200 evaluated turns per seed. Dissolution and ceiling reductions occurred only under the tested penalty. This is a local negative result under one brake design, not a claim that stakes never matter. No

new experiment was run for this chapter. The mill's second notebook is constructed. This chapter states the cross-domain conclusion earned by the earlier mechanisms; the accompanying source walk records the comparison with V1/V2 and the remaining local content-sensitive witness test.

• Additional source detail follows *Hamilton*, §§6, 11-18, 21 and 52-54. Branch rollover depended on completing a trading statement; no in-system facility disputed figures; challenged sums were treated as debts. Fujitsu held ARQ audit data and PEAK/KEL error records, and POL had a contractual right to obtain information. The court reports no known disclosure of those error documents in the forty-two prosecutions before it. Fraser's civil findings concerning about 580 operators provided a factual basis for criminal appeals; CCRC referrals became appeals, with exceptional circumstances accepted despite only two prior appeals. These are the 2021 judgment's procedural findings, not current legal advice or a new claim about every later Horizon case. The till is a teaching image.

• Criminal Cases Review Commission, "Your questions answered", "Making your application", and "Our powers and practices", checked 13 October 2026: fee-free application, no compulsory lawyer, independent review and evidence-gathering powers. Its jurisdiction includes criminal convictions/sentences in England, Wales and Northern Ireland, and specified service courts; Scotland has a separate commission. Public-body material and court-ordered private material follow different powers. Eligibility and referral conditions still apply; these provisions do not establish universal access or successful outcomes.

• Lucy Welsh and Amy Clarke, *Applicant Experience of the Criminal Cases Review Commission (CCRC)*, Nuffield Foundation, 2025, especially pp. 4-6, 13-14, 17-18 and 42-43. The qualitative study interviewed 49 participants in 2024, including applicants, prospective applicants and family members; it also analyzed 47 letters. The sample was not designed to be representative. Accounts of difficulty and of an easy application form describe participants' experiences, not adjudicated errors or a population estimate.

Chapter 14: The Agency Left to Us

• The morning and driving passages are constructed illustrations. The chapter develops research/consciousness_self_free_will_context.md, AGENCY_AND_THE_CONSTRUCTED_SELF.md, AGENCY_OPERATIONAL_PROTOCOL_2026-09-25.md, WHAT_YOU_CAN_ACTUALLY_DO.md, LTP_LTD_LOOPS.md, the three supplied agency texts, and the author's worldview Chapter 10. The three scales organize the argument; they are not a measured ranking of every person's control.

• Libet et al., *Brain* (1983) studied the relation between preparatory activity, reported intention, and movement. The approximately 200 ms interval precedes movement. Johansson et al., *Science* (2005) supplies the separate choice-blindness result. Schurger et al., *PNAS* (2012) tests an accumulator interpretation. Maoz et al., *eLife* (2019) compares arbitrary and deliberate decisions; eighteen participants were analyzed, and real donations were determined through lotteries.

• The molecular walk draws on Giese et al. (1998), Frey and Morris (1998), Barco, Alarcon, and Kandel (2002), and Harward et al. (2016). These interventions address distinct mechanisms of plasticity. CaMKII phosphorylation is not equated with the entire synaptic tag or a permanent one-bit store.

• Purves et al., *Neuroscience*, "Increased Conduction Velocity as a Result of Myelination" (2001) gives fiber-speed ranges; the National Research Council's "Biologic Basis of Neurotoxicity" (1992) describes an increase up to a factor of 100. McKenzie et al., *Science* (2014) tests new oligodendrocyte production in mouse motor learning. These support the conduction contrast and learning mechanism, with different units.

• Lee and Wilson, *Neuron* (2002) records approximately twentyfold temporal compression in rat hippocampal sleep sequences. De Vivo et al., *Science* (2017) measures selective changes in mouse cortical synaptic contact area. Neither study tests an overnight rewrite of human identity.

• Gollwitzer and Sheeran (2006) reports an average goal-attainment effect of $d = .65$ across ninety-four tests. Harkin et al. (2016) reports $d = .40$ across 138 studies and 19,951 participants. These are standardized effects, not percentage gains or guarantees for the book's exact routine. The two-minute trial and dated amendment are developed in Chapter 15. The accompanying source walk records the primary checks and the comparison with earlier versions; no new laboratory experiment was run for this chapter.

• Additional source detail follows Gollwitzer and Brandstätter (1997), Study 2, pp. 191-192: 86 assigned students, 39 returns, three unidentifiable returns, 36 analyzed reports; deadline completion 12/17 versus 6/19 among identifiable returns. Total return frequency did not differ significantly by condition. Earlier postmarks and comparable writing-to-mailing intervals support the reported timing contrast. Keys beside a bag are an author-created teaching image, not a tested intervention.

Chapter 15: The Practice of Amendment

• The seven-day sequence, four-line record, two-minute start, and brief container image are the book's practical design, developed from the supplied agency texts, AGENCY_OPERATIONAL_PROTOCOL_2026-09-25.md, WHAT_YOU_CAN_ACTUALLY_DO.md, UTILITY_OF_TEMPORARY_GULLIBILITY_LIMBIC_CONTAINERIZATION.md, and the author's worldview discussion of practice, perfectionism, and return. The complete package has not been evaluated as an intervention.

• Gollwitzer and Sheeran's implementation-intention meta-analysis supports cue-linked planning. Harkin and colleagues analyzed 138 studies with 19,951 participants and found an average goal-attainment effect of d = 0.40 for monitoring interventions. Wang, Wang, and Gai analyzed mental contrasting with implementation intentions across 21 studies, finding g = 0.336; publication bias could reduce the estimated benefit. These support components of the guide without assigning an exact dose to its routine.

• Kappes and Oettingen's experiments concern idealized positive fantasies and energization. Sirois and Pychyl's review develops the short-term mood-regulation account of procrastination. Lally and colleagues' habit study followed 96 volunteers for twelve weeks; its findings support variable timetables and return after a missed opportunity. Ghazali's religious context is preserved in *On Intention, Sincerity and Truthfulness* and *On Vigilance and Self-Examination*, Books 37-38 of the *Revival*. The worked week supplies illustrative events, not participant data. CH15_PRACTICE_SOURCE_WALK_2026-09-28.md records the source decisions and comparison with the earlier versions.

• The worked illustration extends day six with a constructed request, pending answer and refusal. The message, Thursday deadline, alternative source and key image are worked design choices drawn from the position-versus-attitude distinction in WHAT_YOU_CAN_ACTUALLY_DO.md and the dated amendment in AGENCY_OPERATIONAL_PROTOCOL_2026-09-25.md. These events are not participant observations or a measured efficacy claim.

• NIH, “Selecting a Data Repository”, checked 13 October 2026, identifies sustainability, metadata, access/use guidance, formats, provenance and retention among desirable repository characteristics. These inform the temporary-access illustration; the proposal is not claimed to be NIH-funded or governed by that policy. The Friday access window and handoff are constructed design examples, not participant observations.

Future Outlook: The Witness Gets Built Into Everything

• The scenes, dates, $1 billion thresholds, and forty-word target are the author's forecasts, developed from GRAND_SYNTHESIS_LEDGERS_ESSAY.md, its competing machine horizons, the supplied expanded outlook, and the archive's mapped-nervous-system and digital-substrate notes. They are separated in evidence.sqlite from observed results. The prior outlook is preserved in development/16_future_outlook_the_witness_prior_draft.md; CH16_OUTLOOK_SOURCE_WALK_2026-09-28.md records the fresh checks and editorial decisions.

• The institutional starting points are Munich Re's AI-risk coverage, the European Commission's updated AI Act schedule, and UNCITRAL's 2024 Model Law on Automated Contracting. The Commission reports December 2, 2027 for specified high-risk areas and August 2, 2028 for relevant systems embedded in regulated products, following the 2026 amendment. The proposed witness contracts go beyond those present requirements.

• Weco's AIDE² report supplies the company-reported 63-to-34-percent benchmark comparison. Dream-RSI supplies the discovery-tree replay mechanism; its simulator traverses recorded outcomes and improved policies are redeployed online. This is distinct from independently repeating a physical experiment. TRM Labs' September 2026 analysis applies permissive and strict agent-attribution tests to $25.62 million of screened x402 commerce. Its range is a model-dependent estimate within that dataset.

• The clinical anchors are the individual closed-loop depression report and the recovery-biomarker study. Brain2Qwerty studied typed, briefly memorized sentences; its character errors do not measure the forecast's silent communication rate. The surface-EMG study reported median handwriting input at 20.9 words per minute from muscle signals. MICrONS links structure and recorded function in a small cortical volume; it supplies groundwork for the emulation question. The dated whole-mouse result remains a future test.

• The worked illustration extends the projected procurement meeting with detection, purchase hold, authorized release, an urgent exception and responsibility for an override. This is a forecast illustration and proposed contract design, not a reported deployment, current insurance condition or additional dated forecast. It draws on the essay's competing witness/opacity horizons and the completed chapters' separation of checking, authority and liability. The smoke alarm is a teaching image.

• NIST, *AI RMF Playbook*, Govern 6.1-6.2 and Manage 2.4 and 3.1, checked 13 October 2026, provides voluntary suggestions on third-party risk, independent auditability, contingencies and safe disengagement. It does not mandate the

projected contract clauses. The disconnected-supplier test is an invented extension of the procurement scene; the builder and apprentice continue the book's illustrative closing image.